

|          |                |           |                  |
|----------|----------------|-----------|------------------|
| NOP (00) |                | Cycles: 2 | Size: 1          |
| Implied  |                |           |                  |
| Cycle    | R/W            |           | Desc.            |
| -X.2     | Read <b>PC</b> |           | Store as OpCode. |
| 1.1      |                |           | Inc. <b>PC</b> . |
| 1.2      |                |           |                  |
| 2.1      |                |           |                  |
| 2.2      | Read <b>PC</b> |           | Store as OpCode. |
| +X.1     |                |           |                  |

|            |                                     |                               |
|------------|-------------------------------------|-------------------------------|
| TCALL (01) | Cycles: 8                           | Size: 1                       |
| Implied    |                                     |                               |
| Cycle      | R/W                                 | Desc.                         |
| -X.2       | Read <b>PC</b>                      | Store as OpCode.              |
| 1.1        |                                     | Inc. <b>PC</b> .              |
| 1.2        | Read <b>PC</b>                      | Store as Operand.             |
| 2.1        |                                     |                               |
| 2.2        |                                     |                               |
| 3.1        |                                     |                               |
| 3.2        | Write to u8( <b>SP</b> +0)   \$0100 | Push <b>PC</b> .H onto stack. |
| 4.1        |                                     |                               |
| 4.2        | Write to u8( <b>SP</b> -1)   \$0100 | Push <b>PC</b> .L onto stack. |
| 5.1        |                                     | Dec. <b>SP</b> by 2.          |
| 5.2        |                                     |                               |
| 6.1        |                                     |                               |
| 6.2        | Read \$FFDE                         | Store as <b>PC</b> .L.        |
| 7.1        |                                     |                               |
| 7.2        | Read \$FFDF                         | Store as <b>PC</b> .H.        |
| 8.1        |                                     |                               |
| 8.2        | Read <b>PC</b>                      | Store as OpCode.              |
| +X.1       |                                     |                               |

|                                         |                  |                                                                     |
|-----------------------------------------|------------------|---------------------------------------------------------------------|
| SET1 (02)                               | Cycles: 4        | Size: 2                                                             |
| Direct Page Bit d.b (Read/Modify/Write) |                  |                                                                     |
| Cycle                                   | R/W              | Desc.                                                               |
| -X.2                                    | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                                     |                  | Inc. <b>PC</b> .                                                    |
| 1.2                                     | Read <b>PC</b>   | Store as Operand.                                                   |
| 2.1                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                                     | Read Address     | Store as Operand.                                                   |
| 3.1                                     |                  | Operand = (Operand & ~\$01)   \$01.                                 |
| 3.2                                     | Write to Address | Write Operand.                                                      |
| 4.1                                     |                  |                                                                     |
| 4.2                                     | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                                    |                  |                                                                     |

|                                 |                |                                                                      |
|---------------------------------|----------------|----------------------------------------------------------------------|
| BBS (03)                        | Cycles: 5-7    | Size: 3                                                              |
| Direct Page Bit Relative d.b, r |                |                                                                      |
| Cycle                           | R/W            | Desc.                                                                |
| -X.2                            | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1                             |                | Inc. <b>PC</b> .                                                     |
| 1.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 2.1                             |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).  |
| 2.2                             | Read Address   | Store as Operand.                                                    |
| 3.1                             |                | Set Jump to (Operand & (1 << 0)) == (1 << 0).                        |
| 3.2                             |                |                                                                      |
| 4.1                             |                |                                                                      |
| 4.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 5.1                             |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 7.2). |
| 5.2                             |                |                                                                      |
| 6.1                             |                | Address = <b>PC</b> + i8(Operand).<br><b>PC</b> .L = Address.L.      |
| 6.2                             |                |                                                                      |
| 7.1                             |                | <b>PC</b> .H = Address.H.                                            |
| 7.2                             | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1                            |                |                                                                      |

|                      |                |                                                                                       |
|----------------------|----------------|---------------------------------------------------------------------------------------|
| OR (04)              | Cycles: 3      | Size: 2                                                                               |
| Direct Page d (Read) |                |                                                                                       |
| Cycle                | R/W            | Desc.                                                                                 |
| -X.2                 | Read <b>PC</b> | Store as OpCode.                                                                      |
| 1.1                  |                | Inc. <b>PC</b> .                                                                      |
| 1.2                  | Read <b>PC</b> | Store as Operand.                                                                     |
| 2.1                  |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                   |
| 2.2                  | Read Address   | Store as Operand.                                                                     |
| 3.1                  |                | <b>A</b>  = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 3.2                  | Read <b>PC</b> | Store as OpCode.                                                                      |
| +X.1                 |                |                                                                                       |

|                    |                |                                                                                       |
|--------------------|----------------|---------------------------------------------------------------------------------------|
| OR (05)            | Cycles: 4      | Size: 3                                                                               |
| Absolute !a (Read) |                |                                                                                       |
| Cycle              | R/W            | Desc.                                                                                 |
| -X.2               | Read <b>PC</b> | Store as OpCode.                                                                      |
| 1.1                |                | Inc. <b>PC</b> .                                                                      |
| 1.2                | Read <b>PC</b> | Store as Address.L.                                                                   |
| 2.1                |                | Inc. <b>PC</b> .                                                                      |
| 2.2                | Read <b>PC</b> | Store as Address.H.                                                                   |
| 3.1                |                | Inc. <b>PC</b> .                                                                      |
| 3.2                | Read Address   | Store as Operand.                                                                     |
| 4.1                |                | <b>A</b>  = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 4.2                | Read <b>PC</b> | Store as OpCode.                                                                      |
| +X.1               |                |                                                                                       |

|                       |                                                |                                                                                       |
|-----------------------|------------------------------------------------|---------------------------------------------------------------------------------------|
| OR (06)               | Cycles: 3                                      | Size: 1                                                                               |
| Indirect X (X) (Read) |                                                |                                                                                       |
| Cycle                 | R/W                                            | Desc.                                                                                 |
| -X.2                  | Read <b>PC</b>                                 | Store as OpCode.                                                                      |
| 1.1                   |                                                | Inc. <b>PC</b> .                                                                      |
| 1.2                   |                                                |                                                                                       |
| 2.1                   |                                                |                                                                                       |
| 2.2                   | Read ( <b>X</b>   (( <b>PSW</b> & \$20) << 3)) | Store as Operand.                                                                     |
| 3.1                   |                                                | <b>A</b>  = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 3.2                   | Read <b>PC</b>                                 | Store as OpCode.                                                                      |
| +X.1                  |                                                |                                                                                       |

|                                 |                                                              |                                                                                       |
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| OR (07)                         | Cycles: 6                                                    | Size: 2                                                                               |
| X-Indexed Indirect [d+X] (Read) |                                                              |                                                                                       |
| Cycle                           | R/W                                                          | Desc.                                                                                 |
| -X.2                            | Read <b>PC</b>                                               | Store as OpCode.                                                                      |
| 1.1                             |                                                              | Inc. <b>PC</b> .                                                                      |
| 1.2                             | Read <b>PC</b>                                               | Store as Operand.                                                                     |
| 2.1                             |                                                              | Inc. <b>PC</b> .                                                                      |
| 2.2                             |                                                              |                                                                                       |
| 3.1                             |                                                              | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).               |
| 3.2                             | Read Address                                                 | Store as Pointer.L.                                                                   |
| 4.1                             |                                                              |                                                                                       |
| 4.2                             | Read ((Address + 1) & \$FF)  <br>(( <b>PSW</b> & \$20) << 3) | Store as Pointer.H.                                                                   |
| 5.1                             |                                                              |                                                                                       |
| 5.2                             | Read Pointer                                                 | Store as Operand.                                                                     |
| 6.1                             |                                                              | <b>A</b>  = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 6.2                             | Read <b>PC</b>                                               | Store as OpCode.                                                                      |
| +X.1                            |                                                              |                                                                                       |



|           |                |                                                                                                        |
|-----------|----------------|--------------------------------------------------------------------------------------------------------|
| OR (08)   | Cycles: 2      | Size: 2                                                                                                |
| Immediate |                |                                                                                                        |
| Cycle     | R/W            | Desc.                                                                                                  |
| -X.2      | Read <b>PC</b> | Store as OpCode.                                                                                       |
| 1.1       |                | Inc. <b>PC</b> .                                                                                       |
| 1.2       | Read <b>PC</b> | Store as Operand.                                                                                      |
| 2.1       |                | Inc. <b>PC</b> . <b>A</b>  = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 2.2       | Read <b>PC</b> | Store as OpCode.                                                                                       |
| +X.1      |                |                                                                                                        |

|                                                       |                  |                                                                                     |
|-------------------------------------------------------|------------------|-------------------------------------------------------------------------------------|
| OR (09)                                               | Cycles: 6        | Size: 3                                                                             |
| Direct Page to Direct Page dd, ds (Read/Modify/Write) |                  |                                                                                     |
| Cycle                                                 | R/W              | Desc.                                                                               |
| -X.2                                                  | Read <b>PC</b>   | Store as OpCode.                                                                    |
| 1.1                                                   |                  | Inc. <b>PC</b> .                                                                    |
| 1.2                                                   | Read <b>PC</b>   | Store as Operand.                                                                   |
| 2.1                                                   |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                 |
| 2.2                                                   | Read Address     | Store as Operand0.                                                                  |
| 3.1                                                   |                  |                                                                                     |
| 3.2                                                   | Read <b>PC</b>   | Store as Operand.                                                                   |
| 4.1                                                   |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                 |
| 4.2                                                   | Read Address     | Store as Operand1.                                                                  |
| 5.1                                                   |                  | Operand0  = Operand1. <b>N</b> flag = (Operand0 & \$80), <b>Z</b> flag = !Operand0. |
| 5.2                                                   | Write to Address | Write Operand0.                                                                     |
| 6.1                                                   |                  |                                                                                     |
| 6.2                                                   | Read <b>PC</b>   | Store as OpCode.                                                                    |
| +X.1                                                  |                  |                                                                                     |

|                          |                         |                                                                         |
|--------------------------|-------------------------|-------------------------------------------------------------------------|
| OR1 (0A)                 | Cycles: 5               | Size: 3                                                                 |
| Absolute Boolean Bit m.b |                         |                                                                         |
| Cycle                    | R/W                     | Desc.                                                                   |
| -X.2                     | Read <b>PC</b>          | Store as OpCode.                                                        |
| 1.1                      |                         | Inc. <b>PC</b> .                                                        |
| 1.2                      | Read <b>PC</b>          | Store as Address.L.                                                     |
| 2.1                      |                         | Inc. <b>PC</b> .                                                        |
| 2.2                      | Read <b>PC</b>          | Store as Address.H.                                                     |
| 3.1                      |                         | Inc. <b>PC</b> .                                                        |
| 3.2                      | Read (Address & \$1FFF) | Store as Operand.                                                       |
| 4.1                      |                         |                                                                         |
| 4.2                      |                         |                                                                         |
| 5.1                      |                         | Bit = Address >> 13.<br><b>C</b> flag  = ((Operand & (1 << Bit)) != 0). |
| 5.2                      | Read <b>PC</b>          | Store as OpCode.                                                        |
| +X.1                     |                         |                                                                         |

|                                   |                  |                                                                                                             |
|-----------------------------------|------------------|-------------------------------------------------------------------------------------------------------------|
| ASL (0B)                          | Cycles: 4        | Size: 2                                                                                                     |
| Direct Page d (Read/Modify/Write) |                  |                                                                                                             |
| Cycle                             | R/W              | Desc.                                                                                                       |
| -X.2                              | Read <b>PC</b>   | Store as OpCode.                                                                                            |
| 1.1                               |                  | Inc. <b>PC</b> .                                                                                            |
| 1.2                               | Read <b>PC</b>   | Store as Operand.                                                                                           |
| 2.1                               |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                         |
| 2.2                               | Read Address     | Store as Operand.                                                                                           |
| 3.1                               |                  | <b>C</b> flag = (Operand & \$80). Operand <= 1. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = !Operand. |
| 3.2                               | Write to Address | Write Operand.                                                                                              |
| 4.1                               |                  |                                                                                                             |
| 4.2                               | Read <b>PC</b>   | Store as OpCode.                                                                                            |
| +X.1                              |                  |                                                                                                             |

|                                 |                  |                                                                                                             |
|---------------------------------|------------------|-------------------------------------------------------------------------------------------------------------|
| ASL (0C)                        | Cycles: 5        | Size: 3                                                                                                     |
| Absolute !a (Read/Modify/Write) |                  |                                                                                                             |
| Cycle                           | R/W              | Desc.                                                                                                       |
| -X.2                            | Read <b>PC</b>   | Store as OpCode.                                                                                            |
| 1.1                             |                  | Inc. <b>PC</b> .                                                                                            |
| 1.2                             | Read <b>PC</b>   | Store as Address.L.                                                                                         |
| 2.1                             |                  | Inc. <b>PC</b> .                                                                                            |
| 2.2                             | Read <b>PC</b>   | Store as Address.H.                                                                                         |
| 3.1                             |                  | Inc. <b>PC</b> .                                                                                            |
| 3.2                             | Read Address     | Store as Operand.                                                                                           |
| 4.1                             |                  | <b>C</b> flag = (Operand & \$80). Operand <= 1. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = !Operand. |
| 4.2                             | Write to Address | Write Operand.                                                                                              |
| 5.1                             |                  |                                                                                                             |
| 5.2                             | Read <b>PC</b>   | Store as OpCode.                                                                                            |
| +X.1                            |                  |                                                                                                             |

|           |                                     |                             |
|-----------|-------------------------------------|-----------------------------|
| PUSH (0D) | Cycles: 4                           | Size: 1                     |
| Implied   |                                     |                             |
| Cycle     | R/W                                 | Desc.                       |
| -X.2      | Read <b>PC</b>                      | Store as OpCode.            |
| 1.1       |                                     | Inc. <b>PC</b> .            |
| 1.2       | Read <b>PC</b>                      | Discard.                    |
| 2.1       |                                     |                             |
| 2.2       | Write to u8( <b>SP</b> +0)   \$0100 | Push <b>PSW</b> onto stack. |
| 3.1       |                                     | Dec. <b>SP</b> by 1.        |
| 3.2       |                                     |                             |
| 4.1       |                                     |                             |
| 4.2       | Read <b>PC</b>                      | Store as OpCode.            |
| +X.1      |                                     |                             |

|                                 |                  |                                                                                                                    |
|---------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------|
| TSET1 (0E)                      | Cycles: 6        | Size: 3                                                                                                            |
| Absolute !a (Read/Modify/Write) |                  |                                                                                                                    |
| Cycle                           | R/W              | Desc.                                                                                                              |
| -X.2                            | Read <b>PC</b>   | Store as OpCode.                                                                                                   |
| 1.1                             |                  | Inc. <b>PC</b> .                                                                                                   |
| 1.2                             | Read <b>PC</b>   | Store as Address.L.                                                                                                |
| 2.1                             |                  | Inc. <b>PC</b> .                                                                                                   |
| 2.2                             | Read <b>PC</b>   | Store as Address.H.                                                                                                |
| 3.1                             |                  | Inc. <b>PC</b> .                                                                                                   |
| 3.2                             | Read Address     | Store as Operand.                                                                                                  |
| 4.1                             |                  | Tmp = ( <b>A</b> - Operand).<br>Operand  = <b>A</b> .<br><b>Z</b> flag = (Tmp == 0), <b>N</b> flag = (Tmp & \$80). |
| 4.2                             | Read Address     | Discard.                                                                                                           |
| 5.1                             |                  |                                                                                                                    |
| 5.2                             | Write to Address | Write Operand.                                                                                                     |
| 6.1                             |                  |                                                                                                                    |
| 6.2                             | Read <b>PC</b>   | Store as OpCode.                                                                                                   |
| +X.1                            |                  |                                                                                                                    |

|          |                                     |                                                            |
|----------|-------------------------------------|------------------------------------------------------------|
| BRK (0F) | Cycles: 8                           | Size: 1                                                    |
| Implied  |                                     |                                                            |
| Cycle    | R/W                                 | Desc.                                                      |
| -X.2     | Read <b>PC</b>                      | Store as OpCode.                                           |
| 1.1      |                                     | Inc. <b>PC</b> .                                           |
| 1.2      | Read <b>PC</b>                      | Discard.                                                   |
| 2.1      |                                     |                                                            |
| 2.2      | Write to u8( <b>SP</b> +0)   \$0100 | Push <b>PC</b> .H onto stack.                              |
| 3.1      |                                     |                                                            |
| 3.2      | Write to u8( <b>SP</b> -1)   \$0100 | Push <b>PC</b> .L onto stack.                              |
| 4.1      |                                     |                                                            |
| 4.2      | Write to u8( <b>SP</b> -2)   \$0100 | Push <b>PSW</b> onto stack.                                |
| 5.1      |                                     | Dec. <b>SP</b> by 3. <b>I</b> flag = 0, <b>X</b> flag = 1. |
| 5.2      |                                     |                                                            |
| 6.1      |                                     |                                                            |
| 6.2      | Read \$FFDE                         | Store as <b>PC</b> .L.                                     |
| 7.1      |                                     |                                                            |
| 7.2      | Read \$FFDF                         | Store as <b>PC</b> .H.                                     |
| 8.1      |                                     |                                                            |
| 8.2      | Read <b>PC</b>                      | Store as OpCode.                                           |
| +X.1     |                                     |                                                            |



|          |                |                                                                      |
|----------|----------------|----------------------------------------------------------------------|
| BPL (10) | Cycles: 2-4    | Size: 2                                                              |
| Implied  |                |                                                                      |
| Cycle    | R/W            | Desc.                                                                |
| -X.2     | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1      |                | Inc. <b>PC</b> . Set Jump to ( <b>PSW</b> & \$80) == \$00.           |
| 1.2      |                | Store as Operand.                                                    |
| 2.1      |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 4.2). |
| 2.2      |                |                                                                      |
| 3.1      |                | Address = <b>PC</b> + i8(Operand).                                   |
|          |                | <b>PC</b> .L = Address.L.                                            |
| 3.2      |                |                                                                      |
| 4.1      |                | <b>PC</b> .H = Address.H.                                            |
| 4.2      | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1     |                |                                                                      |

|            |                                     |                               |
|------------|-------------------------------------|-------------------------------|
| TCALL (11) | Cycles: 8                           | Size: 1                       |
| Implied    |                                     |                               |
| Cycle      | R/W                                 | Desc.                         |
| -X.2       | Read <b>PC</b>                      | Store as OpCode.              |
| 1.1        |                                     | Inc. <b>PC</b> .              |
| 1.2        | Read <b>PC</b>                      | Store as Operand.             |
| 2.1        |                                     |                               |
| 2.2        |                                     |                               |
| 3.1        |                                     |                               |
| 3.2        | Write to u8( <b>SP</b> +0)   \$0100 | Push <b>PC</b> .H onto stack. |
| 4.1        |                                     |                               |
| 4.2        | Write to u8( <b>SP</b> -1)   \$0100 | Push <b>PC</b> .L onto stack. |
| 5.1        |                                     | Dec. <b>SP</b> by 2.          |
| 5.2        |                                     |                               |
| 6.1        |                                     |                               |
| 6.2        | Read \$FFDC                         | Store as <b>PC</b> .L.        |
| 7.1        |                                     |                               |
| 7.2        | Read \$FFDD                         | Store as <b>PC</b> .H.        |
| 8.1        |                                     |                               |
| 8.2        | Read <b>PC</b>                      | Store as OpCode.              |
| +X.1       |                                     |                               |

|                                         |                  |                                                                     |
|-----------------------------------------|------------------|---------------------------------------------------------------------|
| CLR1 (12)                               | Cycles: 4        | Size: 2                                                             |
| Direct Page Bit d.b (Read/Modify/Write) |                  |                                                                     |
| Cycle                                   | R/W              | Desc.                                                               |
| -X.2                                    | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                                     |                  | Inc. <b>PC</b> .                                                    |
| 1.2                                     | Read <b>PC</b>   | Store as Operand.                                                   |
| 2.1                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                                     | Read Address     | Store as Operand.                                                   |
| 3.1                                     |                  | Operand = (Operand & ~\$01)   \$00.                                 |
| 3.2                                     | Write to Address | Write Operand.                                                      |
| 4.1                                     |                  |                                                                     |
| 4.2                                     | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                                    |                  |                                                                     |

|                                 |                |                                                                      |
|---------------------------------|----------------|----------------------------------------------------------------------|
| BBC (13)                        | Cycles: 5-7    | Size: 3                                                              |
| Direct Page Bit Relative d.b, r |                |                                                                      |
| Cycle                           | R/W            | Desc.                                                                |
| -X.2                            | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1                             |                | Inc. <b>PC</b> .                                                     |
| 1.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 2.1                             |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).  |
| 2.2                             | Read Address   | Store as Operand.                                                    |
| 3.1                             |                | Set Jump to (Operand & (1 << 0)) == (0 << 0).                        |
| 3.2                             |                |                                                                      |
| 4.1                             |                |                                                                      |
| 4.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 5.1                             |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 7.2). |
| 5.2                             |                |                                                                      |
| 6.1                             |                | Address = <b>PC</b> + i8(Operand).<br><b>PC</b> .L = Address.L.      |
| 6.2                             |                |                                                                      |
| 7.1                             |                | <b>PC</b> .H = Address.H.                                            |
| 7.2                             | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1                            |                |                                                                      |

|                                  |                |                                                                                       |
|----------------------------------|----------------|---------------------------------------------------------------------------------------|
| OR (14)                          | Cycles: 4      | Size: 2                                                                               |
| X-Indexed Direct Page d+X (Read) |                |                                                                                       |
| Cycle                            | R/W            | Desc.                                                                                 |
| -X.2                             | Read <b>PC</b> | Store as OpCode.                                                                      |
| 1.1                              |                | Inc. <b>PC</b> .                                                                      |
| 1.2                              | Read <b>PC</b> | Store as Operand.                                                                     |
| 2.1                              |                | Inc. <b>PC</b> .                                                                      |
| 2.2                              |                |                                                                                       |
| 3.1                              |                | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).               |
| 3.2                              | Read Address   | Store as Operand.                                                                     |
| 4.1                              |                | <b>A</b>  = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 4.2                              | Read <b>PC</b> | Store as OpCode.                                                                      |
| +X.1                             |                |                                                                                       |

|                                |                |                                                                                       |
|--------------------------------|----------------|---------------------------------------------------------------------------------------|
| OR (15)                        | Cycles: 5      | Size: 3                                                                               |
| X-Indexed Absolute !a+X (Read) |                |                                                                                       |
| Cycle                          | R/W            | Desc.                                                                                 |
| -X.2                           | Read <b>PC</b> | Store as OpCode.                                                                      |
| 1.1                            |                | Inc. <b>PC</b> .                                                                      |
| 1.2                            | Read <b>PC</b> | Store as Address.L.                                                                   |
| 2.1                            |                | Inc. <b>PC</b> .                                                                      |
| 2.2                            | Read <b>PC</b> | Store as Address.H.                                                                   |
| 3.1                            |                | Inc. <b>PC</b> .                                                                      |
| 3.2                            |                |                                                                                       |
| 4.1                            |                | Pointer = ( <b>X</b> + Address).                                                      |
| 4.2                            | Read Pointer   | Store as Operand.                                                                     |
| 5.1                            |                | <b>A</b>  = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 5.2                            | Read <b>PC</b> | Store as OpCode.                                                                      |
| +X.1                           |                |                                                                                       |

|                                |                |                                                                                       |
|--------------------------------|----------------|---------------------------------------------------------------------------------------|
| OR (16)                        | Cycles: 5      | Size: 3                                                                               |
| Y-Indexed Absolute !a+Y (Read) |                |                                                                                       |
| Cycle                          | R/W            | Desc.                                                                                 |
| -X.2                           | Read <b>PC</b> | Store as OpCode.                                                                      |
| 1.1                            |                | Inc. <b>PC</b> .                                                                      |
| 1.2                            | Read <b>PC</b> | Store as Address.L.                                                                   |
| 2.1                            |                | Inc. <b>PC</b> .                                                                      |
| 2.2                            | Read <b>PC</b> | Store as Address.H.                                                                   |
| 3.1                            |                | Inc. <b>PC</b> .                                                                      |
| 3.2                            |                |                                                                                       |
| 4.1                            |                | Pointer = ( <b>Y</b> + Address).                                                      |
| 4.2                            | Read Pointer   | Store as Operand.                                                                     |
| 5.1                            |                | <b>A</b>  = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 5.2                            | Read <b>PC</b> | Store as OpCode.                                                                      |
| +X.1                           |                |                                                                                       |

|                                 |                                                           |                                                                                       |
|---------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------|
| OR (17)                         | Cycles: 6                                                 | Size: 2                                                                               |
| Indirect Y-Indexed [d]+Y (Read) |                                                           |                                                                                       |
| Cycle                           | R/W                                                       | Desc.                                                                                 |
| -X.2                            | Read <b>PC</b>                                            | Store as OpCode.                                                                      |
| 1.1                             |                                                           | Inc. <b>PC</b> .                                                                      |
| 1.2                             | Read <b>PC</b>                                            | Store as Operand.                                                                     |
| 2.1                             |                                                           | Inc. <b>PC</b> .                                                                      |
| 2.2                             |                                                           |                                                                                       |
| 3.1                             |                                                           | Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                    |
| 3.2                             | Read Address                                              | Store as Pointer.L.                                                                   |
| 4.1                             |                                                           |                                                                                       |
| 4.2                             | Read ((Address + 1) & \$FF)   (( <b>PSW</b> & \$20) << 3) | Store as Pointer.H.                                                                   |
| 5.1                             |                                                           | Address = ( <b>Y</b> + Pointer).                                                      |
| 5.2                             | Read Address                                              | Store as Operand.                                                                     |
| 6.1                             |                                                           | <b>A</b>  = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 6.2                             | Read <b>PC</b>                                            | Store as OpCode.                                                                      |
| +X.1                            |                                                           |                                                                                       |



|                                                         |                  |                                                                                     |
|---------------------------------------------------------|------------------|-------------------------------------------------------------------------------------|
| OR (18)                                                 | Cycles: 5        | Size: 3                                                                             |
| Immediate Data to Direct Page d, #i (Read/Modify/Write) |                  |                                                                                     |
| Cycle                                                   | R/W              | Desc.                                                                               |
| -X.2                                                    | Read <b>PC</b>   | Store as OpCode.                                                                    |
| 1.1                                                     |                  | Inc. <b>PC</b> .                                                                    |
| 1.2                                                     | Read <b>PC</b>   | Store as Operand0.                                                                  |
| 2.1                                                     |                  | Inc. <b>PC</b> .                                                                    |
| 2.2                                                     | Read <b>PC</b>   | Store as Operand.                                                                   |
| 3.1                                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                 |
| 3.2                                                     | Read Address     | Store as Operand1.                                                                  |
| 4.1                                                     |                  | Operand0  = Operand1. <b>N</b> flag = (Operand0 & \$80), <b>Z</b> flag = !Operand0. |
| 4.2                                                     | Write to Address | Write Operand0.                                                                     |
| 5.1                                                     |                  |                                                                                     |
| 5.2                                                     | Read <b>PC</b>   | Store as OpCode.                                                                    |
| +X.1                                                    |                  |                                                                                     |

|                                |                                                    |                                                                                     |
|--------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------|
| OR (19)                        | Cycles: 5                                          | Size: 1                                                                             |
| Indirect Page to Indirect Page |                                                    |                                                                                     |
| Cycle                          | R/W                                                | Desc.                                                                               |
| -X.2                           | Read <b>PC</b>                                     | Store as OpCode.                                                                    |
| 1.1                            |                                                    | Inc. <b>PC</b> .                                                                    |
| 1.2                            | Read <b>PC</b>                                     | Discard.                                                                            |
| 2.1                            |                                                    |                                                                                     |
| 2.2                            | Read ( <b>Y</b>   (( <b>PSW</b> & \$20) << 3))     | Store as Operand0.                                                                  |
| 3.1                            |                                                    |                                                                                     |
| 3.2                            | Read ( <b>X</b>   (( <b>PSW</b> & \$20) << 3))     | Store as Operand1.                                                                  |
| 4.1                            |                                                    | Operand0  = Operand1. <b>N</b> flag = (Operand0 & \$80), <b>Z</b> flag = !Operand0. |
| 4.2                            | Write to ( <b>X</b>   (( <b>PSW</b> & \$20) << 3)) | Write Operand0.                                                                     |
| 5.1                            |                                                    |                                                                                     |
| 5.2                            | Read <b>PC</b>                                     | Store as OpCode.                                                                    |
| +X.1                           |                                                    |                                                                                     |

|                                      |                                                               |                                                                                                |
|--------------------------------------|---------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| DECW (1A)                            | Cycles: 6                                                     | Size: 2                                                                                        |
| Direct Page d (Read/Modify/Write 16) |                                                               |                                                                                                |
| Cycle                                | R/W                                                           | Desc.                                                                                          |
| -X.2                                 | Read <b>PC</b>                                                | Store as OpCode.                                                                               |
| 1.1                                  |                                                               | Inc. <b>PC</b> .                                                                               |
| 1.2                                  | Read <b>PC</b>                                                | Store as Operand.                                                                              |
| 2.1                                  |                                                               | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                            |
| 2.2                                  | Read Address                                                  | Store as Operand.                                                                              |
| 3.1                                  |                                                               | Underflow = (Operand == 0).<br>Operand -= 1.<br>LowZero = (Operand == 0).                      |
| 3.2                                  | Write to Address                                              | Write Operand.                                                                                 |
| 4.1                                  |                                                               |                                                                                                |
| 4.2                                  | Read ((Address + 1) & \$FF)   (( <b>PSW</b> & \$20) << 3)     | Store as Operand.                                                                              |
| 5.1                                  |                                                               | Operand -= Underflow. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = (LowZero && !Operand). |
| 5.2                                  | Write to ((Address + 1) & \$FF)   (( <b>PSW</b> & \$20) << 3) | Write Operand.                                                                                 |
| 6.1                                  |                                                               |                                                                                                |
| 6.2                                  | Read <b>PC</b>                                                | Store as OpCode.                                                                               |
| +X.1                                 |                                                               |                                                                                                |

|                                               |                  |                                                                                                             |
|-----------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------|
| ASL (1B)                                      | Cycles: 5        | Size: 2                                                                                                     |
| X-Indexed Direct Page d+X (Read/Modify/Write) |                  |                                                                                                             |
| Cycle                                         | R/W              | Desc.                                                                                                       |
| -X.2                                          | Read <b>PC</b>   | Store as OpCode.                                                                                            |
| 1.1                                           |                  | Inc. <b>PC</b> .                                                                                            |
| 1.2                                           | Read <b>PC</b>   | Store as Operand.                                                                                           |
| 2.1                                           |                  | Inc. <b>PC</b> .                                                                                            |
| 2.2                                           |                  |                                                                                                             |
| 3.1                                           |                  | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).                                     |
| 3.2                                           | Read Address     | Store as Operand.                                                                                           |
| 4.1                                           |                  | <b>C</b> flag = (Operand & \$80). Operand <= 1. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = !Operand. |
| 4.2                                           | Write to Address | Write Operand.                                                                                              |
| 5.1                                           |                  |                                                                                                             |
| 5.2                                           | Read <b>PC</b>   | Store as OpCode.                                                                                            |
| +X.1                                          |                  |                                                                                                             |

|               |                |                                                                                                                      |
|---------------|----------------|----------------------------------------------------------------------------------------------------------------------|
| ASL (1C)      | Cycles: 2      | Size: 1                                                                                                              |
| Accumulator A |                |                                                                                                                      |
| Cycle         | R/W            | Desc.                                                                                                                |
| -X.2          | Read <b>PC</b> | Store as OpCode.                                                                                                     |
| 1.1           |                | Inc. <b>PC</b> .                                                                                                     |
| 1.2           | Read <b>PC</b> | Discard.                                                                                                             |
| 2.1           |                | <b>C</b> flag = ( <b>A</b> & \$80). <b>A</b> <<= 1. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 2.2           | Read <b>PC</b> | Store as OpCode.                                                                                                     |
| +X.1          |                |                                                                                                                      |

|          |                |                                                                                 |
|----------|----------------|---------------------------------------------------------------------------------|
| DEC (1D) | Cycles: 2      | Size: 1                                                                         |
| Implied  |                |                                                                                 |
| Cycle    | R/W            | Desc.                                                                           |
| -X.2     | Read <b>PC</b> | Store as OpCode.                                                                |
| 1.1      |                | Inc. <b>PC</b> .                                                                |
| 1.2      |                |                                                                                 |
| 2.1      |                | <b>X</b> -= 1. <b>N</b> flag = ( <b>X</b> & \$80), <b>Z</b> flag = ! <b>X</b> . |
| 2.2      |                | Store as OpCode.                                                                |
| +X.1     |                |                                                                                 |

|                    |                |                                                                                                           |
|--------------------|----------------|-----------------------------------------------------------------------------------------------------------|
| CMP (1E)           | Cycles: 4      | Size: 3                                                                                                   |
| Absolute !a (Read) |                |                                                                                                           |
| Cycle              | R/W            | Desc.                                                                                                     |
| -X.2               | Read <b>PC</b> | Store as OpCode.                                                                                          |
| 1.1                |                | Inc. <b>PC</b> .                                                                                          |
| 1.2                | Read <b>PC</b> | Store as Address.L.                                                                                       |
| 2.1                |                | Inc. <b>PC</b> .                                                                                          |
| 2.2                | Read <b>PC</b> | Store as Address.H.                                                                                       |
| 3.1                |                | Inc. <b>PC</b> .                                                                                          |
| 3.2                | Read Address   | Store as Operand.                                                                                         |
| 4.1                |                | Tmp = <b>X</b> - Operand. <b>C</b> flag = (Tmp >= 0), <b>N</b> flag = (Tmp & \$80), <b>Z</b> flag = !Tmp. |
| 4.2                | Read <b>PC</b> | Store as OpCode.                                                                                          |
| +X.1               |                |                                                                                                           |

|                                    |                  |                                  |
|------------------------------------|------------------|----------------------------------|
| JMP (1F)                           | Cycles: 6        | Size: 3                          |
| Absolute X-Indexed Indirect [!a+X] |                  |                                  |
| Cycle                              | R/W              | Desc.                            |
| -X.2                               | Read <b>PC</b>   | Store as OpCode.                 |
| 1.1                                |                  | Inc. <b>PC</b> .                 |
| 1.2                                | Read <b>PC</b>   | Store as Address.L.              |
| 2.1                                |                  | Inc. <b>PC</b> .                 |
| 2.2                                | Read <b>PC</b>   | Store as Address.H.              |
| 3.1                                |                  |                                  |
| 3.2                                |                  |                                  |
| 4.1                                |                  | Pointer = ( <b>X</b> + Address). |
| 4.2                                | Read Pointer     | Store as <b>PC</b> .L.           |
| 5.1                                |                  |                                  |
| 5.2                                | Read Pointer + 1 | Store as <b>PC</b> .H.           |
| 6.1                                |                  |                                  |
| 6.2                                | Read <b>PC</b>   | Store as OpCode.                 |
| +X.1                               |                  |                                  |



|           |                |                                     |
|-----------|----------------|-------------------------------------|
| CLRP (20) | Cycles: 2      | Size: 1                             |
| Implied   |                |                                     |
| Cycle     | R/W            | Desc.                               |
| -X.2      | Read <b>PC</b> | Store as OpCode.                    |
| 1.1       |                | Inc. <b>PC</b> .                    |
| 1.2       |                |                                     |
| 2.1       |                | <b>PSW</b> = ( <b>PSW</b> & ~\$20). |
| 2.2       |                | Store as OpCode.                    |
| +X.1      |                |                                     |

|            |                                     |                               |
|------------|-------------------------------------|-------------------------------|
| TCALL (21) | Cycles: 8                           | Size: 1                       |
| Implied    |                                     |                               |
| Cycle      | R/W                                 | Desc.                         |
| -X.2       | Read <b>PC</b>                      | Store as OpCode.              |
| 1.1        |                                     | Inc. <b>PC</b> .              |
| 1.2        | Read <b>PC</b>                      | Store as Operand.             |
| 2.1        |                                     |                               |
| 2.2        |                                     |                               |
| 3.1        |                                     |                               |
| 3.2        | Write to u8( <b>SP</b> +0)   \$0100 | Push <b>PC</b> .H onto stack. |
| 4.1        |                                     |                               |
| 4.2        | Write to u8( <b>SP</b> -1)   \$0100 | Push <b>PC</b> .L onto stack. |
| 5.1        |                                     | Dec. <b>SP</b> by 2.          |
| 5.2        |                                     |                               |
| 6.1        |                                     |                               |
| 6.2        | Read \$FFDA                         | Store as <b>PC</b> .L.        |
| 7.1        |                                     |                               |
| 7.2        | Read \$FFDB                         | Store as <b>PC</b> .H.        |
| 8.1        |                                     |                               |
| 8.2        | Read <b>PC</b>                      | Store as OpCode.              |
| +X.1       |                                     |                               |

|                                         |                  |                                                                     |
|-----------------------------------------|------------------|---------------------------------------------------------------------|
| SET1 (22)                               | Cycles: 4        | Size: 2                                                             |
| Direct Page Bit d.b (Read/Modify/Write) |                  |                                                                     |
| Cycle                                   | R/W              | Desc.                                                               |
| -X.2                                    | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                                     |                  | Inc. <b>PC</b> .                                                    |
| 1.2                                     | Read <b>PC</b>   | Store as Operand.                                                   |
| 2.1                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                                     | Read Address     | Store as Operand.                                                   |
| 3.1                                     |                  | Operand = (Operand & ~\$02)   \$02.                                 |
| 3.2                                     | Write to Address | Write Operand.                                                      |
| 4.1                                     |                  |                                                                     |
| 4.2                                     | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                                    |                  |                                                                     |

|                                 |                |                                                                      |
|---------------------------------|----------------|----------------------------------------------------------------------|
| BBS (23)                        | Cycles: 5-7    | Size: 3                                                              |
| Direct Page Bit Relative d.b, r |                |                                                                      |
| Cycle                           | R/W            | Desc.                                                                |
| -X.2                            | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1                             |                | Inc. <b>PC</b> .                                                     |
| 1.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 2.1                             |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).  |
| 2.2                             | Read Address   | Store as Operand.                                                    |
| 3.1                             |                | Set Jump to (Operand & (1 << 1)) == (1 << 1).                        |
| 3.2                             |                |                                                                      |
| 4.1                             |                |                                                                      |
| 4.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 5.1                             |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 7.2). |
| 5.2                             |                |                                                                      |
| 6.1                             |                | Address = <b>PC</b> + i8(Operand).<br><b>PC</b> .L = Address.L.      |
| 6.2                             |                |                                                                      |
| 7.1                             |                | <b>PC</b> .H = Address.H.                                            |
| 7.2                             | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1                            |                |                                                                      |

|                      |                |                                                                                       |
|----------------------|----------------|---------------------------------------------------------------------------------------|
| AND (24)             | Cycles: 3      | Size: 2                                                                               |
| Direct Page d (Read) |                |                                                                                       |
| Cycle                | R/W            | Desc.                                                                                 |
| -X.2                 | Read <b>PC</b> | Store as OpCode.                                                                      |
| 1.1                  |                | Inc. <b>PC</b> .                                                                      |
| 1.2                  | Read <b>PC</b> | Store as Operand.                                                                     |
| 2.1                  |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                   |
| 2.2                  | Read Address   | Store as Operand.                                                                     |
| 3.1                  |                | <b>A</b> &= Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 3.2                  | Read <b>PC</b> | Store as OpCode.                                                                      |
| +X.1                 |                |                                                                                       |

|                    |                |                                                                                       |
|--------------------|----------------|---------------------------------------------------------------------------------------|
| AND (25)           | Cycles: 4      | Size: 3                                                                               |
| Absolute !a (Read) |                |                                                                                       |
| Cycle              | R/W            | Desc.                                                                                 |
| -X.2               | Read <b>PC</b> | Store as OpCode.                                                                      |
| 1.1                |                | Inc. <b>PC</b> .                                                                      |
| 1.2                | Read <b>PC</b> | Store as Address.L.                                                                   |
| 2.1                |                | Inc. <b>PC</b> .                                                                      |
| 2.2                | Read <b>PC</b> | Store as Address.H.                                                                   |
| 3.1                |                | Inc. <b>PC</b> .                                                                      |
| 3.2                | Read Address   | Store as Operand.                                                                     |
| 4.1                |                | <b>A</b> &= Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 4.2                | Read <b>PC</b> | Store as OpCode.                                                                      |
| +X.1               |                |                                                                                       |

|                       |                                                |                                                                                       |
|-----------------------|------------------------------------------------|---------------------------------------------------------------------------------------|
| AND (26)              | Cycles: 3                                      | Size: 1                                                                               |
| Indirect X (X) (Read) |                                                |                                                                                       |
| Cycle                 | R/W                                            | Desc.                                                                                 |
| -X.2                  | Read <b>PC</b>                                 | Store as OpCode.                                                                      |
| 1.1                   |                                                | Inc. <b>PC</b> .                                                                      |
| 1.2                   |                                                |                                                                                       |
| 2.1                   |                                                |                                                                                       |
| 2.2                   | Read ( <b>X</b>   (( <b>PSW</b> & \$20) << 3)) | Store as Operand.                                                                     |
| 3.1                   |                                                | <b>A</b> &= Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 3.2                   | Read <b>PC</b>                                 | Store as OpCode.                                                                      |
| +X.1                  |                                                |                                                                                       |

|                                 |                                                              |                                                                                       |
|---------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------|
| AND (27)                        | Cycles: 6                                                    | Size: 2                                                                               |
| X-Indexed Indirect [d+X] (Read) |                                                              |                                                                                       |
| Cycle                           | R/W                                                          | Desc.                                                                                 |
| -X.2                            | Read <b>PC</b>                                               | Store as OpCode.                                                                      |
| 1.1                             |                                                              | Inc. <b>PC</b> .                                                                      |
| 1.2                             | Read <b>PC</b>                                               | Store as Operand.                                                                     |
| 2.1                             |                                                              | Inc. <b>PC</b> .                                                                      |
| 2.2                             |                                                              |                                                                                       |
| 3.1                             |                                                              | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).               |
| 3.2                             | Read Address                                                 | Store as Pointer.L.                                                                   |
| 4.1                             |                                                              |                                                                                       |
| 4.2                             | Read ((Address + 1) & \$FF)  <br>(( <b>PSW</b> & \$20) << 3) | Store as Pointer.H.                                                                   |
| 5.1                             |                                                              |                                                                                       |
| 5.2                             | Read Pointer                                                 | Store as Operand.                                                                     |
| 6.1                             |                                                              | <b>A</b> &= Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 6.2                             | Read <b>PC</b>                                               | Store as OpCode.                                                                      |
| +X.1                            |                                                              |                                                                                       |



|           |                |                                                                                                        |
|-----------|----------------|--------------------------------------------------------------------------------------------------------|
| AND (28)  | Cycles: 2      | Size: 2                                                                                                |
| Immediate |                |                                                                                                        |
| Cycle     | R/W            | Desc.                                                                                                  |
| -X.2      | Read <b>PC</b> | Store as OpCode.                                                                                       |
| 1.1       |                | Inc. <b>PC</b> .                                                                                       |
| 1.2       | Read <b>PC</b> | Store as Operand.                                                                                      |
| 2.1       |                | Inc. <b>PC</b> . <b>A</b> &= Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 2.2       | Read <b>PC</b> | Store as OpCode.                                                                                       |
| +X.1      |                |                                                                                                        |

|                                                       |                  |                                                                                     |
|-------------------------------------------------------|------------------|-------------------------------------------------------------------------------------|
| AND (29)                                              | Cycles: 6        | Size: 3                                                                             |
| Direct Page to Direct Page dd, ds (Read/Modify/Write) |                  |                                                                                     |
| Cycle                                                 | R/W              | Desc.                                                                               |
| -X.2                                                  | Read <b>PC</b>   | Store as OpCode.                                                                    |
| 1.1                                                   |                  | Inc. <b>PC</b> .                                                                    |
| 1.2                                                   | Read <b>PC</b>   | Store as Operand.                                                                   |
| 2.1                                                   |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                 |
| 2.2                                                   | Read Address     | Store as Operand0.                                                                  |
| 3.1                                                   |                  |                                                                                     |
| 3.2                                                   | Read <b>PC</b>   | Store as Operand.                                                                   |
| 4.1                                                   |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                 |
| 4.2                                                   | Read Address     | Store as Operand1.                                                                  |
| 5.1                                                   |                  | Operand0 &= Operand1. <b>N</b> flag = (Operand0 & \$80), <b>Z</b> flag = !Operand0. |
| 5.2                                                   | Write to Address | Write Operand0.                                                                     |
| 6.1                                                   |                  |                                                                                     |
| 6.2                                                   | Read <b>PC</b>   | Store as OpCode.                                                                    |
| +X.1                                                  |                  |                                                                                     |

|                           |                         |                                                                         |
|---------------------------|-------------------------|-------------------------------------------------------------------------|
| OR1 (2A)                  | Cycles: 5               | Size: 3                                                                 |
| Absolute Boolean Bit /m.b |                         |                                                                         |
| Cycle                     | R/W                     | Desc.                                                                   |
| -X.2                      | Read <b>PC</b>          | Store as OpCode.                                                        |
| 1.1                       |                         | Inc. <b>PC</b> .                                                        |
| 1.2                       | Read <b>PC</b>          | Store as Address.L.                                                     |
| 2.1                       |                         | Inc. <b>PC</b> .                                                        |
| 2.2                       | Read <b>PC</b>          | Store as Address.H.                                                     |
| 3.1                       |                         | Inc. <b>PC</b> .                                                        |
| 3.2                       | Read (Address & \$1FFF) | Store as Operand.                                                       |
| 4.1                       |                         |                                                                         |
| 4.2                       |                         |                                                                         |
| 5.1                       |                         | Bit = Address >> 13.<br><b>C</b> flag  = ((Operand & (1 << Bit)) == 0). |
| 5.2                       | Read <b>PC</b>          | Store as OpCode.                                                        |
| +X.1                      |                         |                                                                         |

|                                   |                  |                                                                                                                                                    |
|-----------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| ROL (2B)                          | Cycles: 4        | Size: 2                                                                                                                                            |
| Direct Page d (Read/Modify/Write) |                  |                                                                                                                                                    |
| Cycle                             | R/W              | Desc.                                                                                                                                              |
| -X.2                              | Read <b>PC</b>   | Store as OpCode.                                                                                                                                   |
| 1.1                               |                  | Inc. <b>PC</b> .                                                                                                                                   |
| 1.2                               | Read <b>PC</b>   | Store as Operand.                                                                                                                                  |
| 2.1                               |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                                                                |
| 2.2                               | Read Address     | Store as Operand.                                                                                                                                  |
| 3.1                               |                  | Tmp = <b>C</b> flag. <b>C</b> flag = (Operand & \$80). Operand = (Operand << 1)   Tmp. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = !Operand. |
| 3.2                               | Write to Address | Write Operand.                                                                                                                                     |
| 4.1                               |                  |                                                                                                                                                    |
| 4.2                               | Read <b>PC</b>   | Store as OpCode.                                                                                                                                   |
| +X.1                              |                  |                                                                                                                                                    |

|                                 |                  |                                                                                                                                                    |
|---------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| ROL (2C)                        | Cycles: 5        | Size: 3                                                                                                                                            |
| Absolute !a (Read/Modify/Write) |                  |                                                                                                                                                    |
| Cycle                           | R/W              | Desc.                                                                                                                                              |
| -X.2                            | Read <b>PC</b>   | Store as OpCode.                                                                                                                                   |
| 1.1                             |                  | Inc. <b>PC</b> .                                                                                                                                   |
| 1.2                             | Read <b>PC</b>   | Store as Address.L.                                                                                                                                |
| 2.1                             |                  | Inc. <b>PC</b> .                                                                                                                                   |
| 2.2                             | Read <b>PC</b>   | Store as Address.H.                                                                                                                                |
| 3.1                             |                  | Inc. <b>PC</b> .                                                                                                                                   |
| 3.2                             | Read Address     | Store as Operand.                                                                                                                                  |
| 4.1                             |                  | Tmp = <b>C</b> flag. <b>C</b> flag = (Operand & \$80). Operand = (Operand << 1)   Tmp. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = !Operand. |
| 4.2                             | Write to Address | Write Operand.                                                                                                                                     |
| 5.1                             |                  |                                                                                                                                                    |
| 5.2                             | Read <b>PC</b>   | Store as OpCode.                                                                                                                                   |
| +X.1                            |                  |                                                                                                                                                    |

|           |                                                           |                           |
|-----------|-----------------------------------------------------------|---------------------------|
| PUSH (2D) | Cycles: 4                                                 | Size: 1                   |
| Implied   |                                                           |                           |
| Cycle     | R/W                                                       | Desc.                     |
| -X.2      | Read PC                                                   | Store as OpCode.          |
| 1.1       | Read <b>PC</b><br><br>Write to u8( <b>SP</b> +0)   \$0100 | Inc. <b>PC</b> .          |
| 1.2       |                                                           | Discard.                  |
| 2.1       |                                                           |                           |
| 2.2       |                                                           | Push <b>A</b> onto stack. |
| 3.1       |                                                           | Dec. <b>SP</b> by 1.      |
| 3.2       |                                                           |                           |
| 4.1       |                                                           |                           |
| 4.2       |                                                           | Store as OpCode.          |
| +X.1      |                                                           |                           |

|                           |                |                                                                      |
|---------------------------|----------------|----------------------------------------------------------------------|
| CBNE (2E)                 | Cycles: 5-7    | Size: 3                                                              |
| Direct Page Relative d, r |                |                                                                      |
| Cycle                     | R/W            | Desc.                                                                |
| -X.2                      | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1                       |                | Inc. <b>PC</b> .                                                     |
| 1.2                       | Read <b>PC</b> | Store as Operand.                                                    |
| 2.1                       |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).  |
| 2.2                       | Read Address   | Store as Operand.                                                    |
| 3.1                       |                | Set Jump to ( <b>A</b> != Operand).                                  |
| 3.2                       |                |                                                                      |
| 4.1                       |                |                                                                      |
| 4.2                       | Read <b>PC</b> | Store as Operand.                                                    |
| 5.1                       |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 7.2). |
| 5.2                       |                |                                                                      |
| 6.1                       |                | Address = <b>PC</b> + i8(Operand).<br><b>PC</b> .L = Address.L.      |
| 6.2                       |                |                                                                      |
| 7.1                       |                | <b>PC</b> .H = Address.H.                                            |
| 7.2                       | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1                      |                |                                                                      |

|            |                |                                                                      |
|------------|----------------|----------------------------------------------------------------------|
| BRA (2F)   | Cycles: 4      | Size: 2                                                              |
| Relative r |                |                                                                      |
| Cycle      | R/W            | Desc.                                                                |
| -X.2       | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1        |                | Inc. <b>PC</b> . Set Jump to true.                                   |
| 1.2        |                | Store as Operand.                                                    |
| 2.1        |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 4.2). |
| 2.2        |                |                                                                      |
| 3.1        |                | Address = <b>PC</b> + i8(Operand).                                   |
|            |                | <b>PC</b> .L = Address.L.                                            |
| 3.2        |                |                                                                      |
| 4.1        |                | <b>PC</b> .H = Address.H.                                            |
| 4.2        | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1       |                |                                                                      |



|            |                |                                                                      |
|------------|----------------|----------------------------------------------------------------------|
| BMI (30)   | Cycles: 2-4    | Size: 2                                                              |
| Relative r |                |                                                                      |
| Cycle      | R/W            | Desc.                                                                |
| -X.2       | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1        |                | Inc. <b>PC</b> . Set Jump to ( <b>PSW</b> & \$80) == \$80.           |
| 1.2        |                | Store as Operand.                                                    |
| 2.1        |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 4.2). |
| 2.2        |                |                                                                      |
| 3.1        |                | Address = <b>PC</b> + i8(Operand).                                   |
|            |                | <b>PC</b> .L = Address.L.                                            |
| 3.2        |                |                                                                      |
| 4.1        |                | <b>PC</b> .H = Address.H.                                            |
| 4.2        | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1       |                |                                                                      |

|            |                                     |                               |
|------------|-------------------------------------|-------------------------------|
| TCALL (31) | Cycles: 8                           | Size: 1                       |
| Implied    |                                     |                               |
| Cycle      | R/W                                 | Desc.                         |
| -X.2       | Read <b>PC</b>                      | Store as OpCode.              |
| 1.1        |                                     | Inc. <b>PC</b> .              |
| 1.2        | Read <b>PC</b>                      | Store as Operand.             |
| 2.1        |                                     |                               |
| 2.2        |                                     |                               |
| 3.1        |                                     |                               |
| 3.2        | Write to u8( <b>SP</b> +0)   \$0100 | Push <b>PC</b> .H onto stack. |
| 4.1        |                                     |                               |
| 4.2        | Write to u8( <b>SP</b> -1)   \$0100 | Push <b>PC</b> .L onto stack. |
| 5.1        |                                     | Dec. <b>SP</b> by 2.          |
| 5.2        |                                     |                               |
| 6.1        |                                     |                               |
| 6.2        | Read \$FFD8                         | Store as <b>PC</b> .L.        |
| 7.1        |                                     |                               |
| 7.2        | Read \$FFD9                         | Store as <b>PC</b> .H.        |
| 8.1        |                                     |                               |
| 8.2        | Read <b>PC</b>                      | Store as OpCode.              |
| +X.1       |                                     |                               |

|                                         |                  |                                                                     |
|-----------------------------------------|------------------|---------------------------------------------------------------------|
| CLR1 (32)                               | Cycles: 4        | Size: 2                                                             |
| Direct Page Bit d.b (Read/Modify/Write) |                  |                                                                     |
| Cycle                                   | R/W              | Desc.                                                               |
| -X.2                                    | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                                     |                  | Inc. <b>PC</b> .                                                    |
| 1.2                                     | Read <b>PC</b>   | Store as Operand.                                                   |
| 2.1                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                                     | Read Address     | Store as Operand.                                                   |
| 3.1                                     |                  | Operand = (Operand & ~\$02)   \$00.                                 |
| 3.2                                     | Write to Address | Write Operand.                                                      |
| 4.1                                     |                  |                                                                     |
| 4.2                                     | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                                    |                  |                                                                     |

|                                 |                |                                                                      |
|---------------------------------|----------------|----------------------------------------------------------------------|
| BBC (33)                        | Cycles: 5-7    | Size: 3                                                              |
| Direct Page Bit Relative d.b, r |                |                                                                      |
| Cycle                           | R/W            | Desc.                                                                |
| -X.2                            | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1                             |                | Inc. <b>PC</b> .                                                     |
| 1.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 2.1                             |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).  |
| 2.2                             | Read Address   | Store as Operand.                                                    |
| 3.1                             |                | Set Jump to (Operand & (1 << 1)) == (0 << 1).                        |
| 3.2                             |                |                                                                      |
| 4.1                             |                |                                                                      |
| 4.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 5.1                             |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 7.2). |
| 5.2                             |                |                                                                      |
| 6.1                             |                | Address = <b>PC</b> + i8(Operand).<br><b>PC</b> .L = Address.L.      |
| 6.2                             |                |                                                                      |
| 7.1                             |                | <b>PC</b> .H = Address.H.                                            |
| 7.2                             | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1                            |                |                                                                      |

|                                      |                |                                                                                       |
|--------------------------------------|----------------|---------------------------------------------------------------------------------------|
| AND (34)      Cycles: 4      Size: 2 |                |                                                                                       |
| X-Indexed Direct Page d+X (Read)     |                |                                                                                       |
| Cycle                                | R/W            | Desc.                                                                                 |
| -X.2                                 | Read <b>PC</b> | Store as OpCode.                                                                      |
| 1.1                                  |                | Inc. <b>PC</b> .                                                                      |
| 1.2                                  | Read <b>PC</b> | Store as Operand.                                                                     |
| 2.1                                  |                | Inc. <b>PC</b> .                                                                      |
| 2.2                                  |                |                                                                                       |
| 3.1                                  |                | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).               |
| 3.2                                  | Read Address   | Store as Operand.                                                                     |
| 4.1                                  |                | <b>A</b> &= Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 4.2                                  | Read <b>PC</b> | Store as OpCode.                                                                      |
| +X.1                                 |                |                                                                                       |

|                                |                |                                                                                       |
|--------------------------------|----------------|---------------------------------------------------------------------------------------|
| AND (35)                       | Cycles: 5      | Size: 3                                                                               |
| X-Indexed Absolute !a+X (Read) |                |                                                                                       |
| Cycle                          | R/W            | Desc.                                                                                 |
| -X.2                           | Read <b>PC</b> | Store as OpCode.                                                                      |
| 1.1                            |                | Inc. <b>PC</b> .                                                                      |
| 1.2                            | Read <b>PC</b> | Store as Address.L.                                                                   |
| 2.1                            |                | Inc. <b>PC</b> .                                                                      |
| 2.2                            | Read <b>PC</b> | Store as Address.H.                                                                   |
| 3.1                            |                | Inc. <b>PC</b> .                                                                      |
| 3.2                            |                |                                                                                       |
| 4.1                            |                | Pointer = ( <b>X</b> + Address).                                                      |
| 4.2                            | Read Pointer   | Store as Operand.                                                                     |
| 5.1                            |                | <b>A</b> &= Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 5.2                            | Read <b>PC</b> | Store as OpCode.                                                                      |
| +X.1                           |                |                                                                                       |

|                                |                |                                                                                       |
|--------------------------------|----------------|---------------------------------------------------------------------------------------|
| AND (36)                       | Cycles: 5      | Size: 3                                                                               |
| Y-Indexed Absolute !a+Y (Read) |                |                                                                                       |
| Cycle                          | R/W            | Desc.                                                                                 |
| -X.2                           | Read <b>PC</b> | Store as OpCode.                                                                      |
| 1.1                            |                | Inc. <b>PC</b> .                                                                      |
| 1.2                            | Read <b>PC</b> | Store as Address.L.                                                                   |
| 2.1                            |                | Inc. <b>PC</b> .                                                                      |
| 2.2                            | Read <b>PC</b> | Store as Address.H.                                                                   |
| 3.1                            |                | Inc. <b>PC</b> .                                                                      |
| 3.2                            |                |                                                                                       |
| 4.1                            |                | Pointer = ( <b>Y</b> + Address).                                                      |
| 4.2                            | Read Pointer   | Store as Operand.                                                                     |
| 5.1                            |                | <b>A</b> &= Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 5.2                            | Read <b>PC</b> | Store as OpCode.                                                                      |
| +X.1                           |                |                                                                                       |

|                                 |                                                           |                                                                              |
|---------------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------|
| AND (37)                        | Cycles: 6                                                 | Size: 2                                                                      |
| Indirect Y-Indexed [d]+Y (Read) |                                                           |                                                                              |
| Cycle                           | R/W                                                       | Desc.                                                                        |
| -X.2                            | Read <b>PC</b>                                            | Store as OpCode.                                                             |
| 1.1                             |                                                           | Inc. <b>PC</b> .                                                             |
| 1.2                             | Read <b>PC</b>                                            | Store as Operand.                                                            |
| 2.1                             |                                                           | Inc. <b>PC</b> .                                                             |
| 2.2                             |                                                           |                                                                              |
| 3.1                             |                                                           | Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                           |
| 3.2                             | Read Address                                              | Store as Pointer.L.                                                          |
| 4.1                             |                                                           |                                                                              |
| 4.2                             | Read ((Address + 1) & \$FF)   (( <b>PSW</b> & \$20) << 3) | Store as Pointer.H.                                                          |
| 5.1                             |                                                           | Address = ( <b>Y</b> + Pointer).                                             |
| 5.2                             | Read Address                                              | Store as Operand.                                                            |
| 6.1                             |                                                           | <b>A</b> &= Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = !A. |
| 6.2                             | Read <b>PC</b>                                            | Store as OpCode.                                                             |
| +X.1                            |                                                           |                                                                              |



|                                                         |                  |                                                                                     |
|---------------------------------------------------------|------------------|-------------------------------------------------------------------------------------|
| AND (38)                                                | Cycles: 5        | Size: 3                                                                             |
| Immediate Data to Direct Page d, #i (Read/Modify/Write) |                  |                                                                                     |
| Cycle                                                   | R/W              | Desc.                                                                               |
| -X.2                                                    | Read <b>PC</b>   | Store as OpCode.                                                                    |
| 1.1                                                     |                  | Inc. <b>PC</b> .                                                                    |
| 1.2                                                     | Read <b>PC</b>   | Store as Operand0.                                                                  |
| 2.1                                                     |                  | Inc. <b>PC</b> .                                                                    |
| 2.2                                                     | Read <b>PC</b>   | Store as Operand.                                                                   |
| 3.1                                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                 |
| 3.2                                                     | Read Address     | Store as Operand1.                                                                  |
| 4.1                                                     |                  | Operand0 &= Operand1. <b>N</b> flag = (Operand0 & \$80), <b>Z</b> flag = !Operand0. |
| 4.2                                                     | Write to Address | Write Operand0.                                                                     |
| 5.1                                                     |                  |                                                                                     |
| 5.2                                                     | Read <b>PC</b>   | Store as OpCode.                                                                    |
| +X.1                                                    |                  |                                                                                     |

|                                                             |                                                    |                                                                                     |
|-------------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------|
| AND (39)                                                    | Cycles: 5                                          | Size: 1                                                                             |
| Indirect Page to Indirect Page (X), (Y) (Read/Modify/Write) |                                                    |                                                                                     |
| Cycle                                                       | R/W                                                | Desc.                                                                               |
| -X.2                                                        | Read <b>PC</b>                                     | Store as OpCode.                                                                    |
| 1.1                                                         |                                                    | Inc. <b>PC</b> .                                                                    |
| 1.2                                                         | Read <b>PC</b>                                     | Discard.                                                                            |
| 2.1                                                         |                                                    |                                                                                     |
| 2.2                                                         | Read ( <b>Y</b>   (( <b>PSW</b> & \$20) << 3))     | Store as Operand0.                                                                  |
| 3.1                                                         |                                                    |                                                                                     |
| 3.2                                                         | Read ( <b>X</b>   (( <b>PSW</b> & \$20) << 3))     | Store as Operand1.                                                                  |
| 4.1                                                         |                                                    | Operand0 &= Operand1. <b>N</b> flag = (Operand0 & \$80), <b>Z</b> flag = !Operand0. |
| 4.2                                                         | Write to ( <b>X</b>   (( <b>PSW</b> & \$20) << 3)) | Write Operand0.                                                                     |
| 5.1                                                         |                                                    |                                                                                     |
| 5.2                                                         | Read <b>PC</b>                                     | Store as OpCode.                                                                    |
| +X.1                                                        |                                                    |                                                                                     |

|                                      |                                                               |                                                                                               |
|--------------------------------------|---------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| INCW (3A)                            | Cycles: 6                                                     | Size: 2                                                                                       |
| Direct Page d (Read/Modify/Write 16) |                                                               |                                                                                               |
| Cycle                                | R/W                                                           | Desc.                                                                                         |
| -X.2                                 | Read <b>PC</b>                                                | Store as OpCode.                                                                              |
| 1.1                                  |                                                               | Inc. <b>PC</b> .                                                                              |
| 1.2                                  | Read <b>PC</b>                                                | Store as Operand.                                                                             |
| 2.1                                  |                                                               | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                           |
| 2.2                                  | Read Address                                                  | Store as Operand.                                                                             |
| 3.1                                  |                                                               | Overflow = (Operand == \$FF).<br>Operand += 1.<br>LowZero = (Operand == 0).                   |
| 3.2                                  | Write to Address                                              | Write Operand.                                                                                |
| 4.1                                  |                                                               |                                                                                               |
| 4.2                                  | Read ((Address + 1) & \$FF)   (( <b>PSW</b> & \$20) << 3)     | Store as Operand.                                                                             |
| 5.1                                  |                                                               | Operand += Overflow. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = (LowZero && !Operand). |
| 5.2                                  | Write to ((Address + 1) & \$FF)   (( <b>PSW</b> & \$20) << 3) | Write Operand.                                                                                |
| 6.1                                  |                                                               |                                                                                               |
| 6.2                                  | Read <b>PC</b>                                                | Store as OpCode.                                                                              |
| +X.1                                 |                                                               |                                                                                               |

|                                               |                  |                                                                                                                                                    |
|-----------------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| ROL (3B)                                      | Cycles: 5        | Size: 2                                                                                                                                            |
| X-Indexed Direct Page d+X (Read/Modify/Write) |                  |                                                                                                                                                    |
| Cycle                                         | R/W              | Desc.                                                                                                                                              |
| -X.2                                          | Read <b>PC</b>   | Store as OpCode.                                                                                                                                   |
| 1.1                                           |                  | Inc. <b>PC</b> .                                                                                                                                   |
| 1.2                                           | Read <b>PC</b>   | Store as Operand.                                                                                                                                  |
| 2.1                                           |                  | Inc. <b>PC</b> .                                                                                                                                   |
| 2.2                                           |                  |                                                                                                                                                    |
| 3.1                                           |                  | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).                                                                            |
| 3.2                                           | Read Address     | Store as Operand.                                                                                                                                  |
| 4.1                                           |                  | Tmp = <b>C</b> flag. <b>C</b> flag = (Operand & \$80). Operand = (Operand << 1)   Tmp. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = !Operand. |
| 4.2                                           | Write to Address | Write Operand.                                                                                                                                     |
| 5.1                                           |                  |                                                                                                                                                    |
| 5.2                                           | Read <b>PC</b>   | Store as OpCode.                                                                                                                                   |
| +X.1                                          |                  |                                                                                                                                                    |

|               |                |                                                                                                                                                              |
|---------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROL (3C)      | Cycles: 2      | Size: 1                                                                                                                                                      |
| Accumulator A |                |                                                                                                                                                              |
| Cycle         | R/W            | Desc.                                                                                                                                                        |
| -X.2          | Read <b>PC</b> | Store as OpCode.                                                                                                                                             |
| 1.1           |                | Inc. <b>PC</b> .                                                                                                                                             |
| 1.2           | Read <b>PC</b> | Discard.                                                                                                                                                     |
| 2.1           |                | Tmp = <b>C</b> flag. <b>C</b> flag = ( <b>A</b> & \$80). <b>A</b> = ( <b>A</b> << 1)   Tmp. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 2.2           | Read <b>PC</b> | Store as OpCode.                                                                                                                                             |
| +X.1          |                |                                                                                                                                                              |

|          |                |           |                                                                                 |
|----------|----------------|-----------|---------------------------------------------------------------------------------|
| INC (3D) |                | Cycles: 2 | Size: 1                                                                         |
| Implied  |                |           |                                                                                 |
| Cycle    | R/W            |           | Desc.                                                                           |
| -X.2     | Read <b>PC</b> |           | Store as OpCode.                                                                |
| 1.1      |                |           | Inc. <b>PC</b> .                                                                |
| 1.2      |                |           |                                                                                 |
| 2.1      |                |           | <b>X</b> += 1. <b>N</b> flag = ( <b>X</b> & \$80), <b>Z</b> flag = ! <b>X</b> . |
| 2.2      | Read <b>PC</b> |           | Store as OpCode.                                                                |
| +X.1     |                |           |                                                                                 |

|                      |                |                                                                                                           |
|----------------------|----------------|-----------------------------------------------------------------------------------------------------------|
| CMP (3E)             | Cycles: 3      | Size: 2                                                                                                   |
| Direct Page d (Read) |                |                                                                                                           |
| Cycle                | R/W            | Desc.                                                                                                     |
| -X.2                 | Read <b>PC</b> | Store as OpCode.                                                                                          |
| 1.1                  |                | Inc. <b>PC</b> .                                                                                          |
| 1.2                  | Read <b>PC</b> | Store as Operand.                                                                                         |
| 2.1                  |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                       |
| 2.2                  | Read Address   | Store as Operand.                                                                                         |
| 3.1                  |                | Tmp = <b>X</b> - Operand. <b>C</b> flag = (Tmp >= 0), <b>N</b> flag = (Tmp & \$80), <b>Z</b> flag = !Tmp. |
| 3.2                  | Read <b>PC</b> | Store as OpCode.                                                                                          |
| +X.1                 |                |                                                                                                           |

|                  |                                     |                               |
|------------------|-------------------------------------|-------------------------------|
| CALL (3F)        | Cycles: 8                           | Size: 3                       |
| Absolute Call !a |                                     |                               |
| Cycle            | R/W                                 | Desc.                         |
| -X.2             | Read <b>PC</b>                      | Store as OpCode.              |
| 1.1              |                                     | Inc. <b>PC</b> .              |
| 1.2              | Read <b>PC</b>                      | Store as Address.L.           |
| 2.1              |                                     | Inc. <b>PC</b> .              |
| 2.2              | Read <b>PC</b>                      | Store as Address.H.           |
| 3.1              |                                     | Inc. <b>PC</b> .              |
| 3.2              |                                     |                               |
| 4.1              |                                     |                               |
| 4.2              | Write to u8( <b>SP</b> +0)   \$0100 | Push <b>PC</b> .H onto stack. |
| 5.1              |                                     |                               |
| 5.2              | Write to u8( <b>SP</b> -1)   \$0100 | Push <b>PC</b> .L onto stack. |
| 6.1              |                                     | Dec. <b>SP</b> by 2.          |
| 6.2              |                                     |                               |
| 7.1              |                                     | <b>PC</b> .L = Address.L.     |
| 7.2              |                                     |                               |
| 8.1              |                                     | <b>PC</b> .H = Address.H.     |
| 8.2              | Read <b>PC</b>                      | Store as OpCode.              |
| +X.1             |                                     |                               |



|           |                |                                    |
|-----------|----------------|------------------------------------|
| SETP (40) | Cycles: 2      | Size: 1                            |
| Implied   |                |                                    |
| Cycle     | R/W            | Desc.                              |
| -X.2      | Read <b>PC</b> | Store as OpCode.                   |
| 1.1       |                | Inc. <b>PC</b> .                   |
| 1.2       |                |                                    |
| 2.1       |                | <b>PSW</b> = ( <b>PSW</b>   \$20). |
| 2.2       |                | Store as OpCode.                   |
| +X.1      |                |                                    |

|            |                                     |                               |
|------------|-------------------------------------|-------------------------------|
| TCALL (41) | Cycles: 8                           | Size: 1                       |
| Implied    |                                     |                               |
| Cycle      | R/W                                 | Desc.                         |
| -X.2       | Read <b>PC</b>                      | Store as OpCode.              |
| 1.1        |                                     | Inc. <b>PC</b> .              |
| 1.2        | Read <b>PC</b>                      | Store as Operand.             |
| 2.1        |                                     |                               |
| 2.2        |                                     |                               |
| 3.1        |                                     |                               |
| 3.2        | Write to u8( <b>SP</b> +0)   \$0100 | Push <b>PC</b> .H onto stack. |
| 4.1        |                                     |                               |
| 4.2        | Write to u8( <b>SP</b> -1)   \$0100 | Push <b>PC</b> .L onto stack. |
| 5.1        |                                     | Dec. <b>SP</b> by 2.          |
| 5.2        |                                     |                               |
| 6.1        |                                     |                               |
| 6.2        | Read \$FFD6                         | Store as <b>PC</b> .L.        |
| 7.1        |                                     |                               |
| 7.2        | Read \$FFD7                         | Store as <b>PC</b> .H.        |
| 8.1        |                                     |                               |
| 8.2        | Read <b>PC</b>                      | Store as OpCode.              |
| +X.1       |                                     |                               |

|                                         |                  |                                                                     |
|-----------------------------------------|------------------|---------------------------------------------------------------------|
| SET1 (42)                               | Cycles: 4        | Size: 2                                                             |
| Direct Page Bit d.b (Read/Modify/Write) |                  |                                                                     |
| Cycle                                   | R/W              | Desc.                                                               |
| -X.2                                    | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                                     |                  | Inc. <b>PC</b> .                                                    |
| 1.2                                     | Read <b>PC</b>   | Store as Operand.                                                   |
| 2.1                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                                     | Read Address     | Store as Operand.                                                   |
| 3.1                                     |                  | Operand = (Operand & ~\$04)   \$04.                                 |
| 3.2                                     | Write to Address | Write Operand.                                                      |
| 4.1                                     |                  |                                                                     |
| 4.2                                     | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                                    |                  |                                                                     |

|                                 |                |                                                                      |
|---------------------------------|----------------|----------------------------------------------------------------------|
| BBS (43)                        | Cycles: 5-7    | Size: 3                                                              |
| Direct Page Bit Relative d.b, r |                |                                                                      |
| Cycle                           | R/W            | Desc.                                                                |
| -X.2                            | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1                             |                | Inc. <b>PC</b> .                                                     |
| 1.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 2.1                             |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).  |
| 2.2                             | Read Address   | Store as Operand.                                                    |
| 3.1                             |                | Set Jump to (Operand & (1 << 2)) == (1 << 2).                        |
| 3.2                             |                |                                                                      |
| 4.1                             |                |                                                                      |
| 4.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 5.1                             |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 7.2). |
| 5.2                             |                |                                                                      |
| 6.1                             |                | Address = <b>PC</b> + i8(Operand).<br><b>PC</b> .L = Address.L.      |
| 6.2                             |                |                                                                      |
| 7.1                             |                | <b>PC</b> .H = Address.H.                                            |
| 7.2                             | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1                            |                |                                                                      |

|                      |                |                                                                                       |
|----------------------|----------------|---------------------------------------------------------------------------------------|
| EOR (44)             | Cycles: 3      | Size: 2                                                                               |
| Direct Page d (Read) |                |                                                                                       |
| Cycle                | R/W            | Desc.                                                                                 |
| -X.2                 | Read <b>PC</b> | Store as OpCode.                                                                      |
| 1.1                  |                | Inc. <b>PC</b> .                                                                      |
| 1.2                  | Read <b>PC</b> | Store as Operand.                                                                     |
| 2.1                  |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                   |
| 2.2                  | Read Address   | Store as Operand.                                                                     |
| 3.1                  |                | <b>A</b> ^= Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 3.2                  | Read <b>PC</b> | Store as OpCode.                                                                      |
| +X.1                 |                |                                                                                       |

|                    |                |                                                                                       |
|--------------------|----------------|---------------------------------------------------------------------------------------|
| EOR (45)           | Cycles: 4      | Size: 3                                                                               |
| Absolute !a (Read) |                |                                                                                       |
| Cycle              | R/W            | Desc.                                                                                 |
| -X.2               | Read <b>PC</b> | Store as OpCode.                                                                      |
| 1.1                |                | Inc. <b>PC</b> .                                                                      |
| 1.2                | Read <b>PC</b> | Store as Address.L.                                                                   |
| 2.1                |                | Inc. <b>PC</b> .                                                                      |
| 2.2                | Read <b>PC</b> | Store as Address.H.                                                                   |
| 3.1                |                | Inc. <b>PC</b> .                                                                      |
| 3.2                | Read Address   | Store as Operand.                                                                     |
| 4.1                |                | <b>A</b> ^= Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 4.2                | Read <b>PC</b> | Store as OpCode.                                                                      |
| +X.1               |                |                                                                                       |

|                       |                                                |                                                                                       |
|-----------------------|------------------------------------------------|---------------------------------------------------------------------------------------|
| EOR (46)              | Cycles: 3                                      | Size: 1                                                                               |
| Indirect X (X) (Read) |                                                |                                                                                       |
| Cycle                 | R/W                                            | Desc.                                                                                 |
| -X.2                  | Read <b>PC</b>                                 | Store as OpCode.                                                                      |
| 1.1                   |                                                | Inc. <b>PC</b> .                                                                      |
| 1.2                   |                                                |                                                                                       |
| 2.1                   |                                                |                                                                                       |
| 2.2                   | Read ( <b>X</b>   (( <b>PSW</b> & \$20) << 3)) | Store as Operand.                                                                     |
| 3.1                   |                                                | <b>A</b> ^= Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 3.2                   | Read <b>PC</b>                                 | Store as OpCode.                                                                      |
| +X.1                  |                                                |                                                                                       |

|                                 |                                                              |                                                                                       |
|---------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------|
| EOR (47)                        | Cycles: 6                                                    | Size: 2                                                                               |
| X-Indexed Indirect [d+X] (Read) |                                                              |                                                                                       |
| Cycle                           | R/W                                                          | Desc.                                                                                 |
| -X.2                            | Read <b>PC</b>                                               | Store as OpCode.                                                                      |
| 1.1                             |                                                              | Inc. <b>PC</b> .                                                                      |
| 1.2                             | Read <b>PC</b>                                               | Store as Operand.                                                                     |
| 2.1                             |                                                              | Inc. <b>PC</b> .                                                                      |
| 2.2                             |                                                              |                                                                                       |
| 3.1                             |                                                              | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).               |
| 3.2                             | Read Address                                                 | Store as Pointer.L.                                                                   |
| 4.1                             |                                                              |                                                                                       |
| 4.2                             | Read ((Address + 1) & \$FF)  <br>(( <b>PSW</b> & \$20) << 3) | Store as Pointer.H.                                                                   |
| 5.1                             |                                                              |                                                                                       |
| 5.2                             | Read Pointer                                                 | Store as Operand.                                                                     |
| 6.1                             |                                                              | <b>A</b> ^= Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 6.2                             | Read <b>PC</b>                                               | Store as OpCode.                                                                      |
| +X.1                            |                                                              |                                                                                       |



|           |                |                                                                                                        |
|-----------|----------------|--------------------------------------------------------------------------------------------------------|
| EOR (48)  | Cycles: 2      | Size: 2                                                                                                |
| Immediate |                |                                                                                                        |
| Cycle     | R/W            | Desc.                                                                                                  |
| -X.2      | Read <b>PC</b> | Store as OpCode.                                                                                       |
| 1.1       |                | Inc. <b>PC</b> .                                                                                       |
| 1.2       | Read <b>PC</b> | Store as Operand.                                                                                      |
| 2.1       |                | Inc. <b>PC</b> . <b>A</b> ^= Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 2.2       | Read <b>PC</b> | Store as OpCode.                                                                                       |
| +X.1      |                |                                                                                                        |

|                                                       |                  |                                                                                     |
|-------------------------------------------------------|------------------|-------------------------------------------------------------------------------------|
| EOR (49)                                              | Cycles: 6        | Size: 3                                                                             |
| Direct Page to Direct Page dd, ds (Read/Modify/Write) |                  |                                                                                     |
| Cycle                                                 | R/W              | Desc.                                                                               |
| -X.2                                                  | Read <b>PC</b>   | Store as OpCode.                                                                    |
| 1.1                                                   |                  | Inc. <b>PC</b> .                                                                    |
| 1.2                                                   | Read <b>PC</b>   | Store as Operand.                                                                   |
| 2.1                                                   |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                 |
| 2.2                                                   | Read Address     | Store as Operand0.                                                                  |
| 3.1                                                   |                  |                                                                                     |
| 3.2                                                   | Read <b>PC</b>   | Store as Operand.                                                                   |
| 4.1                                                   |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                 |
| 4.2                                                   | Read Address     | Store as Operand1.                                                                  |
| 5.1                                                   |                  | Operand0 ^= Operand1. <b>N</b> flag = (Operand0 & \$80), <b>Z</b> flag = !Operand0. |
| 5.2                                                   | Write to Address | Write Operand0.                                                                     |
| 6.1                                                   |                  |                                                                                     |
| 6.2                                                   | Read <b>PC</b>   | Store as OpCode.                                                                    |
| +X.1                                                  |                  |                                                                                     |

|                          |                         |                                                                         |
|--------------------------|-------------------------|-------------------------------------------------------------------------|
| AND1 (4A)                | Cycles: 4               | Size: 3                                                                 |
| Absolute Boolean Bit m.b |                         |                                                                         |
| Cycle                    | R/W                     | Desc.                                                                   |
| -X.2                     | Read <b>PC</b>          | Store as OpCode.                                                        |
| 1.1                      |                         | Inc. <b>PC</b> .                                                        |
| 1.2                      | Read <b>PC</b>          | Store as Address.L.                                                     |
| 2.1                      |                         | Inc. <b>PC</b> .                                                        |
| 2.2                      | Read <b>PC</b>          | Store as Address.H.                                                     |
| 3.1                      |                         | Inc. <b>PC</b> .                                                        |
| 3.2                      | Read (Address & \$1FFF) | Store as Operand.                                                       |
| 4.1                      |                         | Bit = Address >> 13.<br><b>C</b> flag &= ((Operand & (1 << Bit)) != 0). |
| 4.2                      | Read <b>PC</b>          | Store as OpCode.                                                        |
| +X.1                     |                         |                                                                         |

|                                   |                  |                                                                                                            |
|-----------------------------------|------------------|------------------------------------------------------------------------------------------------------------|
| LSR (4B)                          | Cycles: 4        | Size: 2                                                                                                    |
| Direct Page d (Read/Modify/Write) |                  |                                                                                                            |
| Cycle                             | R/W              | Desc.                                                                                                      |
| -X.2                              | Read <b>PC</b>   | Store as OpCode.                                                                                           |
| 1.1                               |                  | Inc. <b>PC</b> .                                                                                           |
| 1.2                               | Read <b>PC</b>   | Store as Operand.                                                                                          |
| 2.1                               |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                        |
| 2.2                               | Read Address     | Store as Operand.                                                                                          |
| 3.1                               |                  | <b>C</b> flag = (Operand & 01). Operand >>= 1. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = !Operand. |
| 3.2                               | Write to Address | Write Operand.                                                                                             |
| 4.1                               |                  |                                                                                                            |
| 4.2                               | Read <b>PC</b>   | Store as OpCode.                                                                                           |
| +X.1                              |                  |                                                                                                            |

|                                 |                  |                                                                                                            |
|---------------------------------|------------------|------------------------------------------------------------------------------------------------------------|
| LSR (4C)                        | Cycles: 5        | Size: 3                                                                                                    |
| Absolute !a (Read/Modify/Write) |                  |                                                                                                            |
| Cycle                           | R/W              | Desc.                                                                                                      |
| -X.2                            | Read <b>PC</b>   | Store as OpCode.                                                                                           |
| 1.1                             |                  | Inc. <b>PC</b> .                                                                                           |
| 1.2                             | Read <b>PC</b>   | Store as Address.L.                                                                                        |
| 2.1                             |                  | Inc. <b>PC</b> .                                                                                           |
| 2.2                             | Read <b>PC</b>   | Store as Address.H.                                                                                        |
| 3.1                             |                  | Inc. <b>PC</b> .                                                                                           |
| 3.2                             | Read Address     | Store as Operand.                                                                                          |
| 4.1                             |                  | <b>C</b> flag = (Operand & 01). Operand >>= 1. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = !Operand. |
| 4.2                             | Write to Address | Write Operand.                                                                                             |
| 5.1                             |                  |                                                                                                            |
| 5.2                             | Read <b>PC</b>   | Store as OpCode.                                                                                           |
| +X.1                            |                  |                                                                                                            |

|           |                                     |                           |         |
|-----------|-------------------------------------|---------------------------|---------|
| PUSH (4D) |                                     | Cycles: 4                 | Size: 1 |
| Implied   |                                     |                           |         |
| Cycle     | R/W                                 | Desc.                     |         |
| -X.2      | Read <b>PC</b>                      | Store as OpCode.          |         |
| 1.1       | Write to u8( <b>SP</b> +0)   \$0100 | Inc. <b>PC</b> .          |         |
| 1.2       |                                     | Discard.                  |         |
| 2.1       |                                     |                           |         |
| 2.2       |                                     | Push <b>X</b> onto stack. |         |
| 3.1       |                                     | Dec. <b>SP</b> by 1.      |         |
| 3.2       |                                     |                           |         |
| 4.1       |                                     |                           |         |
| 4.2       |                                     | Store as OpCode.          |         |
| +X.1      |                                     |                           |         |

|                                 |                  |                                                                                                                                 |
|---------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------|
| TCLR1 (4E)                      | Cycles: 6        | Size: 3                                                                                                                         |
| Absolute !a (Read/Modify/Write) |                  |                                                                                                                                 |
| Cycle                           | R/W              | Desc.                                                                                                                           |
| -X.2                            | Read <b>PC</b>   | Store as OpCode.                                                                                                                |
| 1.1                             |                  | Inc. <b>PC</b> .                                                                                                                |
| 1.2                             | Read <b>PC</b>   | Store as Address.L.                                                                                                             |
| 2.1                             |                  | Inc. <b>PC</b> .                                                                                                                |
| 2.2                             | Read <b>PC</b>   | Store as Address.H.                                                                                                             |
| 3.1                             |                  | Inc. <b>PC</b> .                                                                                                                |
| 3.2                             | Read Address     | Store as Operand.                                                                                                               |
| 4.1                             |                  | Tmp = ( <b>A</b> - Operand).<br>Operand = (Operand & ~ <b>A</b> ).<br><b>Z</b> flag = (Tmp == 0), <b>N</b> flag = (Tmp & \$80). |
| 4.2                             | Read Address     | Discard.                                                                                                                        |
| 5.1                             |                  |                                                                                                                                 |
| 5.2                             | Write to Address | Write Operand.                                                                                                                  |
| 6.1                             |                  |                                                                                                                                 |
| 6.2                             | Read <b>PC</b>   | Store as OpCode.                                                                                                                |
| +X.1                            |                  |                                                                                                                                 |

|                  |                                     |                                                |
|------------------|-------------------------------------|------------------------------------------------|
| PCALL (4F)       | Cycles: 6                           | Size: 2                                        |
| Absolute Call !a |                                     |                                                |
| Cycle            | R/W                                 | Desc.                                          |
| -X.2             | Read <b>PC</b>                      | Store as OpCode.                               |
| 1.1              |                                     | Inc. <b>PC</b> .                               |
| 1.2              | Read <b>PC</b>                      | Store as Address.L.                            |
| 2.1              |                                     | Inc. <b>PC</b> .                               |
| 2.2              |                                     |                                                |
| 3.1              |                                     |                                                |
| 3.2              | Write to u8( <b>SP</b> +0)   \$0100 | Push <b>PC</b> .H onto stack.                  |
| 4.1              |                                     |                                                |
| 4.2              | Write to u8( <b>SP</b> -1)   \$0100 | Push <b>PC</b> .L onto stack.                  |
| 5.1              |                                     | Dec. <b>SP</b> by 2.                           |
| 5.2              |                                     |                                                |
| 6.1              |                                     | <b>PC</b> .L = Address.L, <b>PC</b> .H = \$FF. |
| 6.2              | Read <b>PC</b>                      | Store as OpCode.                               |
| +X.1             |                                     |                                                |



|            |                |                                                                      |
|------------|----------------|----------------------------------------------------------------------|
| BVC (50)   | Cycles: 2-4    | Size: 2                                                              |
| Relative r |                |                                                                      |
| Cycle      | R/W            | Desc.                                                                |
| -X.2       | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1        |                | Inc. <b>PC</b> . Set Jump to ( <b>PSW</b> & \$40) == \$00.           |
| 1.2        |                | Store as Operand.                                                    |
| 2.1        |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 4.2). |
| 2.2        |                |                                                                      |
| 3.1        |                | Address = <b>PC</b> + i8(Operand).                                   |
|            |                | <b>PC</b> .L = Address.L.                                            |
| 3.2        |                |                                                                      |
| 4.1        |                | <b>PC</b> .H = Address.H.                                            |
| 4.2        | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1       |                |                                                                      |

|            |                                     |                               |
|------------|-------------------------------------|-------------------------------|
| TCALL (51) | Cycles: 8                           | Size: 1                       |
| Implied    |                                     |                               |
| Cycle      | R/W                                 | Desc.                         |
| -X.2       | Read <b>PC</b>                      | Store as OpCode.              |
| 1.1        |                                     | Inc. <b>PC</b> .              |
| 1.2        | Read <b>PC</b>                      | Store as Operand.             |
| 2.1        |                                     |                               |
| 2.2        |                                     |                               |
| 3.1        |                                     |                               |
| 3.2        | Write to u8( <b>SP</b> +0)   \$0100 | Push <b>PC</b> .H onto stack. |
| 4.1        |                                     |                               |
| 4.2        | Write to u8( <b>SP</b> -1)   \$0100 | Push <b>PC</b> .L onto stack. |
| 5.1        |                                     | Dec. <b>SP</b> by 2.          |
| 5.2        |                                     |                               |
| 6.1        |                                     |                               |
| 6.2        | Read \$FFD4                         | Store as <b>PC</b> .L.        |
| 7.1        |                                     |                               |
| 7.2        | Read \$FFD5                         | Store as <b>PC</b> .H.        |
| 8.1        |                                     |                               |
| 8.2        | Read <b>PC</b>                      | Store as OpCode.              |
| +X.1       |                                     |                               |

|                                         |                  |                                                                     |
|-----------------------------------------|------------------|---------------------------------------------------------------------|
| CLR1 (52)                               | Cycles: 4        | Size: 2                                                             |
| Direct Page Bit d.b (Read/Modify/Write) |                  |                                                                     |
| Cycle                                   | R/W              | Desc.                                                               |
| -X.2                                    | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                                     |                  | Inc. <b>PC</b> .                                                    |
| 1.2                                     | Read <b>PC</b>   | Store as Operand.                                                   |
| 2.1                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                                     | Read Address     | Store as Operand.                                                   |
| 3.1                                     |                  | Operand = (Operand & ~\$04)   \$00.                                 |
| 3.2                                     | Write to Address | Write Operand.                                                      |
| 4.1                                     |                  |                                                                     |
| 4.2                                     | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                                    |                  |                                                                     |

|                                 |                |                                                                      |
|---------------------------------|----------------|----------------------------------------------------------------------|
| BBC (53)                        | Cycles: 5-7    | Size: 3                                                              |
| Direct Page Bit Relative d.b, r |                |                                                                      |
| Cycle                           | R/W            | Desc.                                                                |
| -X.2                            | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1                             |                | Inc. <b>PC</b> .                                                     |
| 1.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 2.1                             |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).  |
| 2.2                             | Read Address   | Store as Operand.                                                    |
| 3.1                             |                | Set Jump to (Operand & (1 << 2)) == (0 << 2).                        |
| 3.2                             |                |                                                                      |
| 4.1                             |                |                                                                      |
| 4.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 5.1                             |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 7.2). |
| 5.2                             |                |                                                                      |
| 6.1                             |                | Address = <b>PC</b> + i8(Operand).<br><b>PC</b> .L = Address.L.      |
| 6.2                             |                |                                                                      |
| 7.1                             |                | <b>PC</b> .H = Address.H.                                            |
| 7.2                             | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1                            |                |                                                                      |

|                                  |                |                                                                                       |
|----------------------------------|----------------|---------------------------------------------------------------------------------------|
| EOR (54)                         | Cycles: 4      | Size: 2                                                                               |
| X-Indexed Direct Page d+X (Read) |                |                                                                                       |
| Cycle                            | R/W            | Desc.                                                                                 |
| -X.2                             | Read <b>PC</b> | Store as OpCode.                                                                      |
| 1.1                              |                | Inc. <b>PC</b> .                                                                      |
| 1.2                              | Read <b>PC</b> | Store as Operand.                                                                     |
| 2.1                              |                | Inc. <b>PC</b> .                                                                      |
| 2.2                              |                |                                                                                       |
| 3.1                              |                | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).               |
| 3.2                              | Read Address   | Store as Operand.                                                                     |
| 4.1                              |                | <b>A</b> ^= Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 4.2                              | Read <b>PC</b> | Store as OpCode.                                                                      |
| +X.1                             |                |                                                                                       |

|                                |                |                                                                                       |
|--------------------------------|----------------|---------------------------------------------------------------------------------------|
| EOR (55)                       | Cycles: 5      | Size: 3                                                                               |
| X-Indexed Absolute !a+X (Read) |                |                                                                                       |
| Cycle                          | R/W            | Desc.                                                                                 |
| -X.2                           | Read <b>PC</b> | Store as OpCode.                                                                      |
| 1.1                            |                | Inc. <b>PC</b> .                                                                      |
| 1.2                            | Read <b>PC</b> | Store as Address.L.                                                                   |
| 2.1                            |                | Inc. <b>PC</b> .                                                                      |
| 2.2                            | Read <b>PC</b> | Store as Address.H.                                                                   |
| 3.1                            |                | Inc. <b>PC</b> .                                                                      |
| 3.2                            |                |                                                                                       |
| 4.1                            |                | Pointer = ( <b>X</b> + Address).                                                      |
| 4.2                            | Read Pointer   | Store as Operand.                                                                     |
| 5.1                            |                | <b>A</b> ^= Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 5.2                            | Read <b>PC</b> | Store as OpCode.                                                                      |
| +X.1                           |                |                                                                                       |

|                                |                |                                                                                       |
|--------------------------------|----------------|---------------------------------------------------------------------------------------|
| EOR (56)                       | Cycles: 5      | Size: 3                                                                               |
| Y-Indexed Absolute !a+Y (Read) |                |                                                                                       |
| Cycle                          | R/W            | Desc.                                                                                 |
| -X.2                           | Read <b>PC</b> | Store as OpCode.                                                                      |
| 1.1                            |                | Inc. <b>PC</b> .                                                                      |
| 1.2                            | Read <b>PC</b> | Store as Address.L.                                                                   |
| 2.1                            |                | Inc. <b>PC</b> .                                                                      |
| 2.2                            | Read <b>PC</b> | Store as Address.H.                                                                   |
| 3.1                            |                | Inc. <b>PC</b> .                                                                      |
| 3.2                            |                |                                                                                       |
| 4.1                            |                | Pointer = ( <b>Y</b> + Address).                                                      |
| 4.2                            | Read Pointer   | Store as Operand.                                                                     |
| 5.1                            |                | <b>A</b> ^= Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 5.2                            | Read <b>PC</b> | Store as OpCode.                                                                      |
| +X.1                           |                |                                                                                       |

|                                 |                                                           |                                                                                       |
|---------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------|
| EOR (57)                        | Cycles: 6                                                 | Size: 2                                                                               |
| Indirect Y-Indexed [d]+Y (Read) |                                                           |                                                                                       |
| Cycle                           | R/W                                                       | Desc.                                                                                 |
| -X.2                            | Read <b>PC</b>                                            | Store as OpCode.                                                                      |
| 1.1                             |                                                           | Inc. <b>PC</b> .                                                                      |
| 1.2                             | Read <b>PC</b>                                            | Store as Operand.                                                                     |
| 2.1                             |                                                           | Inc. <b>PC</b> .                                                                      |
| 2.2                             |                                                           |                                                                                       |
| 3.1                             |                                                           | Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                    |
| 3.2                             | Read Address                                              | Store as Pointer.L.                                                                   |
| 4.1                             |                                                           |                                                                                       |
| 4.2                             | Read ((Address + 1) & \$FF)   (( <b>PSW</b> & \$20) << 3) | Store as Pointer.H.                                                                   |
| 5.1                             |                                                           | Address = ( <b>Y</b> + Pointer).                                                      |
| 5.2                             | Read Address                                              | Store as Operand.                                                                     |
| 6.1                             |                                                           | <b>A</b> ^= Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 6.2                             | Read <b>PC</b>                                            | Store as OpCode.                                                                      |
| +X.1                            |                                                           |                                                                                       |



|                                                         |                  |                                                                                     |
|---------------------------------------------------------|------------------|-------------------------------------------------------------------------------------|
| EOR (58)                                                | Cycles: 5        | Size: 3                                                                             |
| Immediate Data to Direct Page d, #i (Read/Modify/Write) |                  |                                                                                     |
| Cycle                                                   | R/W              | Desc.                                                                               |
| -X.2                                                    | Read <b>PC</b>   | Store as OpCode.                                                                    |
| 1.1                                                     |                  | Inc. <b>PC</b> .                                                                    |
| 1.2                                                     | Read <b>PC</b>   | Store as Operand0.                                                                  |
| 2.1                                                     |                  | Inc. <b>PC</b> .                                                                    |
| 2.2                                                     | Read <b>PC</b>   | Store as Operand.                                                                   |
| 3.1                                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                 |
| 3.2                                                     | Read Address     | Store as Operand1.                                                                  |
| 4.1                                                     |                  | Operand0 ^= Operand1. <b>N</b> flag = (Operand0 & \$80), <b>Z</b> flag = !Operand0. |
| 4.2                                                     | Write to Address | Write Operand0.                                                                     |
| 5.1                                                     |                  |                                                                                     |
| 5.2                                                     | Read <b>PC</b>   | Store as OpCode.                                                                    |
| +X.1                                                    |                  |                                                                                     |

|                                                             |                                                    |                                                                                     |
|-------------------------------------------------------------|----------------------------------------------------|-------------------------------------------------------------------------------------|
| EOR (59)                                                    | Cycles: 5                                          | Size: 1                                                                             |
| Indirect Page to Indirect Page (X), (Y) (Read/Modify/Write) |                                                    |                                                                                     |
| Cycle                                                       | R/W                                                | Desc.                                                                               |
| -X.2                                                        | Read <b>PC</b>                                     | Store as OpCode.                                                                    |
| 1.1                                                         |                                                    | Inc. <b>PC</b> .                                                                    |
| 1.2                                                         | Read <b>PC</b>                                     | Discard.                                                                            |
| 2.1                                                         |                                                    |                                                                                     |
| 2.2                                                         | Read ( <b>Y</b>   (( <b>PSW</b> & \$20) << 3))     | Store as Operand0.                                                                  |
| 3.1                                                         |                                                    |                                                                                     |
| 3.2                                                         | Read ( <b>X</b>   (( <b>PSW</b> & \$20) << 3))     | Store as Operand1.                                                                  |
| 4.1                                                         |                                                    | Operand0 ^= Operand1. <b>N</b> flag = (Operand0 & \$80), <b>Z</b> flag = !Operand0. |
| 4.2                                                         | Write to ( <b>X</b>   (( <b>PSW</b> & \$20) << 3)) | Write Operand0.                                                                     |
| 5.1                                                         |                                                    |                                                                                     |
| 5.2                                                         | Read <b>PC</b>                                     | Store as OpCode.                                                                    |
| +X.1                                                        |                                                    |                                                                                     |

|                         |                                                           |                                                                                                                |
|-------------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| CMPW (5A)               | Cycles: 4                                                 | Size: 2                                                                                                        |
| Direct Page d (Read 16) |                                                           |                                                                                                                |
| Cycle                   | R/W                                                       | Desc.                                                                                                          |
| -X.2                    | Read <b>PC</b>                                            | Store as OpCode.                                                                                               |
| 1.1                     |                                                           | Inc. <b>PC</b> .                                                                                               |
| 1.2                     | Read <b>PC</b>                                            | Store as Operand.                                                                                              |
| 2.1                     |                                                           | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                            |
| 2.2                     | Read Address                                              | Store as Operand16.L.                                                                                          |
| 3.1                     |                                                           |                                                                                                                |
| 3.2                     | Read ((Address + 1) & \$FF)   (( <b>PSW</b> & \$20) << 3) | Store as Operand16.H.                                                                                          |
| 4.1                     |                                                           | Tmp = <b>YA</b> - Operand16. <b>C</b> flag = (Tmp >= 0), <b>N</b> flag = (Tmp & \$8000), <b>Z</b> flag = !Tmp. |
| 4.2                     | Read <b>PC</b>                                            | Store as OpCode.                                                                                               |
| +X.1                    |                                                           |                                                                                                                |

|                                               |                  |                                                                                                            |
|-----------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------|
| LSR (5B)                                      | Cycles: 5        | Size: 2                                                                                                    |
| X-Indexed Direct Page d+X (Read/Modify/Write) |                  |                                                                                                            |
| Cycle                                         | R/W              | Desc.                                                                                                      |
| -X.2                                          | Read <b>PC</b>   | Store as OpCode.                                                                                           |
| 1.1                                           |                  | Inc. <b>PC</b> .                                                                                           |
| 1.2                                           | Read <b>PC</b>   | Store as Operand.                                                                                          |
| 2.1                                           |                  | Inc. <b>PC</b> .                                                                                           |
| 2.2                                           |                  |                                                                                                            |
| 3.1                                           |                  | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).                                    |
| 3.2                                           | Read Address     | Store as Operand.                                                                                          |
| 4.1                                           |                  | <b>C</b> flag = (Operand & 01). Operand >>= 1. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = !Operand. |
| 4.2                                           | Write to Address | Write Operand.                                                                                             |
| 5.1                                           |                  |                                                                                                            |
| 5.2                                           | Read <b>PC</b>   | Store as OpCode.                                                                                           |
| +X.1                                          |                  |                                                                                                            |

|               |                |                                                                                                                    |
|---------------|----------------|--------------------------------------------------------------------------------------------------------------------|
| LSR (5C)      | Cycles: 2      | Size: 1                                                                                                            |
| Accumulator A |                |                                                                                                                    |
| Cycle         | R/W            | Desc.                                                                                                              |
| -X.2          | Read <b>PC</b> | Store as OpCode.                                                                                                   |
| 1.1           |                | Inc. <b>PC</b> .                                                                                                   |
| 1.2           | Read <b>PC</b> | Discard.                                                                                                           |
| 2.1           |                | <b>C</b> flag = ( <b>A</b> & 01). <b>A</b> >>= 1. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 2.2           | Read <b>PC</b> | Store as OpCode.                                                                                                   |
| +X.1          |                |                                                                                                                    |

|          |                |                                                                                        |
|----------|----------------|----------------------------------------------------------------------------------------|
| MOV (5D) | Cycles: 2      | Size: 1                                                                                |
| Implied  |                |                                                                                        |
| Cycle    | R/W            | Desc.                                                                                  |
| -X.2     | Read <b>PC</b> | Store as OpCode.                                                                       |
| 1.1      |                | Inc. <b>PC</b> .                                                                       |
| 1.2      |                |                                                                                        |
| 2.1      |                | <b>X</b> = <b>A</b> . <b>N</b> flag = ( <b>X</b> & \$80), <b>Z</b> flag = ! <b>X</b> . |
| 2.2      |                | Store as OpCode.                                                                       |
| +X.1     |                |                                                                                        |

|                    |                |                                                                                                           |
|--------------------|----------------|-----------------------------------------------------------------------------------------------------------|
| CMP (5E)           | Cycles: 4      | Size: 3                                                                                                   |
| Absolute !a (Read) |                |                                                                                                           |
| Cycle              | R/W            | Desc.                                                                                                     |
| -X.2               | Read <b>PC</b> | Store as OpCode.                                                                                          |
| 1.1                |                | Inc. <b>PC</b> .                                                                                          |
| 1.2                | Read <b>PC</b> | Store as Address.L.                                                                                       |
| 2.1                |                | Inc. <b>PC</b> .                                                                                          |
| 2.2                | Read <b>PC</b> | Store as Address.H.                                                                                       |
| 3.1                |                | Inc. <b>PC</b> .                                                                                          |
| 3.2                | Read Address   | Store as Operand.                                                                                         |
| 4.1                |                | Tmp = <b>Y</b> - Operand. <b>C</b> flag = (Tmp >= 0), <b>N</b> flag = (Tmp & \$80), <b>Z</b> flag = !Tmp. |
| 4.2                | Read <b>PC</b> | Store as OpCode.                                                                                          |
| +X.1               |                |                                                                                                           |

|             |                |                      |
|-------------|----------------|----------------------|
| JMP (5F)    | Cycles: 3      | Size: 3              |
| Absolute !a |                |                      |
| Cycle       | R/W            | Desc.                |
| -X.2        | Read <b>PC</b> | Store as OpCode.     |
| 1.1         |                | Inc. <b>PC</b> .     |
| 1.2         | Read <b>PC</b> | Store as Address.L.  |
| 2.1         |                | Inc. <b>PC</b> .     |
| 2.2         | Read <b>PC</b> | Store as Address.H.  |
| 3.1         |                | <b>PC</b> = Address. |
| 3.2         | Read <b>PC</b> | Store as OpCode.     |
| +X.1        |                |                      |



|           |                |                                     |
|-----------|----------------|-------------------------------------|
| CLRC (60) | Cycles: 2      | Size: 1                             |
| Implied   |                |                                     |
| Cycle     | R/W            | Desc.                               |
| -X.2      | Read <b>PC</b> | Store as OpCode.                    |
| 1.1       |                | Inc. <b>PC</b> .                    |
| 1.2       |                |                                     |
| 2.1       |                | <b>PSW</b> = ( <b>PSW</b> & ~\$01). |
| 2.2       |                | Store as OpCode.                    |
| +X.1      |                |                                     |

|            |                                     |                               |
|------------|-------------------------------------|-------------------------------|
| TCALL (61) | Cycles: 8                           | Size: 1                       |
| Implied    |                                     |                               |
| Cycle      | R/W                                 | Desc.                         |
| -X.2       | Read <b>PC</b>                      | Store as OpCode.              |
| 1.1        |                                     | Inc. <b>PC</b> .              |
| 1.2        | Read <b>PC</b>                      | Store as Operand.             |
| 2.1        |                                     |                               |
| 2.2        |                                     |                               |
| 3.1        |                                     |                               |
| 3.2        | Write to u8( <b>SP</b> +0)   \$0100 | Push <b>PC</b> .H onto stack. |
| 4.1        |                                     |                               |
| 4.2        | Write to u8( <b>SP</b> -1)   \$0100 | Push <b>PC</b> .L onto stack. |
| 5.1        |                                     | Dec. <b>SP</b> by 2.          |
| 5.2        |                                     |                               |
| 6.1        |                                     |                               |
| 6.2        | Read \$FFD2                         | Store as <b>PC</b> .L.        |
| 7.1        |                                     |                               |
| 7.2        | Read \$FFD3                         | Store as <b>PC</b> .H.        |
| 8.1        |                                     |                               |
| 8.2        | Read <b>PC</b>                      | Store as OpCode.              |
| +X.1       |                                     |                               |

|                                         |                  |                                                                     |
|-----------------------------------------|------------------|---------------------------------------------------------------------|
| SET1 (62)                               | Cycles: 4        | Size: 2                                                             |
| Direct Page Bit d.b (Read/Modify/Write) |                  |                                                                     |
| Cycle                                   | R/W              | Desc.                                                               |
| -X.2                                    | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                                     |                  | Inc. <b>PC</b> .                                                    |
| 1.2                                     | Read <b>PC</b>   | Store as Operand.                                                   |
| 2.1                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                                     | Read Address     | Store as Operand.                                                   |
| 3.1                                     |                  | Operand = (Operand & ~\$08)   \$08.                                 |
| 3.2                                     | Write to Address | Write Operand.                                                      |
| 4.1                                     |                  |                                                                     |
| 4.2                                     | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                                    |                  |                                                                     |

|                                 |                |                                                                      |
|---------------------------------|----------------|----------------------------------------------------------------------|
| BBS (63)                        | Cycles: 5-7    | Size: 3                                                              |
| Direct Page Bit Relative d.b, r |                |                                                                      |
| Cycle                           | R/W            | Desc.                                                                |
| -X.2                            | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1                             |                | Inc. <b>PC</b> .                                                     |
| 1.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 2.1                             |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).  |
| 2.2                             | Read Address   | Store as Operand.                                                    |
| 3.1                             |                | Set Jump to (Operand & (1 << 3)) == (1 << 3).                        |
| 3.2                             |                |                                                                      |
| 4.1                             |                |                                                                      |
| 4.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 5.1                             |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 7.2). |
| 5.2                             |                |                                                                      |
| 6.1                             |                | Address = <b>PC</b> + i8(Operand).<br><b>PC</b> .L = Address.L.      |
| 6.2                             |                |                                                                      |
| 7.1                             |                | <b>PC</b> .H = Address.H.                                            |
| 7.2                             | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1                            |                |                                                                      |

|                      |                |                                                                                                           |
|----------------------|----------------|-----------------------------------------------------------------------------------------------------------|
| CMP (64)             | Cycles: 3      | Size: 2                                                                                                   |
| Direct Page d (Read) |                |                                                                                                           |
| Cycle                | R/W            | Desc.                                                                                                     |
| -X.2                 | Read <b>PC</b> | Store as OpCode.                                                                                          |
| 1.1                  |                | Inc. <b>PC</b> .                                                                                          |
| 1.2                  | Read <b>PC</b> | Store as Operand.                                                                                         |
| 2.1                  |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                       |
| 2.2                  | Read Address   | Store as Operand.                                                                                         |
| 3.1                  |                | Tmp = <b>A</b> - Operand. <b>C</b> flag = (Tmp >= 0), <b>N</b> flag = (Tmp & \$80), <b>Z</b> flag = !Tmp. |
| 3.2                  | Read <b>PC</b> | Store as OpCode.                                                                                          |
| +X.1                 |                |                                                                                                           |

|                    |                |                                                                                                           |
|--------------------|----------------|-----------------------------------------------------------------------------------------------------------|
| CMP (65)           | Cycles: 4      | Size: 3                                                                                                   |
| Absolute !a (Read) |                |                                                                                                           |
| Cycle              | R/W            | Desc.                                                                                                     |
| -X.2               | Read <b>PC</b> | Store as OpCode.                                                                                          |
| 1.1                |                | Inc. <b>PC</b> .                                                                                          |
| 1.2                | Read <b>PC</b> | Store as Address.L.                                                                                       |
| 2.1                |                | Inc. <b>PC</b> .                                                                                          |
| 2.2                | Read <b>PC</b> | Store as Address.H.                                                                                       |
| 3.1                |                | Inc. <b>PC</b> .                                                                                          |
| 3.2                | Read Address   | Store as Operand.                                                                                         |
| 4.1                |                | Tmp = <b>A</b> - Operand. <b>C</b> flag = (Tmp >= 0), <b>N</b> flag = (Tmp & \$80), <b>Z</b> flag = !Tmp. |
| 4.2                | Read <b>PC</b> | Store as OpCode.                                                                                          |
| +X.1               |                |                                                                                                           |

|                       |                                                |                                                                                                           |
|-----------------------|------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| CMP (66)              | Cycles: 3                                      | Size: 1                                                                                                   |
| Indirect X (X) (Read) |                                                |                                                                                                           |
| Cycle                 | R/W                                            | Desc.                                                                                                     |
| -X.2                  | Read <b>PC</b>                                 | Store as OpCode.                                                                                          |
| 1.1                   |                                                | Inc. <b>PC</b> .                                                                                          |
| 1.2                   |                                                |                                                                                                           |
| 2.1                   |                                                |                                                                                                           |
| 2.2                   | Read ( <b>X</b>   (( <b>PSW</b> & \$20) << 3)) | Store as Operand.                                                                                         |
| 3.1                   |                                                | Tmp = <b>A</b> - Operand. <b>C</b> flag = (Tmp >= 0), <b>N</b> flag = (Tmp & \$80), <b>Z</b> flag = !Tmp. |
| 3.2                   | Read <b>PC</b>                                 | Store as OpCode.                                                                                          |
| +X.1                  |                                                |                                                                                                           |

|                                 |                                                              |                                                                                                           |
|---------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| CMP (67)                        | Cycles: 6                                                    | Size: 2                                                                                                   |
| X-Indexed Indirect [d+X] (Read) |                                                              |                                                                                                           |
| Cycle                           | R/W                                                          | Desc.                                                                                                     |
| -X.2                            | Read <b>PC</b>                                               | Store as OpCode.                                                                                          |
| 1.1                             |                                                              | Inc. <b>PC</b> .                                                                                          |
| 1.2                             | Read <b>PC</b>                                               | Store as Operand.                                                                                         |
| 2.1                             |                                                              | Inc. <b>PC</b> .                                                                                          |
| 2.2                             |                                                              |                                                                                                           |
| 3.1                             |                                                              | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).                                   |
| 3.2                             | Read Address                                                 | Store as Pointer.L.                                                                                       |
| 4.1                             |                                                              |                                                                                                           |
| 4.2                             | Read ((Address + 1) & \$FF)  <br>(( <b>PSW</b> & \$20) << 3) | Store as Pointer.H.                                                                                       |
| 5.1                             |                                                              |                                                                                                           |
| 5.2                             | Read Pointer                                                 | Store as Operand.                                                                                         |
| 6.1                             |                                                              | Tmp = <b>A</b> - Operand. <b>C</b> flag = (Tmp >= 0), <b>N</b> flag = (Tmp & \$80), <b>Z</b> flag = !Tmp. |
| 6.2                             | Read <b>PC</b>                                               | Store as OpCode.                                                                                          |
| +X.1                            |                                                              |                                                                                                           |



|           |                |                                                                                                                            |
|-----------|----------------|----------------------------------------------------------------------------------------------------------------------------|
| CMP (68)  | Cycles: 2      | Size: 2                                                                                                                    |
| Immediate |                |                                                                                                                            |
| Cycle     | R/W            | Desc.                                                                                                                      |
| -X.2      | Read <b>PC</b> | Store as OpCode.                                                                                                           |
| 1.1       |                | Inc. <b>PC</b> .                                                                                                           |
| 1.2       | Read <b>PC</b> | Store as Operand.                                                                                                          |
| 2.1       |                | Inc. <b>PC</b> . Tmp = <b>A</b> - Operand. <b>C</b> flag = (Tmp >= 0), <b>N</b> flag = (Tmp & \$80), <b>Z</b> flag = !Tmp. |
| 2.2       | Read <b>PC</b> | Store as OpCode.                                                                                                           |
| +X.1      |                |                                                                                                                            |

|                                          |                |                                                                                                            |
|------------------------------------------|----------------|------------------------------------------------------------------------------------------------------------|
| CMP (69)                                 | Cycles: 6      | Size: 3                                                                                                    |
| Direct Page to Direct Page dd, ds (Read) |                |                                                                                                            |
| Cycle                                    | R/W            | Desc.                                                                                                      |
| -X.2                                     | Read <b>PC</b> | Store as OpCode.                                                                                           |
| 1.1                                      |                | Inc. <b>PC</b> .                                                                                           |
| 1.2                                      | Read <b>PC</b> | Store as Operand.                                                                                          |
| 2.1                                      |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                        |
| 2.2                                      | Read Address   | Store as Operand0.                                                                                         |
| 3.1                                      |                |                                                                                                            |
| 3.2                                      | Read <b>PC</b> | Store as Operand.                                                                                          |
| 4.1                                      |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                        |
| 4.2                                      | Read Address   | Store as Operand1.                                                                                         |
| 5.1                                      |                | Tmp = Operand1 - Operand0. <b>C</b> flag = (Tmp >= 0), <b>N</b> flag = (Tmp & \$80), <b>Z</b> flag = !Tmp. |
| 5.2                                      |                |                                                                                                            |
| 6.1                                      |                |                                                                                                            |
| 6.2                                      | Read <b>PC</b> | Store as OpCode.                                                                                           |
| +X.1                                     |                |                                                                                                            |

|                          |                         |                                                                         |
|--------------------------|-------------------------|-------------------------------------------------------------------------|
| AND1 (6A)                | Cycles: 4               | Size: 3                                                                 |
| Absolute Boolean Bit m.b |                         |                                                                         |
| Cycle                    | R/W                     | Desc.                                                                   |
| -X.2                     | Read <b>PC</b>          | Store as OpCode.                                                        |
| 1.1                      |                         | Inc. <b>PC</b> .                                                        |
| 1.2                      | Read <b>PC</b>          | Store as Address.L.                                                     |
| 2.1                      |                         | Inc. <b>PC</b> .                                                        |
| 2.2                      | Read <b>PC</b>          | Store as Address.H.                                                     |
| 3.1                      |                         | Inc. <b>PC</b> .                                                        |
| 3.2                      | Read (Address & \$1FFF) | Store as Operand.                                                       |
| 4.1                      |                         | Bit = Address >> 13.<br><b>C</b> flag &= ((Operand & (1 << Bit)) == 0). |
| 4.2                      | Read <b>PC</b>          | Store as OpCode.                                                        |
| +X.1                     |                         |                                                                         |

|                                   |                  |                                                                                                                                                            |
|-----------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROR (6B)                          | Cycles: 4        | Size: 2                                                                                                                                                    |
| Direct Page d (Read/Modify/Write) |                  |                                                                                                                                                            |
| Cycle                             | R/W              | Desc.                                                                                                                                                      |
| -X.2                              | Read <b>PC</b>   | Store as OpCode.                                                                                                                                           |
| 1.1                               |                  | Inc. <b>PC</b> .                                                                                                                                           |
| 1.2                               | Read <b>PC</b>   | Store as Operand.                                                                                                                                          |
| 2.1                               |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                                                                        |
| 2.2                               | Read Address     | Store as Operand.                                                                                                                                          |
| 3.1                               |                  | Tmp = ( <b>C</b> flag << 7). <b>C</b> flag = (Operand & \$01). Operand = (Operand << 1)   Tmp. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = !Operand. |
| 3.2                               | Write to Address | Write Operand.                                                                                                                                             |
| 4.1                               |                  |                                                                                                                                                            |
| 4.2                               | Read <b>PC</b>   | Store as OpCode.                                                                                                                                           |
| +X.1                              |                  |                                                                                                                                                            |

|                                 |                  |                                                                                                                                                  |
|---------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| ROR (6C)                        | Cycles: 5        | Size: 3                                                                                                                                          |
| Absolute !a (Read/Modify/Write) |                  |                                                                                                                                                  |
| Cycle                           | R/W              | Desc.                                                                                                                                            |
| -X.2                            | Read <b>PC</b>   | Store as OpCode.                                                                                                                                 |
| 1.1                             |                  | Inc. <b>PC</b> .                                                                                                                                 |
| 1.2                             | Read <b>PC</b>   | Store as Address.L.                                                                                                                              |
| 2.1                             |                  | Inc. <b>PC</b> .                                                                                                                                 |
| 2.2                             | Read <b>PC</b>   | Store as Address.H.                                                                                                                              |
| 3.1                             |                  | Inc. <b>PC</b> .                                                                                                                                 |
| 3.2                             | Read Address     | Store as Operand.                                                                                                                                |
| 4.1                             |                  | Tmp = ( <b>C</b> flag << 7). <b>C</b> flag = (Operand & \$01). Operand = (Operand << 1)   Tmp. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = |
| 4.2                             | Write to Address | Write Operand.                                                                                                                                   |
| 5.1                             |                  |                                                                                                                                                  |
| 5.2                             | Read <b>PC</b>   | Store as OpCode.                                                                                                                                 |
| +X.1                            |                  |                                                                                                                                                  |

|           |                                     |           |                           |
|-----------|-------------------------------------|-----------|---------------------------|
| PUSH (6D) |                                     | Cycles: 4 | Size: 1                   |
| Implied   |                                     |           |                           |
| Cycle     | R/W                                 |           | Desc.                     |
| -X.2      | Read <b>PC</b>                      |           | Store as OpCode.          |
| 1.1       |                                     |           | Inc. <b>PC</b> .          |
| 1.2       | Read <b>PC</b>                      |           | Discard.                  |
| 2.1       |                                     |           |                           |
| 2.2       | Write to u8( <b>SP</b> +0)   \$0100 |           | Push <b>Y</b> onto stack. |
| 3.1       |                                     |           | Dec. <b>SP</b> by 1.      |
| 3.2       |                                     |           |                           |
| 4.1       |                                     |           |                           |
| 4.2       | Read <b>PC</b>                      |           | Store as OpCode.          |
| +X.1      |                                     |           |                           |

|                           |                  |                                                                      |
|---------------------------|------------------|----------------------------------------------------------------------|
| DBNZ (6E)                 | Cycles: 5-7      | Size: 3                                                              |
| Direct Page Relative d, r |                  |                                                                      |
| Cycle                     | R/W              | Desc.                                                                |
| -X.2                      | Read <b>PC</b>   | Store as OpCode.                                                     |
| 1.1                       |                  | Inc. <b>PC</b> .                                                     |
| 1.2                       | Read <b>PC</b>   | Store as Operand.                                                    |
| 2.1                       |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).  |
| 2.2                       | Read Address     | Store as Operand.                                                    |
| 3.1                       |                  | --Operand. Set Jump to (Operand != 0).                               |
| 3.2                       | Write to Address | Write Operand.                                                       |
| 4.1                       |                  |                                                                      |
| 4.2                       | Read <b>PC</b>   | Store as Operand.                                                    |
| 5.1                       |                  | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 7.2). |
| 5.2                       |                  |                                                                      |
| 6.1                       |                  | Address = <b>PC</b> + i8(Operand).<br><b>PC</b> .L = Address.L.      |
| 6.2                       |                  |                                                                      |
| 7.1                       |                  | <b>PC</b> .H = Address.H.                                            |
| 7.2                       | Read <b>PC</b>   | Store as OpCode.                                                     |
| +X.1                      |                  |                                                                      |

|          |                                 |                       |
|----------|---------------------------------|-----------------------|
| RET (6F) | Cycles: 5                       | Size: 1               |
| Implied  |                                 |                       |
| Cycle    | R/W                             | Desc.                 |
| -X.2     | Read <b>PC</b>                  | Store as OpCode.      |
| 1.1      |                                 | Inc. <b>PC</b> .      |
| 1.2      | Read <b>PC</b>                  | Discard.              |
| 2.1      |                                 |                       |
| 2.2      |                                 |                       |
| 3.1      |                                 | Inc. <b>SP</b> by 1.  |
| 3.2      | Read u8( <b>SP</b> +0)   \$0100 | Pull to <b>PC</b> .L. |
| 4.1      |                                 |                       |
| 4.2      | Read u8( <b>SP</b> +1)   \$0100 | Pull to <b>PC</b> .H. |
| 5.1      | Inc. <b>SP</b> by 1.            |                       |
| 5.2      | Read <b>PC</b>                  | Store as OpCode.      |
| +X.1     |                                 |                       |



|            |                |                                                                      |
|------------|----------------|----------------------------------------------------------------------|
| BVS (70)   | Cycles: 2-4    | Size: 2                                                              |
| Relative r |                |                                                                      |
| Cycle      | R/W            | Desc.                                                                |
| -X.2       | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1        |                | Inc. <b>PC</b> . Set Jump to ( <b>PSW</b> & \$40) == \$40.           |
| 1.2        |                | Store as Operand.                                                    |
| 2.1        |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 4.2). |
| 2.2        |                |                                                                      |
| 3.1        |                | Address = <b>PC</b> + i8(Operand).                                   |
|            |                | <b>PC</b> .L = Address.L.                                            |
| 3.2        |                |                                                                      |
| 4.1        |                | <b>PC</b> .H = Address.H.                                            |
| 4.2        | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1       |                |                                                                      |

|            |                                     |                               |
|------------|-------------------------------------|-------------------------------|
| TCALL (71) | Cycles: 8                           | Size: 1                       |
| Implied    |                                     |                               |
| Cycle      | R/W                                 | Desc.                         |
| -X.2       | Read <b>PC</b>                      | Store as OpCode.              |
| 1.1        |                                     | Inc. <b>PC</b> .              |
| 1.2        | Read <b>PC</b>                      | Store as Operand.             |
| 2.1        |                                     |                               |
| 2.2        |                                     |                               |
| 3.1        |                                     |                               |
| 3.2        | Write to u8( <b>SP</b> +0)   \$0100 | Push <b>PC</b> .H onto stack. |
| 4.1        |                                     |                               |
| 4.2        | Write to u8( <b>SP</b> -1)   \$0100 | Push <b>PC</b> .L onto stack. |
| 5.1        |                                     | Dec. <b>SP</b> by 2.          |
| 5.2        |                                     |                               |
| 6.1        |                                     |                               |
| 6.2        | Read \$FFD0                         | Store as <b>PC</b> .L.        |
| 7.1        |                                     |                               |
| 7.2        | Read \$FFD1                         | Store as <b>PC</b> .H.        |
| 8.1        |                                     |                               |
| 8.2        | Read <b>PC</b>                      | Store as OpCode.              |
| +X.1       |                                     |                               |

|                                         |                  |                                                                     |
|-----------------------------------------|------------------|---------------------------------------------------------------------|
| CLR1 (72)                               | Cycles: 4        | Size: 2                                                             |
| Direct Page Bit d.b (Read/Modify/Write) |                  |                                                                     |
| Cycle                                   | R/W              | Desc.                                                               |
| -X.2                                    | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                                     |                  | Inc. <b>PC</b> .                                                    |
| 1.2                                     | Read <b>PC</b>   | Store as Operand.                                                   |
| 2.1                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                                     | Read Address     | Store as Operand.                                                   |
| 3.1                                     |                  | Operand = (Operand & ~\$08)   \$00.                                 |
| 3.2                                     | Write to Address | Write Operand.                                                      |
| 4.1                                     |                  |                                                                     |
| 4.2                                     | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                                    |                  |                                                                     |

|                                 |                |                                                                      |
|---------------------------------|----------------|----------------------------------------------------------------------|
| BBC (73)                        | Cycles: 5-7    | Size: 3                                                              |
| Direct Page Bit Relative d.b, r |                |                                                                      |
| Cycle                           | R/W            | Desc.                                                                |
| -X.2                            | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1                             |                | Inc. <b>PC</b> .                                                     |
| 1.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 2.1                             |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).  |
| 2.2                             | Read Address   | Store as Operand.                                                    |
| 3.1                             |                | Set Jump to (Operand & (1 << 3)) == (0 << 3).                        |
| 3.2                             |                |                                                                      |
| 4.1                             |                |                                                                      |
| 4.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 5.1                             |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 7.2). |
| 5.2                             |                |                                                                      |
| 6.1                             |                | Address = <b>PC</b> + i8(Operand).<br><b>PC</b> .L = Address.L.      |
| 6.2                             |                |                                                                      |
| 7.1                             |                | <b>PC</b> .H = Address.H.                                            |
| 7.2                             | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1                            |                |                                                                      |

|                                  |                |                                                                                                           |
|----------------------------------|----------------|-----------------------------------------------------------------------------------------------------------|
| CMP (74)                         | Cycles: 4      | Size: 2                                                                                                   |
| X-Indexed Direct Page d+X (Read) |                |                                                                                                           |
| Cycle                            | R/W            | Desc.                                                                                                     |
| -X.2                             | Read <b>PC</b> | Store as OpCode.                                                                                          |
| 1.1                              |                | Inc. <b>PC</b> .                                                                                          |
| 1.2                              | Read <b>PC</b> | Store as Operand.                                                                                         |
| 2.1                              |                | Inc. <b>PC</b> .                                                                                          |
| 2.2                              |                |                                                                                                           |
| 3.1                              |                | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).                                   |
| 3.2                              | Read Address   | Store as Operand.                                                                                         |
| 4.1                              |                | Tmp = <b>A</b> - Operand. <b>C</b> flag = (Tmp >= 0), <b>N</b> flag = (Tmp & \$80), <b>Z</b> flag = !Tmp. |
| 4.2                              | Read <b>PC</b> | Store as OpCode.                                                                                          |
| +X.1                             |                |                                                                                                           |

|                                |                |                                                                                                           |
|--------------------------------|----------------|-----------------------------------------------------------------------------------------------------------|
| CMP (75)                       | Cycles: 5      | Size: 3                                                                                                   |
| X-Indexed Absolute !a+X (Read) |                |                                                                                                           |
| Cycle                          | R/W            | Desc.                                                                                                     |
| -X.2                           | Read <b>PC</b> | Store as OpCode.                                                                                          |
| 1.1                            |                | Inc. <b>PC</b> .                                                                                          |
| 1.2                            | Read <b>PC</b> | Store as Address.L.                                                                                       |
| 2.1                            |                | Inc. <b>PC</b> .                                                                                          |
| 2.2                            | Read <b>PC</b> | Store as Address.H.                                                                                       |
| 3.1                            |                | Inc. <b>PC</b> .                                                                                          |
| 3.2                            |                |                                                                                                           |
| 4.1                            |                | Pointer = ( <b>X</b> + Address).                                                                          |
| 4.2                            | Read Pointer   | Store as Operand.                                                                                         |
| 5.1                            |                | Tmp = <b>A</b> - Operand. <b>C</b> flag = (Tmp >= 0), <b>N</b> flag = (Tmp & \$80), <b>Z</b> flag = !Tmp. |
| 5.2                            | Read <b>PC</b> | Store as OpCode.                                                                                          |
| +X.1                           |                |                                                                                                           |

|                                |                |                                                                                                           |
|--------------------------------|----------------|-----------------------------------------------------------------------------------------------------------|
| CMP (76)                       | Cycles: 5      | Size: 3                                                                                                   |
| Y-Indexed Absolute !a+Y (Read) |                |                                                                                                           |
| Cycle                          | R/W            | Desc.                                                                                                     |
| -X.2                           | Read <b>PC</b> | Store as OpCode.                                                                                          |
| 1.1                            |                | Inc. <b>PC</b> .                                                                                          |
| 1.2                            | Read <b>PC</b> | Store as Address.L.                                                                                       |
| 2.1                            |                | Inc. <b>PC</b> .                                                                                          |
| 2.2                            | Read <b>PC</b> | Store as Address.H.                                                                                       |
| 3.1                            |                | Inc. <b>PC</b> .                                                                                          |
| 3.2                            |                |                                                                                                           |
| 4.1                            |                | Pointer = ( <b>Y</b> + Address).                                                                          |
| 4.2                            | Read Pointer   | Store as Operand.                                                                                         |
| 5.1                            |                | Tmp = <b>A</b> - Operand. <b>C</b> flag = (Tmp >= 0), <b>N</b> flag = (Tmp & \$80), <b>Z</b> flag = !Tmp. |
| 5.2                            | Read <b>PC</b> | Store as OpCode.                                                                                          |
| +X.1                           |                |                                                                                                           |

|                                 |                                                              |                                                                                                           |
|---------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| CMP (77)                        | Cycles: 6                                                    | Size: 2                                                                                                   |
| Indirect Y-Indexed [d]+Y (Read) |                                                              |                                                                                                           |
| Cycle                           | R/W                                                          | Desc.                                                                                                     |
| -X.2                            | Read <b>PC</b>                                               | Store as OpCode.                                                                                          |
| 1.1                             |                                                              | Inc. <b>PC</b> .                                                                                          |
| 1.2                             | Read <b>PC</b>                                               | Store as Operand.                                                                                         |
| 2.1                             |                                                              | Inc. <b>PC</b> .                                                                                          |
| 2.2                             |                                                              |                                                                                                           |
| 3.1                             |                                                              | Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                                        |
| 3.2                             | Read Address                                                 | Store as Pointer.L.                                                                                       |
| 4.1                             |                                                              |                                                                                                           |
| 4.2                             | Read ((Address + 1) & \$FF)  <br>(( <b>PSW</b> & \$20) << 3) | Store as Pointer.H.                                                                                       |
| 5.1                             |                                                              | Address = ( <b>Y</b> + Pointer).                                                                          |
| 5.2                             | Read Address                                                 | Store as Operand.                                                                                         |
| 6.1                             |                                                              | Tmp = <b>A</b> - Operand. <b>C</b> flag = (Tmp >= 0), <b>N</b> flag = (Tmp & \$80), <b>Z</b> flag = !Tmp. |
| 6.2                             | Read <b>PC</b>                                               | Store as OpCode.                                                                                          |
| +X.1                            |                                                              |                                                                                                           |



|                                            |                |                                                                                                            |
|--------------------------------------------|----------------|------------------------------------------------------------------------------------------------------------|
| CMP (78)                                   | Cycles: 5      | Size: 3                                                                                                    |
| Immediate Data to Direct Page d, #i (Read) |                |                                                                                                            |
| Cycle                                      | R/W            | Desc.                                                                                                      |
| -X.2                                       | Read <b>PC</b> | Store as OpCode.                                                                                           |
| 1.1                                        |                | Inc. <b>PC</b> .                                                                                           |
| 1.2                                        | Read <b>PC</b> | Store as Operand0.                                                                                         |
| 2.1                                        |                | Inc. <b>PC</b> .                                                                                           |
| 2.2                                        | Read <b>PC</b> | Store as Operand.                                                                                          |
| 3.1                                        |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                        |
| 3.2                                        | Read Address   | Store as Operand1.                                                                                         |
| 4.1                                        |                | Tmp = Operand1 - Operand0. <b>C</b> flag = (Tmp >= 0), <b>N</b> flag = (Tmp & \$80), <b>Z</b> flag = !Tmp. |
| 4.2                                        |                |                                                                                                            |
| 5.1                                        |                |                                                                                                            |
| 5.2                                        | Read <b>PC</b> | Store as OpCode.                                                                                           |
| +X.1                                       |                |                                                                                                            |

|                                                |                                                |                                                                                                            |
|------------------------------------------------|------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| CMP (79)                                       | Cycles: 5                                      | Size: 1                                                                                                    |
| Indirect Page to Indirect Page (X), (Y) (Read) |                                                |                                                                                                            |
| Cycle                                          | R/W                                            | Desc.                                                                                                      |
| -X.2                                           | Read <b>PC</b>                                 | Store as OpCode.                                                                                           |
| 1.1                                            |                                                | Inc. <b>PC</b> .                                                                                           |
| 1.2                                            | Read <b>PC</b>                                 | Discard.                                                                                                   |
| 2.1                                            |                                                |                                                                                                            |
| 2.2                                            | Read ( <b>Y</b>   (( <b>PSW</b> & \$20) << 3)) | Store as Operand0.                                                                                         |
| 3.1                                            |                                                |                                                                                                            |
| 3.2                                            | Read ( <b>X</b>   (( <b>PSW</b> & \$20) << 3)) | Store as Operand1.                                                                                         |
| 4.1                                            |                                                | Tmp = Operand1 - Operand0. <b>C</b> flag = (Tmp >= 0), <b>N</b> flag = (Tmp & \$80), <b>Z</b> flag = !Tmp. |
| 4.2                                            |                                                |                                                                                                            |
| 5.1                                            |                                                |                                                                                                            |
| 5.2                                            | Read <b>PC</b>                                 | Store as OpCode.                                                                                           |
| +X.1                                           |                                                |                                                                                                            |

|                         |                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |
|-------------------------|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| ADDW (7A)               |                                                           | Cycles: 5                                                                                                                                                                                                                                                                                                                                                                                                                                            | Size: 2 |
| Direct Page d (Read 16) |                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |
| Cycle                   | R/W                                                       | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |
| -X.2                    | Read <b>PC</b>                                            | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |
| 1.1                     |                                                           | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |
| 1.2                     | Read <b>PC</b>                                            | Store as Operand.                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |
| 2.1                     |                                                           | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                                                                                                                                                                                                                                                                                                                                                                  |         |
| 2.2                     | Read Address                                              | Store as Operand16.L.                                                                                                                                                                                                                                                                                                                                                                                                                                |         |
| 3.1                     |                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |
| 3.2                     |                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |
| 4.1                     |                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |
| 4.2                     | Read ((Address + 1) & \$FF)   (( <b>PSW</b> & \$20) << 3) | Store as Operand16.H.                                                                                                                                                                                                                                                                                                                                                                                                                                |         |
| 5.1                     |                                                           | Tmp = u32( <b>A</b> ) + u32(Operand16.L).<br><b>C</b> flag = (Tmp > \$FF).<br><b>A</b> = u8(Tmp).<br>Tmp = u32( <b>Y</b> ) + u32(Operand16.H) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>H</b> flag = (( <b>Y</b> ^ Operand16.H ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~( <b>Y</b> ^ Operand16.H) & ( <b>Y</b> ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br><b>Y</b> = u8(Tmp).<br><b>Z</b> flag = (YA == \$0000). |         |
| 5.2                     | Read <b>PC</b>                                            | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |
| +X.1                    |                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |

|                                               |                  |                                                                                                                                                            |
|-----------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROR (7B)                                      | Cycles: 5        | Size: 2                                                                                                                                                    |
| X-Indexed Direct Page d+X (Read/Modify/Write) |                  |                                                                                                                                                            |
| Cycle                                         | R/W              | Desc.                                                                                                                                                      |
| -X.2                                          | Read <b>PC</b>   | Store as OpCode.                                                                                                                                           |
| 1.1                                           |                  | Inc. <b>PC</b> .                                                                                                                                           |
| 1.2                                           | Read <b>PC</b>   | Store as Operand.                                                                                                                                          |
| 2.1                                           |                  | Inc. <b>PC</b> .                                                                                                                                           |
| 2.2                                           |                  |                                                                                                                                                            |
| 3.1                                           |                  | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).                                                                                    |
| 3.2                                           | Read Address     | Store as Operand.                                                                                                                                          |
| 4.1                                           |                  | Tmp = ( <b>C</b> flag << 7). <b>C</b> flag = (Operand & \$01). Operand = (Operand << 1)   Tmp. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = !Operand. |
| 4.2                                           | Write to Address | Write Operand.                                                                                                                                             |
| 5.1                                           |                  |                                                                                                                                                            |
| 5.2                                           | Read <b>PC</b>   | Store as OpCode.                                                                                                                                           |
| +X.1                                          |                  |                                                                                                                                                            |

|               |                |                                                                                                                                                                      |
|---------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROR (7C)      | Cycles: 2      | Size: 1                                                                                                                                                              |
| Accumulator A |                |                                                                                                                                                                      |
| Cycle         | R/W            | Desc.                                                                                                                                                                |
| -X.2          | Read <b>PC</b> | Store as OpCode.                                                                                                                                                     |
| 1.1           |                | Inc. <b>PC</b> .                                                                                                                                                     |
| 1.2           | Read <b>PC</b> | Discard.                                                                                                                                                             |
| 2.1           |                | Tmp = ( <b>C</b> flag << 7). <b>C</b> flag = ( <b>A</b> & \$01). <b>A</b> = ( <b>A</b> >> 1)   Tmp. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 2.2           | Read <b>PC</b> | Store as OpCode.                                                                                                                                                     |
| +X.1          |                |                                                                                                                                                                      |

|                                                      |                |                                                                                        |
|------------------------------------------------------|----------------|----------------------------------------------------------------------------------------|
| MOV (7D)      Cycles: 2                      Size: 1 |                |                                                                                        |
| Implied                                              |                |                                                                                        |
| Cycle                                                | R/W            | Desc.                                                                                  |
| -X.2                                                 | Read <b>PC</b> | Store as OpCode.                                                                       |
| 1.1                                                  |                | Inc. <b>PC</b> .                                                                       |
| 1.2                                                  |                |                                                                                        |
| 2.1                                                  |                | <b>A</b> = <b>X</b> . <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 2.2                                                  |                | Store as OpCode.                                                                       |
| +X.1                                                 |                |                                                                                        |

|                      |                |                                                                                                           |
|----------------------|----------------|-----------------------------------------------------------------------------------------------------------|
| CMP (7E)             | Cycles: 3      | Size: 2                                                                                                   |
| Direct Page d (Read) |                |                                                                                                           |
| Cycle                | R/W            | Desc.                                                                                                     |
| -X.2                 | Read <b>PC</b> | Store as OpCode.                                                                                          |
| 1.1                  |                | Inc. <b>PC</b> .                                                                                          |
| 1.2                  | Read <b>PC</b> | Store as Operand.                                                                                         |
| 2.1                  |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                       |
| 2.2                  | Read Address   | Store as Operand.                                                                                         |
| 3.1                  |                | Tmp = <b>Y</b> - Operand. <b>C</b> flag = (Tmp >= 0), <b>N</b> flag = (Tmp & \$80), <b>Z</b> flag = !Tmp. |
| 3.2                  | Read <b>PC</b> | Store as OpCode.                                                                                          |
| +X.1                 |                |                                                                                                           |

|           |                                 |                       |
|-----------|---------------------------------|-----------------------|
| RETI (7F) | Cycles: 6                       | Size: 1               |
| Implied   |                                 |                       |
| Cycle     | R/W                             | Desc.                 |
| -X.2      | Read <b>PC</b>                  | Store as OpCode.      |
| 1.1       |                                 | Inc. <b>PC</b> .      |
| 1.2       | Read <b>PC</b>                  | Discard.              |
| 2.1       |                                 |                       |
| 2.2       |                                 |                       |
| 3.1       |                                 | Inc. <b>SP</b> by 1.  |
| 3.2       | Read u8( <b>SP</b> +0)   \$0100 | Pull to <b>PSW</b> .  |
| 4.1       |                                 |                       |
| 4.2       | Read u8( <b>SP</b> +1)   \$0100 | Pull to <b>PC</b> .L. |
| 5.1       |                                 |                       |
| 5.2       | Read u8( <b>SP</b> +2)   \$0100 | Pull to <b>PC</b> .H. |
| 6.1       | Inc. <b>SP</b> by 2.            |                       |
| 6.2       | Read <b>PC</b>                  | Store as OpCode.      |
| +X.1      |                                 |                       |



|           |                |                                    |
|-----------|----------------|------------------------------------|
| SETC (80) | Cycles: 2      | Size: 1                            |
| Implied   |                |                                    |
| Cycle     | R/W            | Desc.                              |
| -X.2      | Read <b>PC</b> | Store as OpCode.                   |
| 1.1       |                | Inc. <b>PC</b> .                   |
| 1.2       |                |                                    |
| 2.1       |                | <b>PSW</b> = ( <b>PSW</b>   \$01). |
| 2.2       |                | Store as OpCode.                   |
| +X.1      |                |                                    |

|            |                                     |                               |
|------------|-------------------------------------|-------------------------------|
| TCALL (81) | Cycles: 8                           | Size: 1                       |
| Implied    |                                     |                               |
| Cycle      | R/W                                 | Desc.                         |
| -X.2       | Read <b>PC</b>                      | Store as OpCode.              |
| 1.1        |                                     | Inc. <b>PC</b> .              |
| 1.2        | Read <b>PC</b>                      | Store as Operand.             |
| 2.1        |                                     |                               |
| 2.2        |                                     |                               |
| 3.1        |                                     |                               |
| 3.2        | Write to u8( <b>SP</b> +0)   \$0100 | Push <b>PC</b> .H onto stack. |
| 4.1        |                                     |                               |
| 4.2        | Write to u8( <b>SP</b> -1)   \$0100 | Push <b>PC</b> .L onto stack. |
| 5.1        |                                     | Dec. <b>SP</b> by 2.          |
| 5.2        |                                     |                               |
| 6.1        |                                     |                               |
| 6.2        | Read \$FFCE                         | Store as <b>PC</b> .L.        |
| 7.1        |                                     |                               |
| 7.2        | Read \$FFCF                         | Store as <b>PC</b> .H.        |
| 8.1        |                                     |                               |
| 8.2        | Read <b>PC</b>                      | Store as OpCode.              |
| +X.1       |                                     |                               |

|                                         |                  |                                                                     |
|-----------------------------------------|------------------|---------------------------------------------------------------------|
| SET1 (82)                               | Cycles: 4        | Size: 2                                                             |
| Direct Page Bit d.b (Read/Modify/Write) |                  |                                                                     |
| Cycle                                   | R/W              | Desc.                                                               |
| -X.2                                    | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                                     |                  | Inc. <b>PC</b> .                                                    |
| 1.2                                     | Read <b>PC</b>   | Store as Operand.                                                   |
| 2.1                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                                     | Read Address     | Store as Operand.                                                   |
| 3.1                                     |                  | Operand = (Operand & ~\$10)   \$10.                                 |
| 3.2                                     | Write to Address | Write Operand.                                                      |
| 4.1                                     |                  |                                                                     |
| 4.2                                     | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                                    |                  |                                                                     |

|                                 |                |                                                                      |
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| BBS (83)                        | Cycles: 5-7    | Size: 3                                                              |
| Direct Page Bit Relative d.b, r |                |                                                                      |
| Cycle                           | R/W            | Desc.                                                                |
| -X.2                            | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1                             |                | Inc. <b>PC</b> .                                                     |
| 1.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 2.1                             |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).  |
| 2.2                             | Read Address   | Store as Operand.                                                    |
| 3.1                             |                | Set Jump to (Operand & (1 << 4)) == (1 << 4).                        |
| 3.2                             |                |                                                                      |
| 4.1                             |                |                                                                      |
| 4.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 5.1                             |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 7.2). |
| 5.2                             |                |                                                                      |
| 6.1                             |                | Address = <b>PC</b> + i8(Operand).<br><b>PC</b> .L = Address.L.      |
| 6.2                             |                |                                                                      |
| 7.1                             |                | <b>PC</b> .H = Address.H.                                            |
| 7.2                             | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1                            |                |                                                                      |

|                      |                |                                                                                                                                                                                                                                                                                                                                        |
|----------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADC (84)             | Cycles: 3      | Size: 2                                                                                                                                                                                                                                                                                                                                |
| Direct Page d (Read) |                |                                                                                                                                                                                                                                                                                                                                        |
| Cycle                | R/W            | Desc.                                                                                                                                                                                                                                                                                                                                  |
| -X.2                 | Read <b>PC</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| 1.1                  |                | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 1.2                  | Read <b>PC</b> | Store as Operand.                                                                                                                                                                                                                                                                                                                      |
| 2.1                  |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                                                                                                                                                                                                                                                    |
| 2.2                  | Read Address   | Store as Operand.                                                                                                                                                                                                                                                                                                                      |
| 3.1                  |                | Tmp = u32( <b>A</b> ) + u32(Operand) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = (( <b>A</b> ^ Operand ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~( <b>A</b> ^ Operand) & ( <b>A</b> ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br><b>A</b> = u8(Tmp). |
| 3.2                  | Read <b>PC</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| +X.1                 |                |                                                                                                                                                                                                                                                                                                                                        |

|                    |                                                                                |                                                                                                                                                                                                                                                                                                                                        |
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| ADC (85)           | Cycles: 4                                                                      | Size: 3                                                                                                                                                                                                                                                                                                                                |
| Absolute !a (Read) |                                                                                |                                                                                                                                                                                                                                                                                                                                        |
| Cycle              | R/W                                                                            | Desc.                                                                                                                                                                                                                                                                                                                                  |
| -X.2               | Read PC                                                                        | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| 1.1                | Read <b>PC</b><br><br>Read <b>PC</b><br><br>Read <b>PC</b><br><br>Read Address | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 1.2                |                                                                                | Store as Address.L.                                                                                                                                                                                                                                                                                                                    |
| 2.1                |                                                                                | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 2.2                |                                                                                | Store as Address.H.                                                                                                                                                                                                                                                                                                                    |
| 3.1                |                                                                                | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 3.2                |                                                                                | Store as Operand.                                                                                                                                                                                                                                                                                                                      |
| 4.1                |                                                                                | Tmp = u32( <b>A</b> ) + u32(Operand) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = (( <b>A</b> ^ Operand ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~( <b>A</b> ^ Operand) & ( <b>A</b> ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br><b>A</b> = u8(Tmp). |
| 4.2                |                                                                                | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| +X.1               |                                                                                |                                                                                                                                                                                                                                                                                                                                        |

|                       |                                                |                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADC (86)              | Cycles: 3                                      | Size: 1                                                                                                                                                                                                                                                                                                                                |
| Indirect X (X) (Read) |                                                |                                                                                                                                                                                                                                                                                                                                        |
| Cycle                 | R/W                                            | Desc.                                                                                                                                                                                                                                                                                                                                  |
| -X.2                  | Read <b>PC</b>                                 | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| 1.1                   | Read ( <b>X</b>   (( <b>PSW</b> & \$20) << 3)) | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 1.2                   |                                                |                                                                                                                                                                                                                                                                                                                                        |
| 2.1                   |                                                |                                                                                                                                                                                                                                                                                                                                        |
| 2.2                   |                                                | Store as Operand.                                                                                                                                                                                                                                                                                                                      |
| 3.1                   |                                                | Tmp = u32( <b>A</b> ) + u32(Operand) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = (( <b>A</b> ^ Operand ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~( <b>A</b> ^ Operand) & ( <b>A</b> ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br><b>A</b> = u8(Tmp). |
| 3.2                   | Read <b>PC</b>                                 | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| +X.1                  |                                                |                                                                                                                                                                                                                                                                                                                                        |

|                                 |                                                              |                                                                                                                                                                                                                                                                                                                                        |
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| ADC (87)                        | Cycles: 6                                                    | Size: 2                                                                                                                                                                                                                                                                                                                                |
| X-Indexed Indirect [d+X] (Read) |                                                              |                                                                                                                                                                                                                                                                                                                                        |
| Cycle                           | R/W                                                          | Desc.                                                                                                                                                                                                                                                                                                                                  |
| -X.2                            | Read <b>PC</b>                                               | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| 1.1                             |                                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 1.2                             | Read <b>PC</b>                                               | Store as Operand.                                                                                                                                                                                                                                                                                                                      |
| 2.1                             |                                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 2.2                             |                                                              |                                                                                                                                                                                                                                                                                                                                        |
| 3.1                             |                                                              | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).                                                                                                                                                                                                                                                                |
| 3.2                             | Read Address                                                 | Store as Pointer.L.                                                                                                                                                                                                                                                                                                                    |
| 4.1                             |                                                              |                                                                                                                                                                                                                                                                                                                                        |
| 4.2                             | Read ((Address + 1) & \$FF)  <br>(( <b>PSW</b> & \$20) << 3) | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                    |
| 5.1                             |                                                              |                                                                                                                                                                                                                                                                                                                                        |
| 5.2                             | Read Pointer                                                 | Store as Operand.                                                                                                                                                                                                                                                                                                                      |
| 6.1                             |                                                              | Tmp = u32( <b>A</b> ) + u32(Operand) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = (( <b>A</b> ^ Operand ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~( <b>A</b> ^ Operand) & ( <b>A</b> ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br><b>A</b> = u8(Tmp). |
| 6.2                             | Read <b>PC</b>                                               | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| +X.1                            |                                                              |                                                                                                                                                                                                                                                                                                                                        |



|           |                |                                                                                                                                                                                                                                                                                                                                                         |
|-----------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADC (88)  | Cycles: 2      | Size: 2                                                                                                                                                                                                                                                                                                                                                 |
| Immediate |                |                                                                                                                                                                                                                                                                                                                                                         |
| Cycle     | R/W            | Desc.                                                                                                                                                                                                                                                                                                                                                   |
| -X.2      | Read PC        | Store as OpCode.                                                                                                                                                                                                                                                                                                                                        |
| 1.1       | Read PC        | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                        |
| 1.2       |                | Store as Operand.                                                                                                                                                                                                                                                                                                                                       |
| 2.1       |                | Inc. <b>PC</b> . Tmp = u32( <b>A</b> ) + u32(Operand) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = (( <b>A</b> ^ Operand ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~( <b>A</b> ^ Operand) & ( <b>A</b> ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br><b>A</b> = u8(Tmp). |
| 2.2       | Read <b>PC</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                        |
| +X.1      |                |                                                                                                                                                                                                                                                                                                                                                         |

|                                                       |                  |                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADC (89)                                              | Cycles: 6        | Size: 3                                                                                                                                                                                                                                                                                                                              |
| Direct Page to Direct Page dd, ds (Read/Modify/Write) |                  |                                                                                                                                                                                                                                                                                                                                      |
| Cycle                                                 | R/W              | Desc.                                                                                                                                                                                                                                                                                                                                |
| -X.2                                                  | Read <b>PC</b>   | Store as OpCode.                                                                                                                                                                                                                                                                                                                     |
| 1.1                                                   |                  | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                     |
| 1.2                                                   | Read <b>PC</b>   | Store as Operand.                                                                                                                                                                                                                                                                                                                    |
| 2.1                                                   |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                                                                                                                                                                                                                                                  |
| 2.2                                                   | Read Address     | Store as Operand0.                                                                                                                                                                                                                                                                                                                   |
| 3.1                                                   |                  |                                                                                                                                                                                                                                                                                                                                      |
| 3.2                                                   | Read <b>PC</b>   | Store as Operand.                                                                                                                                                                                                                                                                                                                    |
| 4.1                                                   |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                                                                                                                                                                                                                                                  |
| 4.2                                                   | Read Address     | Store as Operand1.                                                                                                                                                                                                                                                                                                                   |
| 5.1                                                   |                  | Tmp = u32(Operand0) + u32(Operand1) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = ((Operand0 ^ Operand1 ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~(Operand0 ^ Operand1) & (Operand0 ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br>Operand0 = u8(Tmp). |
| 5.2                                                   | Write to Address | Write Operand0.                                                                                                                                                                                                                                                                                                                      |
| 6.1                                                   |                  |                                                                                                                                                                                                                                                                                                                                      |
| 6.2                                                   | Read <b>PC</b>   | Store as OpCode.                                                                                                                                                                                                                                                                                                                     |
| +X.1                                                  |                  |                                                                                                                                                                                                                                                                                                                                      |

|                          |                         |                                                                         |
|--------------------------|-------------------------|-------------------------------------------------------------------------|
| EOR1 (8A)                | Cycles: 5               | Size: 3                                                                 |
| Absolute Boolean Bit m.b |                         |                                                                         |
| Cycle                    | R/W                     | Desc.                                                                   |
| -X.2                     | Read <b>PC</b>          | Store as OpCode.                                                        |
| 1.1                      |                         | Inc. <b>PC</b> .                                                        |
| 1.2                      | Read <b>PC</b>          | Store as Address.L.                                                     |
| 2.1                      |                         | Inc. <b>PC</b> .                                                        |
| 2.2                      | Read <b>PC</b>          | Store as Address.H.                                                     |
| 3.1                      |                         | Inc. <b>PC</b> .                                                        |
| 3.2                      | Read (Address & \$1FFF) | Store as Operand.                                                       |
| 4.1                      |                         |                                                                         |
| 4.2                      |                         |                                                                         |
| 5.1                      |                         | Bit = Address >> 13.<br><b>C</b> flag ^= ((Operand & (1 << Bit)) != 0). |
| 5.2                      | Read <b>PC</b>          | Store as OpCode.                                                        |
| +X.1                     |                         |                                                                         |

|                                   |                  |                                                                           |
|-----------------------------------|------------------|---------------------------------------------------------------------------|
| DEC (8B)                          | Cycles: 4        | Size: 2                                                                   |
| Direct Page d (Read/Modify/Write) |                  |                                                                           |
| Cycle                             | R/W              | Desc.                                                                     |
| -X.2                              | Read <b>PC</b>   | Store as OpCode.                                                          |
| 1.1                               |                  | Inc. <b>PC</b> .                                                          |
| 1.2                               | Read <b>PC</b>   | Store as Operand.                                                         |
| 2.1                               |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).       |
| 2.2                               | Read Address     | Store as Operand.                                                         |
| 3.1                               |                  | Operand -= 1. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = !Operand. |
| 3.2                               | Write to Address | Write Operand.                                                            |
| 4.1                               |                  |                                                                           |
| 4.2                               | Read <b>PC</b>   | Store as OpCode.                                                          |
| +X.1                              |                  |                                                                           |

|                                 |                  |                                                                           |
|---------------------------------|------------------|---------------------------------------------------------------------------|
| DEC (8C)                        | Cycles: 5        | Size: 3                                                                   |
| Absolute !a (Read/Modify/Write) |                  |                                                                           |
| Cycle                           | R/W              | Desc.                                                                     |
| -X.2                            | Read <b>PC</b>   | Store as OpCode.                                                          |
| 1.1                             |                  | Inc. <b>PC</b> .                                                          |
| 1.2                             | Read <b>PC</b>   | Store as Address.L.                                                       |
| 2.1                             |                  | Inc. <b>PC</b> .                                                          |
| 2.2                             | Read <b>PC</b>   | Store as Address.H.                                                       |
| 3.1                             |                  | Inc. <b>PC</b> .                                                          |
| 3.2                             | Read Address     | Store as Operand.                                                         |
| 4.1                             |                  | Operand -= 1. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = !Operand. |
| 4.2                             | Write to Address | Write Operand.                                                            |
| 5.1                             |                  |                                                                           |
| 5.2                             | Read <b>PC</b>   | Store as OpCode.                                                          |
| +X.1                            |                  |                                                                           |

|           |                |                                                                                                  |
|-----------|----------------|--------------------------------------------------------------------------------------------------|
| MOV (8D)  | Cycles: 2      | Size: 2                                                                                          |
| Immediate |                |                                                                                                  |
| Cycle     | R/W            | Desc.                                                                                            |
| -X.2      | Read <b>PC</b> | Store as OpCode.                                                                                 |
| 1.1       |                | Inc. <b>PC</b> .                                                                                 |
| 1.2       | Read <b>PC</b> | Store as Operand.                                                                                |
| 2.1       |                | Inc. <b>PC</b> . <b>Y</b> = Operand. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = !Operand. |
| 2.2       | Read <b>PC</b> | Store as OpCode.                                                                                 |
| +X.1      |                |                                                                                                  |

|          |                |                      |
|----------|----------------|----------------------|
| POP (8E) | Cycles: 4      | Size: 1              |
| Implied  |                |                      |
| Cycle    | R/W            | Desc.                |
| -X.2     | Read <b>PC</b> | Store as OpCode.     |
| 1.1      | Read <b>PC</b> | Inc. <b>PC</b> .     |
| 1.2      |                | Discard.             |
| 2.1      |                |                      |
| 2.2      |                |                      |
| 3.1      |                | Inc. <b>SP</b> by 1. |
| 3.2      |                | Pull to <b>PSW</b> . |
| 4.1      |                |                      |
| 4.2      |                | Store as OpCode.     |
| +X.1     |                |                      |

|                                                         |                  |                                                                     |
|---------------------------------------------------------|------------------|---------------------------------------------------------------------|
| MOV (8F)                                                | Cycles: 5        | Size: 3                                                             |
| Immediate Data to Direct Page d, #i (Read/Modify/Write) |                  |                                                                     |
| Cycle                                                   | R/W              | Desc.                                                               |
| -X.2                                                    | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                                                     |                  | Inc. <b>PC</b> .                                                    |
| 1.2                                                     | Read <b>PC</b>   | Store as Operand0.                                                  |
| 2.1                                                     |                  | Inc. <b>PC</b> .                                                    |
| 2.2                                                     | Read <b>PC</b>   | Store as Operand.                                                   |
| 3.1                                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 3.2                                                     | Read Address     | Store as Operand1.                                                  |
| 4.1                                                     |                  |                                                                     |
| 4.2                                                     | Write to Address | Write Operand0.                                                     |
| 5.1                                                     |                  |                                                                     |
| 5.2                                                     | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                                                    |                  |                                                                     |



|            |                |                                                                      |
|------------|----------------|----------------------------------------------------------------------|
| BCC (90)   | Cycles: 2-4    | Size: 2                                                              |
| Relative r |                |                                                                      |
| Cycle      | R/W            | Desc.                                                                |
| -X.2       | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1        |                | Inc. <b>PC</b> . Set Jump to ( <b>PSW</b> & \$01) == \$00.           |
| 1.2        |                | Store as Operand.                                                    |
| 2.1        |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 4.2). |
| 2.2        |                |                                                                      |
| 3.1        |                | Address = <b>PC</b> + i8(Operand).                                   |
|            |                | <b>PC</b> .L = Address.L.                                            |
| 3.2        |                |                                                                      |
| 4.1        |                | <b>PC</b> .H = Address.H.                                            |
| 4.2        | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1       |                |                                                                      |

|            |                                     |                               |
|------------|-------------------------------------|-------------------------------|
| TCALL (91) | Cycles: 8                           | Size: 1                       |
| Implied    |                                     |                               |
| Cycle      | R/W                                 | Desc.                         |
| -X.2       | Read <b>PC</b>                      | Store as OpCode.              |
| 1.1        |                                     | Inc. <b>PC</b> .              |
| 1.2        | Read <b>PC</b>                      | Store as Operand.             |
| 2.1        |                                     |                               |
| 2.2        |                                     |                               |
| 3.1        |                                     |                               |
| 3.2        | Write to u8( <b>SP</b> +0)   \$0100 | Push <b>PC</b> .H onto stack. |
| 4.1        |                                     |                               |
| 4.2        | Write to u8( <b>SP</b> -1)   \$0100 | Push <b>PC</b> .L onto stack. |
| 5.1        |                                     | Dec. <b>SP</b> by 2.          |
| 5.2        |                                     |                               |
| 6.1        |                                     |                               |
| 6.2        | Read \$FFCC                         | Store as <b>PC</b> .L.        |
| 7.1        |                                     |                               |
| 7.2        | Read \$FFCD                         | Store as <b>PC</b> .H.        |
| 8.1        |                                     |                               |
| 8.2        | Read <b>PC</b>                      | Store as OpCode.              |
| +X.1       |                                     |                               |

|                                         |                  |                                                                     |
|-----------------------------------------|------------------|---------------------------------------------------------------------|
| CLR1 (92)                               | Cycles: 4        | Size: 2                                                             |
| Direct Page Bit d.b (Read/Modify/Write) |                  |                                                                     |
| Cycle                                   | R/W              | Desc.                                                               |
| -X.2                                    | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                                     |                  | Inc. <b>PC</b> .                                                    |
| 1.2                                     | Read <b>PC</b>   | Store as Operand.                                                   |
| 2.1                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                                     | Read Address     | Store as Operand.                                                   |
| 3.1                                     |                  | Operand = (Operand & ~\$10)   \$00.                                 |
| 3.2                                     | Write to Address | Write Operand.                                                      |
| 4.1                                     |                  |                                                                     |
| 4.2                                     | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                                    |                  |                                                                     |

|                                 |                |                                                                      |
|---------------------------------|----------------|----------------------------------------------------------------------|
| BBC (93)                        | Cycles: 5-7    | Size: 3                                                              |
| Direct Page Bit Relative d.b, r |                |                                                                      |
| Cycle                           | R/W            | Desc.                                                                |
| -X.2                            | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1                             |                | Inc. <b>PC</b> .                                                     |
| 1.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 2.1                             |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).  |
| 2.2                             | Read Address   | Store as Operand.                                                    |
| 3.1                             |                | Set Jump to (Operand & (1 << 4)) == (0 << 4).                        |
| 3.2                             |                |                                                                      |
| 4.1                             |                |                                                                      |
| 4.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 5.1                             |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 7.2). |
| 5.2                             |                |                                                                      |
| 6.1                             |                | Address = <b>PC</b> + i8(Operand).<br><b>PC</b> .L = Address.L.      |
| 6.2                             |                |                                                                      |
| 7.1                             |                | <b>PC</b> .H = Address.H.                                            |
| 7.2                             | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1                            |                |                                                                      |

|                           |                |                                                                                                                                                                                                                                                                                                                                        |
|---------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADC (94)                  | Cycles: 4      | Size: 2                                                                                                                                                                                                                                                                                                                                |
| X-Indexed Direct Page d+X |                |                                                                                                                                                                                                                                                                                                                                        |
| Cycle                     | R/W            | Desc.                                                                                                                                                                                                                                                                                                                                  |
| -X.2                      | Read <b>PC</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| 1.1                       |                | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 1.2                       | Read <b>PC</b> | Store as Operand.                                                                                                                                                                                                                                                                                                                      |
| 2.1                       |                | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 2.2                       |                |                                                                                                                                                                                                                                                                                                                                        |
| 3.1                       |                | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).                                                                                                                                                                                                                                                                |
| 3.2                       | Read Address   | Store as Operand.                                                                                                                                                                                                                                                                                                                      |
| 4.1                       |                | Tmp = u32( <b>A</b> ) + u32(Operand) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = (( <b>A</b> ^ Operand ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~( <b>A</b> ^ Operand) & ( <b>A</b> ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br><b>A</b> = u8(Tmp). |
| 4.2                       | Read <b>PC</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| +X.1                      |                |                                                                                                                                                                                                                                                                                                                                        |

|                                |                |                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADC (95)                       | Cycles: 5      | Size: 3                                                                                                                                                                                                                                                                                                                                |
| X-Indexed Absolute !a+X (Read) |                |                                                                                                                                                                                                                                                                                                                                        |
| Cycle                          | R/W            | Desc.                                                                                                                                                                                                                                                                                                                                  |
| -X.2                           | Read <b>PC</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| 1.1                            |                | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 1.2                            | Read <b>PC</b> | Store as Address.L.                                                                                                                                                                                                                                                                                                                    |
| 2.1                            |                | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 2.2                            | Read <b>PC</b> | Store as Address.H.                                                                                                                                                                                                                                                                                                                    |
| 3.1                            |                | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 3.2                            |                |                                                                                                                                                                                                                                                                                                                                        |
| 4.1                            |                | Pointer = ( <b>X</b> + Address).                                                                                                                                                                                                                                                                                                       |
| 4.2                            | Read Pointer   | Store as Operand.                                                                                                                                                                                                                                                                                                                      |
| 5.1                            |                | Tmp = u32( <b>A</b> ) + u32(Operand) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = (( <b>A</b> ^ Operand ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~( <b>A</b> ^ Operand) & ( <b>A</b> ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br><b>A</b> = u8(Tmp). |
| 5.2                            | Read <b>PC</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| +X.1                           |                |                                                                                                                                                                                                                                                                                                                                        |

|                                |                |                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADC (96)                       | Cycles: 5      | Size: 3                                                                                                                                                                                                                                                                                                                                 |
| Y-Indexed Absolute !a+Y (Read) |                |                                                                                                                                                                                                                                                                                                                                         |
| Cycle                          | R/W            | Desc.                                                                                                                                                                                                                                                                                                                                   |
| -X.2                           | Read <b>PC</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                        |
| 1.1                            | Read <b>PC</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                        |
| 1.2                            |                | Store as Address.L.                                                                                                                                                                                                                                                                                                                     |
| 2.1                            |                | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                        |
| 2.2                            |                | Store as Address.H.                                                                                                                                                                                                                                                                                                                     |
| 3.1                            |                | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                        |
| 3.2                            |                |                                                                                                                                                                                                                                                                                                                                         |
| 4.1                            |                | Pointer = ( <b>Y</b> + Address).                                                                                                                                                                                                                                                                                                        |
| 4.2                            |                | Store as Operand.                                                                                                                                                                                                                                                                                                                       |
| 5.1                            |                | Tmp = u32( <b>A</b> ) + u32(Operand) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = (( <b>A</b> ^ Operand ^ u8(Tmp)) & \$10).<br><b>V</b> flag = (~( <b>A</b> ^ Operand) & ( <b>A</b> ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br><b>A</b> = u8(Tmp). |
| 5.2                            |                | Read <b>PC</b>                                                                                                                                                                                                                                                                                                                          |
| +X.1                           |                | Store as OpCode.                                                                                                                                                                                                                                                                                                                        |

| ADC (97)                        | Cycles: 6                                                 | Size: 2                                                                                                                                                                                                                                                                                                                                |
|---------------------------------|-----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Indirect Y-Indexed [d]+Y (Read) |                                                           |                                                                                                                                                                                                                                                                                                                                        |
| Cycle                           | R/W                                                       | Desc.                                                                                                                                                                                                                                                                                                                                  |
| -X.2                            | Read <b>PC</b>                                            | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| 1.1                             |                                                           | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 1.2                             | Read <b>PC</b>                                            | Store as Operand.                                                                                                                                                                                                                                                                                                                      |
| 2.1                             |                                                           | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                       |
| 2.2                             |                                                           |                                                                                                                                                                                                                                                                                                                                        |
| 3.1                             |                                                           | Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                                                                                                                                                                                                                                                                     |
| 3.2                             | Read Address                                              | Store as Pointer.L.                                                                                                                                                                                                                                                                                                                    |
| 4.1                             |                                                           |                                                                                                                                                                                                                                                                                                                                        |
| 4.2                             | Read ((Address + 1) & \$FF)   (( <b>PSW</b> & \$20) << 3) | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                    |
| 5.1                             |                                                           | Address = ( <b>Y</b> + Pointer).                                                                                                                                                                                                                                                                                                       |
| 5.2                             | Read Address                                              | Store as Operand.                                                                                                                                                                                                                                                                                                                      |
| 6.1                             |                                                           | Tmp = u32( <b>A</b> ) + u32(Operand) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = (( <b>A</b> ^ Operand ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~( <b>A</b> ^ Operand) & ( <b>A</b> ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br><b>A</b> = u8(Tmp). |
| 6.2                             | Read <b>PC</b>                                            | Store as OpCode.                                                                                                                                                                                                                                                                                                                       |
| +X.1                            |                                                           |                                                                                                                                                                                                                                                                                                                                        |



|                                                         |                  |                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADC (98)                                                | Cycles: 5        | Size: 3                                                                                                                                                                                                                                                                                                                              |
| Immediate Data to Direct Page d, #i (Read/Modify/Write) |                  |                                                                                                                                                                                                                                                                                                                                      |
| Cycle                                                   | R/W              | Desc.                                                                                                                                                                                                                                                                                                                                |
| -X.2                                                    | Read <b>PC</b>   | Store as OpCode.                                                                                                                                                                                                                                                                                                                     |
| 1.1                                                     |                  | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                     |
| 1.2                                                     | Read <b>PC</b>   | Store as Operand0.                                                                                                                                                                                                                                                                                                                   |
| 2.1                                                     |                  | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                     |
| 2.2                                                     | Read <b>PC</b>   | Store as Operand.                                                                                                                                                                                                                                                                                                                    |
| 3.1                                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                                                                                                                                                                                                                                                  |
| 3.2                                                     | Read Address     | Store as Operand1.                                                                                                                                                                                                                                                                                                                   |
| 4.1                                                     |                  | Tmp = u32(Operand0) + u32(Operand1) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = ((Operand0 ^ Operand1 ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~(Operand0 ^ Operand1) & (Operand0 ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br>Operand0 = u8(Tmp). |
| 4.2                                                     | Write to Address | Write Operand0.                                                                                                                                                                                                                                                                                                                      |
| 5.1                                                     |                  |                                                                                                                                                                                                                                                                                                                                      |
| 5.2                                                     | Read <b>PC</b>   | Store as OpCode.                                                                                                                                                                                                                                                                                                                     |
| +X.1                                                    |                  |                                                                                                                                                                                                                                                                                                                                      |

|                                                             |                                                    |                                                                                                                                                                                                                                                                                                                                      |
|-------------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ADC (99)                                                    | Cycles: 5                                          | Size: 1                                                                                                                                                                                                                                                                                                                              |
| Indirect Page to Indirect Page (X), (Y) (Read/Modify/Write) |                                                    |                                                                                                                                                                                                                                                                                                                                      |
| Cycle                                                       | R/W                                                | Desc.                                                                                                                                                                                                                                                                                                                                |
| -X.2                                                        | Read <b>PC</b>                                     | Store as OpCode.                                                                                                                                                                                                                                                                                                                     |
| 1.1                                                         |                                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                     |
| 1.2                                                         | Read <b>PC</b>                                     | Discard.                                                                                                                                                                                                                                                                                                                             |
| 2.1                                                         |                                                    |                                                                                                                                                                                                                                                                                                                                      |
| 2.2                                                         | Read ( <b>Y</b>   (( <b>PSW</b> & \$20) << 3))     | Store as Operand0.                                                                                                                                                                                                                                                                                                                   |
| 3.1                                                         |                                                    |                                                                                                                                                                                                                                                                                                                                      |
| 3.2                                                         | Read ( <b>X</b>   (( <b>PSW</b> & \$20) << 3))     | Store as Operand1.                                                                                                                                                                                                                                                                                                                   |
| 4.1                                                         |                                                    | Tmp = u32(Operand0) + u32(Operand1) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = ((Operand0 ^ Operand1 ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~(Operand0 ^ Operand1) & (Operand0 ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br>Operand0 = u8(Tmp). |
| 4.2                                                         | Write to ( <b>X</b>   (( <b>PSW</b> & \$20) << 3)) | Write Operand0.                                                                                                                                                                                                                                                                                                                      |
| 5.1                                                         |                                                    |                                                                                                                                                                                                                                                                                                                                      |
| 5.2                                                         | Read <b>PC</b>                                     | Store as OpCode.                                                                                                                                                                                                                                                                                                                     |
| +X.1                                                        |                                                    |                                                                                                                                                                                                                                                                                                                                      |

|                         |                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |
|-------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| SUBW (9A)               |                                                           | Cycles: 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Size: 2 |
| Direct Page d (Read 16) |                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |
| Cycle                   | R/W                                                       | Desc.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |
| -X.2                    | Read <b>PC</b>                                            | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |
| 1.1                     |                                                           | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |
| 1.2                     | Read <b>PC</b>                                            | Store as Operand.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |
| 2.1                     |                                                           | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                                                                                                                                                                                                                                                                                                                                                                                                  |         |
| 2.2                     | Read Address                                              | Store as Operand16.L.                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |
| 3.1                     |                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |
| 3.2                     |                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |
| 4.1                     |                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |
| 4.2                     | Read ((Address + 1) & \$FF)   (( <b>PSW</b> & \$20) << 3) | Store as Operand16.H.                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |
| 5.1                     |                                                           | TmpR = ~Operand16.L.<br>Tmp = u32( <b>A</b> ) + u32(TmpR) + 1.<br><b>C</b> flag = (Tmp > \$FF).<br><b>A</b> = u8(Tmp).<br>TmpR = ~Operand16.H.<br>Tmp = u32( <b>Y</b> ) + u32(TmpR) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>H</b> flag = (( <b>Y</b> ^ TmpR ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~( <b>Y</b> ^ TmpR) & ( <b>Y</b> ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br><b>Y</b> = u8(Tmp).<br><b>Z</b> flag = ( <b>YA</b> == \$0000). |         |
| 5.2                     | Read <b>PC</b>                                            | Store as OpCode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |
| +X.1                    |                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |

|                                               |                  |                                                                           |
|-----------------------------------------------|------------------|---------------------------------------------------------------------------|
| DEC (9B)                                      | Cycles: 5        | Size: 2                                                                   |
| X-Indexed Direct Page d+X (Read/Modify/Write) |                  |                                                                           |
| Cycle                                         | R/W              | Desc.                                                                     |
| -X.2                                          | Read <b>PC</b>   | Store as OpCode.                                                          |
| 1.1                                           |                  | Inc. <b>PC</b> .                                                          |
| 1.2                                           | Read <b>PC</b>   | Store as Operand.                                                         |
| 2.1                                           |                  | Inc. <b>PC</b> .                                                          |
| 2.2                                           |                  |                                                                           |
| 3.1                                           |                  | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).   |
| 3.2                                           | Read Address     | Store as Operand.                                                         |
| 4.1                                           |                  | Operand -= 1. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = !Operand. |
| 4.2                                           | Write to Address | Write Operand.                                                            |
| 5.1                                           |                  |                                                                           |
| 5.2                                           | Read <b>PC</b>   | Store as OpCode.                                                          |
| +X.1                                          |                  |                                                                           |

|               |                |                                                                                 |
|---------------|----------------|---------------------------------------------------------------------------------|
| DEC (9C)      | Cycles: 2      | Size: 1                                                                         |
| Accumulator A |                |                                                                                 |
| Cycle         | R/W            | Desc.                                                                           |
| -X.2          | Read <b>PC</b> | Store as OpCode.                                                                |
| 1.1           |                | Inc. <b>PC</b> .                                                                |
| 1.2           | Read <b>PC</b> | Discard.                                                                        |
| 2.1           |                | <b>A</b> -= 1. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 2.2           | Read <b>PC</b> | Store as OpCode.                                                                |
| +X.1          |                |                                                                                 |

|                                                      |                |                                                                                         |
|------------------------------------------------------|----------------|-----------------------------------------------------------------------------------------|
| MOV (9D)      Cycles: 2                      Size: 1 |                |                                                                                         |
| Implied                                              |                |                                                                                         |
| Cycle                                                | R/W            | Desc.                                                                                   |
| -X.2                                                 | Read <b>PC</b> | Store as OpCode.                                                                        |
| 1.1                                                  |                | Inc. <b>PC</b> .                                                                        |
| 1.2                                                  |                |                                                                                         |
| 2.1                                                  |                | <b>X</b> = <b>SP</b> . <b>N</b> flag = ( <b>X</b> & \$80), <b>Z</b> flag = ! <b>X</b> . |
| 2.2                                                  |                | Store as OpCode.                                                                        |
| +X.1                                                 |                |                                                                                         |

| DIV (9E) |                | Cycles: 12                                                                                                                                                      | Size: 1 |
|----------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| Implied  |                |                                                                                                                                                                 |         |
| Cycle    | R/W            | Desc.                                                                                                                                                           |         |
| -X.2     | Read <b>PC</b> | Store as OpCode.                                                                                                                                                |         |
| 1.1      |                | Inc. <b>PC</b> .                                                                                                                                                |         |
| 1.2      |                |                                                                                                                                                                 |         |
| 2.1      |                | <b>H</b> flag = (( <b>Y</b> & 15) >= ( <b>X</b> & 15)), <b>V</b> flag = ( <b>Y</b> >= <b>X</b> ).<br>DivYa = u32( <b>YA</b> ), ShiftedX = u32( <b>X</b> ) << 9. |         |
| 2.2      |                |                                                                                                                                                                 |         |
| 3.1      |                | DivYa <= 1.<br>If DivYa & \$20000: DivYa ^= \$20001.<br>If DivYa >= ShiftedX: DivYa ^= 1.<br>If DivYa & 1: DivYa = (DivYa - ShiftedX) & \$1FFFF.                |         |
| 3.2      |                |                                                                                                                                                                 |         |
| 4.1      |                | DivYa <= 1.<br>If DivYa & \$20000: DivYa ^= \$20001.<br>If DivYa >= ShiftedX: DivYa ^= 1.<br>If DivYa & 1: DivYa = (DivYa - ShiftedX) & \$1FFFF.                |         |
| 4.2      |                |                                                                                                                                                                 |         |
| 5.1      |                | DivYa <= 1.<br>If DivYa & \$20000: DivYa ^= \$20001.<br>If DivYa >= ShiftedX: DivYa ^= 1.<br>If DivYa & 1: DivYa = (DivYa - ShiftedX) & \$1FFFF.                |         |
| 5.2      |                |                                                                                                                                                                 |         |
| 6.1      |                | DivYa <= 1.<br>If DivYa & \$20000: DivYa ^= \$20001.<br>If DivYa >= ShiftedX: DivYa ^= 1.<br>If DivYa & 1: DivYa = (DivYa - ShiftedX) & \$1FFFF.                |         |
| 6.2      |                |                                                                                                                                                                 |         |
| 7.1      |                | DivYa <= 1.<br>If DivYa & \$20000: DivYa ^= \$20001.<br>If DivYa >= ShiftedX: DivYa ^= 1.<br>If DivYa & 1: DivYa = (DivYa - ShiftedX) & \$1FFFF.                |         |
| 7.2      |                |                                                                                                                                                                 |         |
| 8.1      |                | DivYa <= 1.<br>If DivYa & \$20000: DivYa ^= \$20001.<br>If DivYa >= ShiftedX: DivYa ^= 1.<br>If DivYa & 1: DivYa = (DivYa - ShiftedX) & \$1FFFF.                |         |
| 8.2      |                |                                                                                                                                                                 |         |
| 9.1      |                | DivYa <= 1.<br>If DivYa & \$20000: DivYa ^= \$20001.<br>If DivYa >= ShiftedX: DivYa ^= 1.<br>If DivYa & 1: DivYa = (DivYa - ShiftedX) & \$1FFFF.                |         |
| 9.2      |                |                                                                                                                                                                 |         |
| 10.1     |                | DivYa <= 1.<br>If DivYa & \$20000: DivYa ^= \$20001.<br>If DivYa >= ShiftedX: DivYa ^= 1.<br>If DivYa & 1: DivYa = (DivYa - ShiftedX) & \$1FFFF.                |         |
| 10.2     |                |                                                                                                                                                                 |         |
| 11.1     |                | DivYa <= 1.<br>If DivYa & \$20000: DivYa ^= \$20001.<br>If DivYa >= ShiftedX: DivYa ^= 1.<br>If DivYa & 1: DivYa = (DivYa - ShiftedX) & \$1FFFF.                |         |
| 11.2     |                |                                                                                                                                                                 |         |
| 12.1     |                | <b>Y</b> = u8(DivYa >> 9), <b>A</b> = u8(DivYa).<br><b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> .                                            |         |
| 12.2     | Read <b>PC</b> | Store as OpCode.                                                                                                                                                |         |
| +X.1     |                |                                                                                                                                                                 |         |

|               |                |                                                                                                                  |
|---------------|----------------|------------------------------------------------------------------------------------------------------------------|
| XCN (9F)      | Cycles: 5      | Size: 1                                                                                                          |
| Accumulator A |                |                                                                                                                  |
| Cycle         | R/W            | Desc.                                                                                                            |
| -X.2          | Read <b>PC</b> | Store as OpCode.                                                                                                 |
| 1.1           |                | Inc. <b>PC</b> .                                                                                                 |
| 1.2           |                |                                                                                                                  |
| 2.1           |                | <b>A</b> = ( <b>A</b> >> 7)   ( <b>A</b> << 1).                                                                  |
| 2.2           |                |                                                                                                                  |
| 3.1           |                | <b>A</b> = ( <b>A</b> >> 7)   ( <b>A</b> << 1).                                                                  |
| 3.2           |                |                                                                                                                  |
| 4.1           |                | <b>A</b> = ( <b>A</b> >> 7)   ( <b>A</b> << 1).                                                                  |
| 4.2           |                |                                                                                                                  |
| 5.1           |                | <b>A</b> = ( <b>A</b> >> 7)   ( <b>A</b> << 1). <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 5.2           |                | Read <b>PC</b>                                                                                                   |
|               |                | Store as OpCode.                                                                                                 |
| +X.1          |                |                                                                                                                  |



|         |                |                                    |
|---------|----------------|------------------------------------|
| El (A0) | Cycles: 3      | Size: 1                            |
| Implied |                |                                    |
| Cycle   | R/W            | Desc.                              |
| -X.2    | Read <b>PC</b> | Store as OpCode.                   |
| 1.1     |                | Inc. <b>PC</b> .                   |
| 1.2     |                |                                    |
| 2.1     |                |                                    |
| 2.2     |                |                                    |
| 3.1     |                | <b>PSW</b> = ( <b>PSW</b>   \$04). |
| 3.2     |                | Store as OpCode.                   |
| +X.1    |                |                                    |

|            |                                     |                               |
|------------|-------------------------------------|-------------------------------|
| TCALL (A1) | Cycles: 8                           | Size: 1                       |
| Implied    |                                     |                               |
| Cycle      | R/W                                 | Desc.                         |
| -X.2       | Read <b>PC</b>                      | Store as OpCode.              |
| 1.1        |                                     | Inc. <b>PC</b> .              |
| 1.2        | Read <b>PC</b>                      | Store as Operand.             |
| 2.1        |                                     |                               |
| 2.2        |                                     |                               |
| 3.1        |                                     |                               |
| 3.2        | Write to u8( <b>SP</b> +0)   \$0100 | Push <b>PC</b> .H onto stack. |
| 4.1        |                                     |                               |
| 4.2        | Write to u8( <b>SP</b> -1)   \$0100 | Push <b>PC</b> .L onto stack. |
| 5.1        |                                     | Dec. <b>SP</b> by 2.          |
| 5.2        |                                     |                               |
| 6.1        |                                     |                               |
| 6.2        | Read \$FFCA                         | Store as <b>PC</b> .L.        |
| 7.1        |                                     |                               |
| 7.2        | Read \$FFCB                         | Store as <b>PC</b> .H.        |
| 8.1        |                                     |                               |
| 8.2        | Read <b>PC</b>                      | Store as OpCode.              |
| +X.1       |                                     |                               |

|                                         |                  |                                                                     |
|-----------------------------------------|------------------|---------------------------------------------------------------------|
| SET1 (A2)                               | Cycles: 4        | Size: 2                                                             |
| Direct Page Bit d.b (Read/Modify/Write) |                  |                                                                     |
| Cycle                                   | R/W              | Desc.                                                               |
| -X.2                                    | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                                     |                  | Inc. <b>PC</b> .                                                    |
| 1.2                                     | Read <b>PC</b>   | Store as Operand.                                                   |
| 2.1                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                                     | Read Address     | Store as Operand.                                                   |
| 3.1                                     |                  | Operand = (Operand & ~\$20)   \$20.                                 |
| 3.2                                     | Write to Address | Write Operand.                                                      |
| 4.1                                     |                  |                                                                     |
| 4.2                                     | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                                    |                  |                                                                     |

|                                 |                |                                                                      |
|---------------------------------|----------------|----------------------------------------------------------------------|
| BBS (A3)                        | Cycles: 5-7    | Size: 3                                                              |
| Direct Page Bit Relative d.b, r |                |                                                                      |
| Cycle                           | R/W            | Desc.                                                                |
| -X.2                            | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1                             |                | Inc. <b>PC</b> .                                                     |
| 1.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 2.1                             |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).  |
| 2.2                             | Read Address   | Store as Operand.                                                    |
| 3.1                             |                | Set Jump to (Operand & (1 << 5)) == (1 << 5).                        |
| 3.2                             |                |                                                                      |
| 4.1                             |                |                                                                      |
| 4.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 5.1                             |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 7.2). |
| 5.2                             |                |                                                                      |
| 6.1                             |                | Address = <b>PC</b> + i8(Operand).<br><b>PC</b> .L = Address.L.      |
| 6.2                             |                |                                                                      |
| 7.1                             |                | <b>PC</b> .H = Address.H.                                            |
| 7.2                             | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1                            |                |                                                                      |

|                      |                |                                                                                                                                                                                                                                                                                                                                                   |
|----------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SBC (A4)             | Cycles: 3      | Size: 2                                                                                                                                                                                                                                                                                                                                           |
| Direct Page d (Read) |                |                                                                                                                                                                                                                                                                                                                                                   |
| Cycle                | R/W            | Desc.                                                                                                                                                                                                                                                                                                                                             |
| -X.2                 | Read <b>PC</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                  |
| 1.1                  |                | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                  |
| 1.2                  | Read <b>PC</b> | Store as Operand.                                                                                                                                                                                                                                                                                                                                 |
| 2.1                  |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                                                                                                                                                                                                                                                               |
| 2.2                  | Read Address   | Store as Operand.                                                                                                                                                                                                                                                                                                                                 |
| 3.1                  |                | TmpR = ~Operand.<br>Tmp = u32( <b>A</b> ) + u32(TmpR) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = (( <b>A</b> ^ TmpR ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~( <b>A</b> ^ TmpR) & ( <b>A</b> ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br><b>A</b> = u8(Tmp). |
| 3.2                  | Read <b>PC</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                  |
| +X.1                 |                |                                                                                                                                                                                                                                                                                                                                                   |

|                    |                |                                                                                                                                                                                                                                                                                                                                                   |
|--------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SBC (A5)           | Cycles: 4      | Size: 3                                                                                                                                                                                                                                                                                                                                           |
| Absolute !a (Read) |                |                                                                                                                                                                                                                                                                                                                                                   |
| Cycle              | R/W            | Desc.                                                                                                                                                                                                                                                                                                                                             |
| -X.2               | Read <b>PC</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                  |
| 1.1                | Read <b>PC</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                  |
| 1.2                |                | Store as Address.L.                                                                                                                                                                                                                                                                                                                               |
| 2.1                |                | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                  |
| 2.2                |                | Store as Address.H.                                                                                                                                                                                                                                                                                                                               |
| 3.1                |                | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                  |
| 3.2                |                | Store as Operand.                                                                                                                                                                                                                                                                                                                                 |
| 4.1                |                | TmpR = ~Operand.<br>Tmp = u32( <b>A</b> ) + u32(TmpR) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = (( <b>A</b> ^ TmpR ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~( <b>A</b> ^ TmpR) & ( <b>A</b> ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br><b>A</b> = u8(Tmp). |
| 4.2                |                | Read <b>PC</b>                                                                                                                                                                                                                                                                                                                                    |
|                    |                | Store as OpCode.                                                                                                                                                                                                                                                                                                                                  |
| +X.1               |                |                                                                                                                                                                                                                                                                                                                                                   |

|                       |                                                |                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SBC (A6)              | Cycles: 3                                      | Size: 1                                                                                                                                                                                                                                                                                                                                           |
| Indirect X (X) (Read) |                                                |                                                                                                                                                                                                                                                                                                                                                   |
| Cycle                 | R/W                                            | Desc.                                                                                                                                                                                                                                                                                                                                             |
| -X.2                  | Read <b>PC</b>                                 | Store as OpCode.                                                                                                                                                                                                                                                                                                                                  |
| 1.1                   | Read ( <b>X</b>   (( <b>PSW</b> & \$20) << 3)) | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                  |
| 1.2                   |                                                |                                                                                                                                                                                                                                                                                                                                                   |
| 2.1                   |                                                |                                                                                                                                                                                                                                                                                                                                                   |
| 2.2                   |                                                | Store as Operand.                                                                                                                                                                                                                                                                                                                                 |
| 3.1                   |                                                | TmpR = ~Operand.<br>Tmp = u32( <b>A</b> ) + u32(TmpR) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = (( <b>A</b> ^ TmpR ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~( <b>A</b> ^ TmpR) & ( <b>A</b> ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br><b>A</b> = u8(Tmp). |
| 3.2                   | Read <b>PC</b>                                 | Store as OpCode.                                                                                                                                                                                                                                                                                                                                  |
| +X.1                  |                                                |                                                                                                                                                                                                                                                                                                                                                   |

|                                 |                                                              |                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SBC (A7)                        | Cycles: 6                                                    | Size: 2                                                                                                                                                                                                                                                                                                                                            |
| X-Indexed Indirect [d+X] (Read) |                                                              |                                                                                                                                                                                                                                                                                                                                                    |
| Cycle                           | R/W                                                          | Desc.                                                                                                                                                                                                                                                                                                                                              |
| -X.2                            | Read <b>PC</b>                                               | Store as OpCode.                                                                                                                                                                                                                                                                                                                                   |
| 1.1                             |                                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                   |
| 1.2                             | Read <b>PC</b>                                               | Store as Operand.                                                                                                                                                                                                                                                                                                                                  |
| 2.1                             |                                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                   |
| 2.2                             |                                                              |                                                                                                                                                                                                                                                                                                                                                    |
| 3.1                             |                                                              | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).                                                                                                                                                                                                                                                                            |
| 3.2                             | Read Address                                                 | Store as Pointer.L.                                                                                                                                                                                                                                                                                                                                |
| 4.1                             |                                                              |                                                                                                                                                                                                                                                                                                                                                    |
| 4.2                             | Read ((Address + 1) & \$FF)  <br>(( <b>PSW</b> & \$20) << 3) | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                                |
| 5.1                             |                                                              |                                                                                                                                                                                                                                                                                                                                                    |
| 5.2                             | Read Pointer                                                 | Store as Operand.                                                                                                                                                                                                                                                                                                                                  |
| 6.1                             |                                                              | TmpR = ~Operand.<br>Tmp = u32( <b>A</b> ) + u32(TmpR) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = (( <b>A</b> ^ TmpR ^ u8(Tmp)) & \$10).<br><b>V</b> flag = (~( <b>A</b> ^ TmpR) & ( <b>A</b> ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br><b>A</b> = u8(Tmp). |
| 6.2                             | Read <b>PC</b>                                               | Store as OpCode.                                                                                                                                                                                                                                                                                                                                   |
| +X.1                            |                                                              |                                                                                                                                                                                                                                                                                                                                                    |



|           |                |                                                                                                                                                                                                                                                                                                                                |
|-----------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SBC (A8)  | Cycles: 2      | Size: 2                                                                                                                                                                                                                                                                                                                        |
| Immediate |                |                                                                                                                                                                                                                                                                                                                                |
| Cycle     | R/W            | Desc.                                                                                                                                                                                                                                                                                                                          |
| -X.2      | Read <b>PC</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                               |
| 1.1       | Read <b>PC</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                               |
| 1.2       |                | Store as Operand.                                                                                                                                                                                                                                                                                                              |
| 2.1       |                | Inc. <b>PC</b> . TmpR = ~Operand.                                                                                                                                                                                                                                                                                              |
|           |                | Tmp = u32( <b>A</b> ) + u32(TmpR) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = (( <b>A</b> ^ TmpR ^ u8(Tmp)) & \$10).<br><b>V</b> flag = (~( <b>A</b> ^ TmpR) & ( <b>A</b> ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br><b>A</b> = u8(Tmp). |
| 2.2       | Read <b>PC</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                               |
| +X.1      |                |                                                                                                                                                                                                                                                                                                                                |

|                                                       |                  |                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SBC (A9)                                              | Cycles: 6        | Size: 3                                                                                                                                                                                                                                                                                                                                       |
| Direct Page to Direct Page dd, ds (Read/Modify/Write) |                  |                                                                                                                                                                                                                                                                                                                                               |
| Cycle                                                 | R/W              | Desc.                                                                                                                                                                                                                                                                                                                                         |
| -X.2                                                  | Read <b>PC</b>   | Store as OpCode.                                                                                                                                                                                                                                                                                                                              |
| 1.1                                                   |                  | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                              |
| 1.2                                                   | Read <b>PC</b>   | Store as Operand.                                                                                                                                                                                                                                                                                                                             |
| 2.1                                                   |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                                                                                                                                                                                                                                                           |
| 2.2                                                   | Read Address     | Store as Operand0.                                                                                                                                                                                                                                                                                                                            |
| 3.1                                                   |                  |                                                                                                                                                                                                                                                                                                                                               |
| 3.2                                                   | Read <b>PC</b>   | Store as Operand.                                                                                                                                                                                                                                                                                                                             |
| 4.1                                                   |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                                                                                                                                                                                                                                                           |
| 4.2                                                   | Read Address     | Store as Operand1.                                                                                                                                                                                                                                                                                                                            |
| 5.1                                                   |                  | TmpR = ~Operand0.<br>Tmp = u32(Operand1) + u32(TmpR) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = ((Operand1 ^ TmpR ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~(Operand1 ^ TmpR) & (Operand1 ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br>Operand0 = u8(Tmp). |
| 5.2                                                   | Write to Address | Write Operand0.                                                                                                                                                                                                                                                                                                                               |
| 6.1                                                   |                  |                                                                                                                                                                                                                                                                                                                                               |
| 6.2                                                   | Read <b>PC</b>   | Store as OpCode.                                                                                                                                                                                                                                                                                                                              |
| +X.1                                                  |                  |                                                                                                                                                                                                                                                                                                                                               |

|                          |                         |                                                                        |
|--------------------------|-------------------------|------------------------------------------------------------------------|
| MOV1 (AA)                | Cycles: 4               | Size: 3                                                                |
| Absolute Boolean Bit m.b |                         |                                                                        |
| Cycle                    | R/W                     | Desc.                                                                  |
| -X.2                     | Read <b>PC</b>          | Store as OpCode.                                                       |
| 1.1                      |                         | Inc. <b>PC</b> .                                                       |
| 1.2                      | Read <b>PC</b>          | Store as Address.L.                                                    |
| 2.1                      |                         | Inc. <b>PC</b> .                                                       |
| 2.2                      | Read <b>PC</b>          | Store as Address.H.                                                    |
| 3.1                      |                         | Inc. <b>PC</b> .                                                       |
| 3.2                      | Read (Address & \$1FFF) | Store as Operand.                                                      |
| 4.1                      |                         | Bit = Address >> 13.<br><b>C</b> flag = ((Operand & (1 << Bit)) != 0). |
| 4.2                      | Read <b>PC</b>          | Store as OpCode.                                                       |
| +X.1                     |                         |                                                                        |

|                                   |                  |                                                                           |
|-----------------------------------|------------------|---------------------------------------------------------------------------|
| INC (AB)                          | Cycles: 4        | Size: 2                                                                   |
| Direct Page d (Read/Modify/Write) |                  |                                                                           |
| Cycle                             | R/W              | Desc.                                                                     |
| -X.2                              | Read <b>PC</b>   | Store as OpCode.                                                          |
| 1.1                               |                  | Inc. <b>PC</b> .                                                          |
| 1.2                               | Read <b>PC</b>   | Store as Operand.                                                         |
| 2.1                               |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).       |
| 2.2                               | Read Address     | Store as Operand.                                                         |
| 3.1                               |                  | Operand += 1. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = !Operand. |
| 3.2                               | Write to Address | Write Operand.                                                            |
| 4.1                               |                  |                                                                           |
| 4.2                               | Read <b>PC</b>   | Store as OpCode.                                                          |
| +X.1                              |                  |                                                                           |

|                                 |                  |                                                                           |
|---------------------------------|------------------|---------------------------------------------------------------------------|
| INC (AC)                        | Cycles: 5        | Size: 3                                                                   |
| Absolute !a (Read/Modify/Write) |                  |                                                                           |
| Cycle                           | R/W              | Desc.                                                                     |
| -X.2                            | Read <b>PC</b>   | Store as OpCode.                                                          |
| 1.1                             |                  | Inc. <b>PC</b> .                                                          |
| 1.2                             | Read <b>PC</b>   | Store as Address.L.                                                       |
| 2.1                             |                  | Inc. <b>PC</b> .                                                          |
| 2.2                             | Read <b>PC</b>   | Store as Address.H.                                                       |
| 3.1                             |                  | Inc. <b>PC</b> .                                                          |
| 3.2                             | Read Address     | Store as Operand.                                                         |
| 4.1                             |                  | Operand += 1. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = !Operand. |
| 4.2                             | Write to Address | Write Operand.                                                            |
| 5.1                             |                  |                                                                           |
| 5.2                             | Read <b>PC</b>   | Store as OpCode.                                                          |
| +X.1                            |                  |                                                                           |

|           |                |                                                                                                                            |
|-----------|----------------|----------------------------------------------------------------------------------------------------------------------------|
| CMP (AD)  | Cycles: 2      | Size: 2                                                                                                                    |
| Immediate |                |                                                                                                                            |
| Cycle     | R/W            | Desc.                                                                                                                      |
| -X.2      | Read <b>PC</b> | Store as OpCode.                                                                                                           |
| 1.1       |                | Inc. <b>PC</b> .                                                                                                           |
| 1.2       | Read <b>PC</b> | Store as Operand.                                                                                                          |
| 2.1       |                | Inc. <b>PC</b> . Tmp = <b>Y</b> - Operand. <b>C</b> flag = (Tmp >= 0), <b>N</b> flag = (Tmp & \$80), <b>Z</b> flag = !Tmp. |
| 2.2       | Read <b>PC</b> | Store as OpCode.                                                                                                           |
| +X.1      |                |                                                                                                                            |

|          |                |                      |
|----------|----------------|----------------------|
| POP (AE) | Cycles: 4      | Size: 1              |
| Implied  |                |                      |
| Cycle    | R/W            | Desc.                |
| -X.2     | Read <b>PC</b> | Store as OpCode.     |
| 1.1      | Read <b>PC</b> | Inc. <b>PC</b> .     |
| 1.2      |                | Discard.             |
| 2.1      |                |                      |
| 2.2      |                |                      |
| 3.1      |                | Inc. <b>SP</b> by 1. |
| 3.2      |                | Pull to <b>A</b> .   |
| 4.1      |                |                      |
| 4.2      |                | Store as OpCode.     |
| +X.1     |                |                      |

|                                      |                  |                                                      |
|--------------------------------------|------------------|------------------------------------------------------|
| MOV (AF)                             | Cycles: 4        | Size: 1                                              |
| Indirect Auto-Increment (X)+ (Write) |                  |                                                      |
| Cycle                                | R/W              | Desc.                                                |
| -X.2                                 | Read <b>PC</b>   | Store as OpCode.                                     |
| 1.1                                  |                  | Inc. <b>PC</b> .                                     |
| 1.2                                  | Read <b>PC</b>   | Discard.                                             |
| 2.1                                  |                  | Address = ( <b>X</b>   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                                  |                  |                                                      |
| 3.1                                  |                  |                                                      |
| 3.2                                  | Write to Address | Write <b>A</b> .                                     |
| 4.1                                  |                  | <b>X</b> += 1.                                       |
| 4.2                                  | Read <b>PC</b>   | Store as OpCode.                                     |
| +X.1                                 |                  |                                                      |



|            |                |                                                                      |
|------------|----------------|----------------------------------------------------------------------|
| BCS (B0)   | Cycles: 2-4    | Size: 2                                                              |
| Relative r |                |                                                                      |
| Cycle      | R/W            | Desc.                                                                |
| -X.2       | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1        |                | Inc. <b>PC</b> . Set Jump to ( <b>PSW</b> & \$01) == \$01.           |
| 1.2        |                | Store as Operand.                                                    |
| 2.1        |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 4.2). |
| 2.2        |                |                                                                      |
| 3.1        |                | Address = <b>PC</b> + i8(Operand).                                   |
|            |                | <b>PC</b> .L = Address.L.                                            |
| 3.2        |                |                                                                      |
| 4.1        |                | <b>PC</b> .H = Address.H.                                            |
| 4.2        | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1       |                |                                                                      |

|            |                                     |                               |
|------------|-------------------------------------|-------------------------------|
| TCALL (B1) | Cycles: 8                           | Size: 1                       |
| Implied    |                                     |                               |
| Cycle      | R/W                                 | Desc.                         |
| -X.2       | Read <b>PC</b>                      | Store as OpCode.              |
| 1.1        |                                     | Inc. <b>PC</b> .              |
| 1.2        | Read <b>PC</b>                      | Store as Operand.             |
| 2.1        |                                     |                               |
| 2.2        |                                     |                               |
| 3.1        |                                     |                               |
| 3.2        | Write to u8( <b>SP</b> +0)   \$0100 | Push <b>PC</b> .H onto stack. |
| 4.1        |                                     |                               |
| 4.2        | Write to u8( <b>SP</b> -1)   \$0100 | Push <b>PC</b> .L onto stack. |
| 5.1        |                                     | Dec. <b>SP</b> by 2.          |
| 5.2        |                                     |                               |
| 6.1        |                                     |                               |
| 6.2        | Read \$FFC8                         | Store as <b>PC</b> .L.        |
| 7.1        |                                     |                               |
| 7.2        | Read \$FFC9                         | Store as <b>PC</b> .H.        |
| 8.1        |                                     |                               |
| 8.2        | Read <b>PC</b>                      | Store as OpCode.              |
| +X.1       |                                     |                               |

|                                         |                  |                                                                     |
|-----------------------------------------|------------------|---------------------------------------------------------------------|
| CLR1 (B2)                               | Cycles: 4        | Size: 2                                                             |
| Direct Page Bit d.b (Read/Modify/Write) |                  |                                                                     |
| Cycle                                   | R/W              | Desc.                                                               |
| -X.2                                    | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                                     |                  | Inc. <b>PC</b> .                                                    |
| 1.2                                     | Read <b>PC</b>   | Store as Operand.                                                   |
| 2.1                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                                     | Read Address     | Store as Operand.                                                   |
| 3.1                                     |                  | Operand = (Operand & ~\$20)   \$00.                                 |
| 3.2                                     | Write to Address | Write Operand.                                                      |
| 4.1                                     |                  |                                                                     |
| 4.2                                     | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                                    |                  |                                                                     |

|                                 |                |                                                                      |
|---------------------------------|----------------|----------------------------------------------------------------------|
| BBC (B3)                        | Cycles: 5-7    | Size: 3                                                              |
| Direct Page Bit Relative d.b, r |                |                                                                      |
| Cycle                           | R/W            | Desc.                                                                |
| -X.2                            | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1                             |                | Inc. <b>PC</b> .                                                     |
| 1.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 2.1                             |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).  |
| 2.2                             | Read Address   | Store as Operand.                                                    |
| 3.1                             |                | Set Jump to (Operand & (1 << 5)) == (0 << 5).                        |
| 3.2                             |                |                                                                      |
| 4.1                             |                |                                                                      |
| 4.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 5.1                             |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 7.2). |
| 5.2                             |                |                                                                      |
| 6.1                             |                | Address = <b>PC</b> + i8(Operand).<br><b>PC</b> .L = Address.L.      |
| 6.2                             |                |                                                                      |
| 7.1                             |                | <b>PC</b> .H = Address.H.                                            |
| 7.2                             | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1                            |                |                                                                      |

|                                  |                |                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SBC (B4)                         | Cycles: 4      | Size: 2                                                                                                                                                                                                                                                                                                                                            |
| X-Indexed Direct Page d+X (Read) |                |                                                                                                                                                                                                                                                                                                                                                    |
| Cycle                            | R/W            | Desc.                                                                                                                                                                                                                                                                                                                                              |
| -X.2                             | Read <b>PC</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                   |
| 1.1                              | Read <b>PC</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                   |
| 1.2                              |                | Store as Operand.                                                                                                                                                                                                                                                                                                                                  |
| 2.1                              |                | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                   |
| 2.2                              |                |                                                                                                                                                                                                                                                                                                                                                    |
| 3.1                              |                | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).                                                                                                                                                                                                                                                                            |
| 3.2                              |                | Store as Operand.                                                                                                                                                                                                                                                                                                                                  |
| 4.1                              |                | TmpR = ~Operand.<br>Tmp = u32( <b>A</b> ) + u32(TmpR) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = (( <b>A</b> ^ TmpR ^ u8(Tmp)) & \$10).<br><b>V</b> flag = (~( <b>A</b> ^ TmpR) & ( <b>A</b> ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br><b>A</b> = u8(Tmp). |
| 4.2                              |                | Read <b>PC</b>                                                                                                                                                                                                                                                                                                                                     |
|                                  |                | Store as OpCode.                                                                                                                                                                                                                                                                                                                                   |
| +X.1                             |                |                                                                                                                                                                                                                                                                                                                                                    |

|                                |                |                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SBC (B5)                       | Cycles: 5      | Size: 3                                                                                                                                                                                                                                                                                                                                           |
| X-Indexed Absolute !a+X (Read) |                |                                                                                                                                                                                                                                                                                                                                                   |
| Cycle                          | R/W            | Desc.                                                                                                                                                                                                                                                                                                                                             |
| -X.2                           | Read <b>PC</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                  |
| 1.1                            | Read <b>PC</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                  |
| 1.2                            |                | Store as Address.L.                                                                                                                                                                                                                                                                                                                               |
| 2.1                            |                | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                  |
| 2.2                            |                | Store as Address.H.                                                                                                                                                                                                                                                                                                                               |
| 3.1                            |                | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                  |
| 3.2                            |                |                                                                                                                                                                                                                                                                                                                                                   |
| 4.1                            |                | Pointer = ( <b>X</b> + Address).                                                                                                                                                                                                                                                                                                                  |
| 4.2                            |                | Store as Operand.                                                                                                                                                                                                                                                                                                                                 |
| 5.1                            |                | TmpR = ~Operand.<br>Tmp = u32( <b>A</b> ) + u32(TmpR) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = (( <b>A</b> ^ TmpR ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~( <b>A</b> ^ TmpR) & ( <b>A</b> ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br><b>A</b> = u8(Tmp). |
| 5.2                            |                | Read <b>PC</b>                                                                                                                                                                                                                                                                                                                                    |
|                                |                | Store as OpCode.                                                                                                                                                                                                                                                                                                                                  |
| +X.1                           |                |                                                                                                                                                                                                                                                                                                                                                   |

|                                |                |                                                                                                                                                                                                                                                                                                                                                   |
|--------------------------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SBC (B6)                       | Cycles: 5      | Size: 3                                                                                                                                                                                                                                                                                                                                           |
| Y-Indexed Absolute !a+Y (Read) |                |                                                                                                                                                                                                                                                                                                                                                   |
| Cycle                          | R/W            | Desc.                                                                                                                                                                                                                                                                                                                                             |
| -X.2                           | Read <b>PC</b> | Store as OpCode.                                                                                                                                                                                                                                                                                                                                  |
| 1.1                            | Read <b>PC</b> | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                  |
| 1.2                            |                | Store as Address.L.                                                                                                                                                                                                                                                                                                                               |
| 2.1                            |                | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                  |
| 2.2                            |                | Store as Address.H.                                                                                                                                                                                                                                                                                                                               |
| 3.1                            |                | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                  |
| 3.2                            |                |                                                                                                                                                                                                                                                                                                                                                   |
| 4.1                            |                | Pointer = ( <b>Y</b> + Address).                                                                                                                                                                                                                                                                                                                  |
| 4.2                            |                | Store as Operand.                                                                                                                                                                                                                                                                                                                                 |
| 5.1                            |                | TmpR = ~Operand.<br>Tmp = u32( <b>A</b> ) + u32(TmpR) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = (( <b>A</b> ^ TmpR ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~( <b>A</b> ^ TmpR) & ( <b>A</b> ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br><b>A</b> = u8(Tmp). |
| 5.2                            |                | Read <b>PC</b>                                                                                                                                                                                                                                                                                                                                    |
|                                |                | Store as OpCode.                                                                                                                                                                                                                                                                                                                                  |
| +X.1                           |                |                                                                                                                                                                                                                                                                                                                                                   |

|                                 |                                                              |                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SBC (B7)                        | Cycles: 6                                                    | Size: 2                                                                                                                                                                                                                                                                                                                                           |
| Indirect Y-Indexed [d]+Y (Read) |                                                              |                                                                                                                                                                                                                                                                                                                                                   |
| Cycle                           | R/W                                                          | Desc.                                                                                                                                                                                                                                                                                                                                             |
| -X.2                            | Read <b>PC</b>                                               | Store as OpCode.                                                                                                                                                                                                                                                                                                                                  |
| 1.1                             |                                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                  |
| 1.2                             | Read <b>PC</b>                                               | Store as Operand.                                                                                                                                                                                                                                                                                                                                 |
| 2.1                             |                                                              | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                  |
| 2.2                             |                                                              |                                                                                                                                                                                                                                                                                                                                                   |
| 3.1                             |                                                              | Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                                                                                                                                                                                                                                                                                |
| 3.2                             | Read Address                                                 | Store as Pointer.L.                                                                                                                                                                                                                                                                                                                               |
| 4.1                             |                                                              |                                                                                                                                                                                                                                                                                                                                                   |
| 4.2                             | Read ((Address + 1) & \$FF)  <br>(( <b>PSW</b> & \$20) << 3) | Store as Pointer.H.                                                                                                                                                                                                                                                                                                                               |
| 5.1                             |                                                              | Address = ( <b>Y</b> + Pointer).                                                                                                                                                                                                                                                                                                                  |
| 5.2                             | Read Address                                                 | Store as Operand.                                                                                                                                                                                                                                                                                                                                 |
| 6.1                             |                                                              | TmpR = ~Operand.<br>Tmp = u32( <b>A</b> ) + u32(TmpR) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = (( <b>A</b> ^ TmpR ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~( <b>A</b> ^ TmpR) & ( <b>A</b> ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br><b>A</b> = u8(Tmp). |
| 6.2                             | Read <b>PC</b>                                               | Store as OpCode.                                                                                                                                                                                                                                                                                                                                  |
| +X.1                            |                                                              |                                                                                                                                                                                                                                                                                                                                                   |



|                                                         |                  |                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SBC (B8)                                                | Cycles: 5        | Size: 3                                                                                                                                                                                                                                                                                                                                         |
| Immediate Data to Direct Page d, #i (Read/Modify/Write) |                  |                                                                                                                                                                                                                                                                                                                                                 |
| Cycle                                                   | R/W              | Desc.                                                                                                                                                                                                                                                                                                                                           |
| -X.2                                                    | Read <b>PC</b>   | Store as OpCode.                                                                                                                                                                                                                                                                                                                                |
| 1.1                                                     |                  | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                |
| 1.2                                                     | Read <b>PC</b>   | Store as Operand0.                                                                                                                                                                                                                                                                                                                              |
| 2.1                                                     |                  | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                                |
| 2.2                                                     | Read <b>PC</b>   | Store as Operand.                                                                                                                                                                                                                                                                                                                               |
| 3.1                                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                                                                                                                                                                                                                                                             |
| 3.2                                                     | Read Address     | Store as Operand1.                                                                                                                                                                                                                                                                                                                              |
| 4.1                                                     |                  | TmpR = ~Operand0.<br>Tmp = u32(Operand1) + u32(TmpR) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = ((Operand1 ^ TmpR ^ u8(Tmp)) & \$10).<br><b>V</b> flag = (~ (Operand1 ^ TmpR) & (Operand1 ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br>Operand0 = u8(Tmp). |
| 4.2                                                     | Write to Address | Write Operand0.                                                                                                                                                                                                                                                                                                                                 |
| 5.1                                                     |                  |                                                                                                                                                                                                                                                                                                                                                 |
| 5.2                                                     | Read <b>PC</b>   | Store as OpCode.                                                                                                                                                                                                                                                                                                                                |
| +X.1                                                    |                  |                                                                                                                                                                                                                                                                                                                                                 |

|                                                             |                                                    |                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SBC (B9)                                                    | Cycles: 5                                          | Size: 1                                                                                                                                                                                                                                                                                                                                       |
| Indirect Page to Indirect Page (X), (Y) (Read/Modify/Write) |                                                    |                                                                                                                                                                                                                                                                                                                                               |
| Cycle                                                       | R/W                                                | Desc.                                                                                                                                                                                                                                                                                                                                         |
| -X.2                                                        | Read <b>PC</b>                                     | Store as OpCode.                                                                                                                                                                                                                                                                                                                              |
| 1.1                                                         |                                                    | Inc. <b>PC</b> .                                                                                                                                                                                                                                                                                                                              |
| 1.2                                                         | Read <b>PC</b>                                     | Discard.                                                                                                                                                                                                                                                                                                                                      |
| 2.1                                                         |                                                    |                                                                                                                                                                                                                                                                                                                                               |
| 2.2                                                         | Read ( <b>Y</b>   (( <b>PSW</b> & \$20) << 3))     | Store as Operand0.                                                                                                                                                                                                                                                                                                                            |
| 3.1                                                         |                                                    |                                                                                                                                                                                                                                                                                                                                               |
| 3.2                                                         | Read ( <b>X</b>   (( <b>PSW</b> & \$20) << 3))     | Store as Operand1.                                                                                                                                                                                                                                                                                                                            |
| 4.1                                                         |                                                    | TmpR = ~Operand0.<br>Tmp = u32(Operand1) + u32(TmpR) + u32( <b>C</b> flag).<br><b>C</b> flag = (Tmp > \$FF).<br><b>Z</b> flag = (u8(Tmp) == \$00).<br><b>H</b> flag = ((Operand1 ^ TmpR ^ u8(Tmp)) & \$10).<br><b>V</b> flag = ~(Operand1 ^ TmpR) & (Operand1 ^ u8(Tmp)) & \$80).<br><b>N</b> flag = (u8(Tmp) & \$80).<br>Operand0 = u8(Tmp). |
| 4.2                                                         | Write to ( <b>X</b>   (( <b>PSW</b> & \$20) << 3)) | Write Operand0.                                                                                                                                                                                                                                                                                                                               |
| 5.1                                                         |                                                    |                                                                                                                                                                                                                                                                                                                                               |
| 5.2                                                         | Read <b>PC</b>                                     | Store as OpCode.                                                                                                                                                                                                                                                                                                                              |
| +X.1                                                        |                                                    |                                                                                                                                                                                                                                                                                                                                               |

|                         |                                                           |                                                                      |         |
|-------------------------|-----------------------------------------------------------|----------------------------------------------------------------------|---------|
| MOVW (BA)               |                                                           | Cycles: 5                                                            | Size: 2 |
| Direct Page d (Read 16) |                                                           |                                                                      |         |
| Cycle                   | R/W                                                       | Desc.                                                                |         |
| -X.2                    | Read <b>PC</b>                                            | Store as OpCode.                                                     |         |
| 1.1                     |                                                           | Inc. <b>PC</b> .                                                     |         |
| 1.2                     | Read <b>PC</b>                                            | Store as Operand.                                                    |         |
| 2.1                     |                                                           | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).  |         |
| 2.2                     | Read Address                                              | Store as <b>A</b> .                                                  |         |
| 3.1                     |                                                           |                                                                      |         |
| 3.2                     |                                                           |                                                                      |         |
| 4.1                     |                                                           |                                                                      |         |
| 4.2                     | Read ((Address + 1) & \$FF)   (( <b>PSW</b> & \$20) << 3) | Store as <b>Y</b> .                                                  |         |
| 5.1                     |                                                           | <b>N</b> flag = ( <b>YA</b> & \$8000), <b>Z</b> flag = ! <b>YA</b> . |         |
| 5.2                     | Read <b>PC</b>                                            | Store as OpCode.                                                     |         |
| +X.1                    |                                                           |                                                                      |         |

|                                               |                  |                                                                           |
|-----------------------------------------------|------------------|---------------------------------------------------------------------------|
| INC (BB)                                      | Cycles: 5        | Size: 2                                                                   |
| X-Indexed Direct Page d+X (Read/Modify/Write) |                  |                                                                           |
| Cycle                                         | R/W              | Desc.                                                                     |
| -X.2                                          | Read <b>PC</b>   | Store as OpCode.                                                          |
| 1.1                                           |                  | Inc. <b>PC</b> .                                                          |
| 1.2                                           | Read <b>PC</b>   | Store as Operand.                                                         |
| 2.1                                           |                  | Inc. <b>PC</b> .                                                          |
| 2.2                                           |                  |                                                                           |
| 3.1                                           |                  | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).   |
| 3.2                                           | Read Address     | Store as Operand.                                                         |
| 4.1                                           |                  | Operand += 1. <b>N</b> flag = (Operand & \$80), <b>Z</b> flag = !Operand. |
| 4.2                                           | Write to Address | Write Operand.                                                            |
| 5.1                                           |                  |                                                                           |
| 5.2                                           | Read <b>PC</b>   | Store as OpCode.                                                          |
| +X.1                                          |                  |                                                                           |

|               |                |                                                                                 |
|---------------|----------------|---------------------------------------------------------------------------------|
| INC (BC)      | Cycles: 2      | Size: 1                                                                         |
| Accumulator A |                |                                                                                 |
| Cycle         | R/W            | Desc.                                                                           |
| -X.2          | Read <b>PC</b> | Store as OpCode.                                                                |
| 1.1           |                | Inc. <b>PC</b> .                                                                |
| 1.2           | Read <b>PC</b> | Discard.                                                                        |
| 2.1           |                | <b>A</b> += 1. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 2.2           | Read <b>PC</b> | Store as OpCode.                                                                |
| +X.1          |                |                                                                                 |

|          |                |                  |
|----------|----------------|------------------|
| MOV (BD) | Cycles: 2      | Size: 1          |
| Implied  |                |                  |
| Cycle    | R/W            | Desc.            |
| -X.2     | Read <b>PC</b> | Store as OpCode. |
| 1.1      |                | Inc. <b>PC</b> . |
| 1.2      |                |                  |
| 2.1      |                | <b>SP = X.</b>   |
| 2.2      |                | Store as OpCode. |
| +X.1     |                |                  |

|          |                |                                                                                                                                             |
|----------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| DAS (BE) | Cycles: 3      | Size: 1                                                                                                                                     |
| Implied  |                |                                                                                                                                             |
| Cycle    | R/W            | Desc.                                                                                                                                       |
| -X.2     | Read <b>PC</b> | Store as OpCode.                                                                                                                            |
| 1.1      |                | Inc. <b>PC</b> .                                                                                                                            |
| 1.2      |                |                                                                                                                                             |
| 2.1      |                | If (! <b>C</b> flag    <b>A</b> > \$99):<br><b>A</b> -= \$60, <b>C</b> flag = 0.                                                            |
| 2.2      |                |                                                                                                                                             |
| 3.1      |                | If (! <b>H</b> flag    ( <b>A</b> & \$0F) > \$09):<br><b>A</b> -= \$06.<br><b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 3.2      |                | <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> .<br>Store as OpCode.                                                        |
| +X.1     | Read <b>PC</b> |                                                                                                                                             |

|                                     |                |                                                                  |
|-------------------------------------|----------------|------------------------------------------------------------------|
| MOV (BF)                            | Cycles: 4      | Size: 1                                                          |
| Indirect Auto-Increment (X)+ (Read) |                |                                                                  |
| Cycle                               | R/W            | Desc.                                                            |
| -X.2                                | Read <b>PC</b> | Store as OpCode.                                                 |
| 1.1                                 |                | Inc. <b>PC</b> .                                                 |
| 1.2                                 | Read <b>PC</b> | Discard.                                                         |
| 2.1                                 |                | Address = ( <b>X</b>   (( <b>PSW</b> & \$20) << 3)).             |
| 2.2                                 | Read Address   | Store as <b>A</b> .                                              |
| 3.1                                 |                | <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 3.2                                 |                |                                                                  |
| 4.1                                 |                | <b>X</b> += 1.                                                   |
| 4.2                                 | Read <b>PC</b> | Store as OpCode.                                                 |
| +X.1                                |                |                                                                  |



|         |                |                                     |
|---------|----------------|-------------------------------------|
| DI (C0) | Cycles: 3      | Size: 1                             |
| Implied |                |                                     |
| Cycle   | R/W            | Desc.                               |
| -X.2    | Read <b>PC</b> | Store as OpCode.                    |
| 1.1     |                | Inc. <b>PC</b> .                    |
| 1.2     |                |                                     |
| 2.1     |                |                                     |
| 2.2     |                |                                     |
| 3.1     |                | <b>PSW</b> = ( <b>PSW</b> & ~\$04). |
| 3.2     |                | Store as OpCode.                    |
| +X.1    |                |                                     |

|            |                                     |                               |
|------------|-------------------------------------|-------------------------------|
| TCALL (C1) | Cycles: 8                           | Size: 1                       |
| Implied    |                                     |                               |
| Cycle      | R/W                                 | Desc.                         |
| -X.2       | Read <b>PC</b>                      | Store as OpCode.              |
| 1.1        |                                     | Inc. <b>PC</b> .              |
| 1.2        | Read <b>PC</b>                      | Store as Operand.             |
| 2.1        |                                     |                               |
| 2.2        |                                     |                               |
| 3.1        |                                     |                               |
| 3.2        | Write to u8( <b>SP</b> +0)   \$0100 | Push <b>PC</b> .H onto stack. |
| 4.1        |                                     |                               |
| 4.2        | Write to u8( <b>SP</b> -1)   \$0100 | Push <b>PC</b> .L onto stack. |
| 5.1        |                                     | Dec. <b>SP</b> by 2.          |
| 5.2        |                                     |                               |
| 6.1        |                                     |                               |
| 6.2        | Read \$FFC6                         | Store as <b>PC</b> .L.        |
| 7.1        |                                     |                               |
| 7.2        | Read \$FFC7                         | Store as <b>PC</b> .H.        |
| 8.1        |                                     |                               |
| 8.2        | Read <b>PC</b>                      | Store as OpCode.              |
| +X.1       |                                     |                               |

|                                         |                  |                                                                     |
|-----------------------------------------|------------------|---------------------------------------------------------------------|
| SET1 (C2)                               | Cycles: 4        | Size: 2                                                             |
| Direct Page Bit d.b (Read/Modify/Write) |                  |                                                                     |
| Cycle                                   | R/W              | Desc.                                                               |
| -X.2                                    | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                                     |                  | Inc. <b>PC</b> .                                                    |
| 1.2                                     | Read <b>PC</b>   | Store as Operand.                                                   |
| 2.1                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                                     | Read Address     | Store as Operand.                                                   |
| 3.1                                     |                  | Operand = (Operand & ~\$40)   \$40.                                 |
| 3.2                                     | Write to Address | Write Operand.                                                      |
| 4.1                                     |                  |                                                                     |
| 4.2                                     | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                                    |                  |                                                                     |

|                                 |                |                                                                      |
|---------------------------------|----------------|----------------------------------------------------------------------|
| BBS (C3)                        | Cycles: 5-7    | Size: 3                                                              |
| Direct Page Bit Relative d.b, r |                |                                                                      |
| Cycle                           | R/W            | Desc.                                                                |
| -X.2                            | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1                             |                | Inc. <b>PC</b> .                                                     |
| 1.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 2.1                             |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).  |
| 2.2                             | Read Address   | Store as Operand.                                                    |
| 3.1                             |                | Set Jump to (Operand & (1 << 6)) == (1 << 6).                        |
| 3.2                             |                |                                                                      |
| 4.1                             |                |                                                                      |
| 4.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 5.1                             |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 7.2). |
| 5.2                             |                |                                                                      |
| 6.1                             |                | Address = <b>PC</b> + i8(Operand).<br><b>PC</b> .L = Address.L.      |
| 6.2                             |                |                                                                      |
| 7.1                             |                | <b>PC</b> .H = Address.H.                                            |
| 7.2                             | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1                            |                |                                                                      |

|                       |                  |                                                                     |
|-----------------------|------------------|---------------------------------------------------------------------|
| MOV (C4)              | Cycles: 4        | Size: 2                                                             |
| Direct Page d (Write) |                  |                                                                     |
| Cycle                 | R/W              | Desc.                                                               |
| -X.2                  | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                   |                  | Inc. <b>PC</b> .                                                    |
| 1.2                   | Read <b>PC</b>   | Store as Operand.                                                   |
| 2.1                   |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                   | Read Address     | Store as Operand.                                                   |
| 3.1                   |                  |                                                                     |
| 3.2                   | Write to Address | Write <b>A</b> .                                                    |
| 4.1                   |                  |                                                                     |
| 4.2                   | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                  |                  |                                                                     |

|                     |                  |                     |
|---------------------|------------------|---------------------|
| MOV (C5)            | Cycles: 5        | Size: 3             |
| Absolute !a (Write) |                  |                     |
| Cycle               | R/W              | Desc.               |
| -X.2                | Read <b>PC</b>   | Store as OpCode.    |
| 1.1                 |                  | Inc. <b>PC</b> .    |
| 1.2                 | Read <b>PC</b>   | Store as Address.L. |
| 2.1                 |                  | Inc. <b>PC</b> .    |
| 2.2                 | Read <b>PC</b>   | Store as Address.H. |
| 3.1                 |                  | Inc. <b>PC</b> .    |
| 3.2                 | Read Address     | Discard.            |
| 4.1                 |                  |                     |
| 4.2                 | Write to Address | Write <b>A</b> .    |
| 5.1                 |                  |                     |
| 5.2                 | Read <b>PC</b>   | Store as OpCode.    |
| +X.1                |                  |                     |

|                        |                                                    |                   |
|------------------------|----------------------------------------------------|-------------------|
| MOV (C6)               | Cycles: 4                                          | Size: 1           |
| Indirect X (X) (Write) |                                                    |                   |
| Cycle                  | R/W                                                | Desc.             |
| -X.2                   | Read <b>PC</b>                                     | Store as OpCode.  |
| 1.1                    |                                                    | Inc. <b>PC</b> .  |
| 1.2                    |                                                    |                   |
| 2.1                    |                                                    |                   |
| 2.2                    | Read ( <b>X</b>   (( <b>PSW</b> & \$20) << 3))     | Store as Discard. |
| 3.1                    |                                                    |                   |
| 3.2                    | Write to ( <b>X</b>   (( <b>PSW</b> & \$20) << 3)) | Write <b>A</b> .  |
| 4.1                    |                                                    |                   |
| 4.2                    | Read <b>PC</b>                                     | Store as OpCode.  |
| +X.1                   |                                                    |                   |

|                                  |                                                              |                                                                         |
|----------------------------------|--------------------------------------------------------------|-------------------------------------------------------------------------|
| MOV (C7)                         | Cycles: 7                                                    | Size: 2                                                                 |
| X-Indexed Indirect [d+X] (Write) |                                                              |                                                                         |
| Cycle                            | R/W                                                          | Desc.                                                                   |
| -X.2                             | Read <b>PC</b>                                               | Store as OpCode.                                                        |
| 1.1                              |                                                              | Inc. <b>PC</b> .                                                        |
| 1.2                              | Read <b>PC</b>                                               | Store as Operand.                                                       |
| 2.1                              |                                                              | Inc. <b>PC</b> .                                                        |
| 2.2                              |                                                              |                                                                         |
| 3.1                              |                                                              | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3). |
| 3.2                              | Read Address                                                 | Store as Pointer.L.                                                     |
| 4.1                              |                                                              |                                                                         |
| 4.2                              | Read ((Address + 1) & \$FF)  <br>(( <b>PSW</b> & \$20) << 3) | Store as Pointer.H.                                                     |
| 5.1                              |                                                              |                                                                         |
| 5.2                              | Read Pointer                                                 | Discard.                                                                |
| 6.1                              |                                                              |                                                                         |
| 6.2                              | Write to Pointer                                             | Write <b>A</b> .                                                        |
| 7.1                              |                                                              |                                                                         |
| 7.2                              | Read <b>PC</b>                                               | Store as OpCode.                                                        |
| +X.1                             |                                                              |                                                                         |



|           |                |                                                                                                                            |
|-----------|----------------|----------------------------------------------------------------------------------------------------------------------------|
| CMP (C8)  | Cycles: 2      | Size: 2                                                                                                                    |
| Immediate |                |                                                                                                                            |
| Cycle     | R/W            | Desc.                                                                                                                      |
| -X.2      | Read <b>PC</b> | Store as OpCode.                                                                                                           |
| 1.1       |                | Inc. <b>PC</b> .                                                                                                           |
| 1.2       | Read <b>PC</b> | Store as Operand.                                                                                                          |
| 2.1       |                | Inc. <b>PC</b> . Tmp = <b>X</b> - Operand. <b>C</b> flag = (Tmp >= 0), <b>N</b> flag = (Tmp & \$80), <b>Z</b> flag = !Tmp. |
| 2.2       | Read <b>PC</b> | Store as OpCode.                                                                                                           |
| +X.1      |                |                                                                                                                            |

|                     |                  |                     |
|---------------------|------------------|---------------------|
| MOV (C9)            | Cycles: 5        | Size: 3             |
| Absolute !a (Write) |                  |                     |
| Cycle               | R/W              | Desc.               |
| -X.2                | Read <b>PC</b>   | Store as OpCode.    |
| 1.1                 |                  | Inc. <b>PC</b> .    |
| 1.2                 | Read <b>PC</b>   | Store as Address.L. |
| 2.1                 |                  | Inc. <b>PC</b> .    |
| 2.2                 | Read <b>PC</b>   | Store as Address.H. |
| 3.1                 |                  | Inc. <b>PC</b> .    |
| 3.2                 | Read Address     | Discard.            |
| 4.1                 |                  |                     |
| 4.2                 | Write to Address | Write <b>X</b> .    |
| 5.1                 |                  |                     |
| 5.2                 | Read <b>PC</b>   | Store as OpCode.    |
| +X.1                |                  |                     |

| MOV1 (CA)                   | Cycles: 6                   | Size: 3                                         |
|-----------------------------|-----------------------------|-------------------------------------------------|
| Absolute Boolean Bit m.b, C |                             |                                                 |
| Cycle                       | R/W                         | Desc.                                           |
| -X.2                        | Read <b>PC</b>              | Store as OpCode.                                |
| 1.1                         |                             | Inc. <b>PC</b> .                                |
| 1.2                         | Read <b>PC</b>              | Store as Address.L.                             |
| 2.1                         |                             | Inc. <b>PC</b> .                                |
| 2.2                         | Read <b>PC</b>              | Store as Address.H.                             |
| 3.1                         |                             | Inc. <b>PC</b> .                                |
| 3.2                         | Read (Address & \$1FFF)     | Store as Operand.                               |
| 4.1                         |                             |                                                 |
| 4.2                         |                             |                                                 |
| 5.1                         |                             | Bit = Address >> 13.<br>Operand &= ~(1 << Bit). |
| 5.2                         | Write to (Address & \$1FFF) | Operand  = ( <b>C</b> flag << Bit).             |
| 6.1                         |                             | Write Operand.                                  |
| 6.2                         | Read <b>PC</b>              | Store as OpCode.                                |
| +X.1                        |                             |                                                 |

|                       |                  |                                                                     |
|-----------------------|------------------|---------------------------------------------------------------------|
| MOV (CB)              | Cycles: 4        | Size: 2                                                             |
| Direct Page d (Write) |                  |                                                                     |
| Cycle                 | R/W              | Desc.                                                               |
| -X.2                  | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                   |                  | Inc. <b>PC</b> .                                                    |
| 1.2                   | Read <b>PC</b>   | Store as Operand.                                                   |
| 2.1                   |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                   | Read Address     | Store as Operand.                                                   |
| 3.1                   |                  |                                                                     |
| 3.2                   | Write to Address | Write <b>Y</b> .                                                    |
| 4.1                   |                  |                                                                     |
| 4.2                   | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                  |                  |                                                                     |

|                     |                  |                     |
|---------------------|------------------|---------------------|
| MOV (CC)            | Cycles: 5        | Size: 3             |
| Absolute !a (Write) |                  |                     |
| Cycle               | R/W              | Desc.               |
| -X.2                | Read <b>PC</b>   | Store as OpCode.    |
| 1.1                 |                  | Inc. <b>PC</b> .    |
| 1.2                 | Read <b>PC</b>   | Store as Address.L. |
| 2.1                 |                  | Inc. <b>PC</b> .    |
| 2.2                 | Read <b>PC</b>   | Store as Address.H. |
| 3.1                 |                  | Inc. <b>PC</b> .    |
| 3.2                 | Read Address     | Discard.            |
| 4.1                 |                  |                     |
| 4.2                 | Write to Address | Write <b>Y</b> .    |
| 5.1                 |                  |                     |
| 5.2                 | Read <b>PC</b>   | Store as OpCode.    |
| +X.1                |                  |                     |

|           |                |                                                                                                       |
|-----------|----------------|-------------------------------------------------------------------------------------------------------|
| MOV (CD)  | Cycles: 2      | Size: 2                                                                                               |
| Immediate |                |                                                                                                       |
| Cycle     | R/W            | Desc.                                                                                                 |
| -X.2      | Read <b>PC</b> | Store as OpCode.                                                                                      |
| 1.1       |                | Inc. <b>PC</b> .                                                                                      |
| 1.2       | Read <b>PC</b> | Store as Operand.                                                                                     |
| 2.1       |                | Inc. <b>PC</b> . <b>X</b> = Operand. <b>N</b> flag = ( <b>X</b> & \$80), <b>Z</b> flag = ! <b>X</b> . |
| 2.2       | Read <b>PC</b> | Store as OpCode.                                                                                      |
| +X.1      |                |                                                                                                       |

|          |                |                      |
|----------|----------------|----------------------|
| POP (CE) | Cycles: 4      | Size: 1              |
| Implied  |                |                      |
| Cycle    | R/W            | Desc.                |
| -X.2     | Read <b>PC</b> | Store as OpCode.     |
| 1.1      |                | Inc. <b>PC</b> .     |
| 1.2      |                | Discard.             |
| 2.1      |                |                      |
| 2.2      |                |                      |
| 3.1      |                | Inc. <b>SP</b> by 1. |
| 3.2      |                | Pull to <b>X</b> .   |
| 4.1      |                |                      |
| 4.2      |                | Store as OpCode.     |
| +X.1     |                |                      |

|          |                |                                                                                                                                                                            |
|----------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MUL (CF) | Cycles: 9      | Size: 1                                                                                                                                                                    |
| Implied  |                |                                                                                                                                                                            |
| Cycle    | R/W            | Desc.                                                                                                                                                                      |
| -X.2     | Read <b>PC</b> | Store as OpCode.                                                                                                                                                           |
| 1.1      |                | Inc. <b>PC</b> .<br>MulYa = u32( <b>A</b> ), ShiftedY = (u32( <b>Y</b> ) << 8).                                                                                            |
| 1.2      |                |                                                                                                                                                                            |
| 2.1      |                | If MulYa & \$1: MulYa += ShiftedY.<br>MulYa >>= 1.                                                                                                                         |
| 2.2      |                |                                                                                                                                                                            |
| 3.1      |                | If MulYa & \$1: MulYa += ShiftedY.<br>MulYa >>= 1.                                                                                                                         |
| 3.2      |                |                                                                                                                                                                            |
| 4.1      |                | If MulYa & \$1: MulYa += ShiftedY.<br>MulYa >>= 1.                                                                                                                         |
| 4.2      |                |                                                                                                                                                                            |
| 5.1      |                | If MulYa & \$1: MulYa += ShiftedY.<br>MulYa >>= 1.                                                                                                                         |
| 5.2      |                |                                                                                                                                                                            |
| 6.1      |                | If MulYa & \$1: MulYa += ShiftedY.<br>MulYa >>= 1.                                                                                                                         |
| 6.2      |                |                                                                                                                                                                            |
| 7.1      |                | If MulYa & \$1: MulYa += ShiftedY.<br>MulYa >>= 1.                                                                                                                         |
| 7.2      |                |                                                                                                                                                                            |
| 8.1      |                | If MulYa & \$1: MulYa += ShiftedY.<br>MulYa >>= 1.                                                                                                                         |
| 8.2      |                |                                                                                                                                                                            |
| 9.1      |                | If MulYa & \$1: MulYa += ShiftedY.<br>MulYa >>= 1.<br><b>Y</b> = u8(MulYa >> 8), <b>A</b> = u8(MulYa).<br><b>N</b> flag = ( <b>Y</b> & \$80), <b>Z</b> flag = ! <b>Y</b> . |
| 9.2      | Read <b>PC</b> | Store as OpCode.                                                                                                                                                           |
| +X.1     |                |                                                                                                                                                                            |



|            |                |                                                                      |
|------------|----------------|----------------------------------------------------------------------|
| BNE (D0)   | Cycles: 2-4    | Size: 2                                                              |
| Relative r |                |                                                                      |
| Cycle      | R/W            | Desc.                                                                |
| -X.2       | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1        |                | Inc. <b>PC</b> . Set Jump to ( <b>PSW</b> & \$02) == \$00.           |
| 1.2        |                | Store as Operand.                                                    |
| 2.1        |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 4.2). |
| 2.2        |                |                                                                      |
| 3.1        |                | Address = <b>PC</b> + i8(Operand).                                   |
|            |                | <b>PC</b> .L = Address.L.                                            |
| 3.2        |                |                                                                      |
| 4.1        |                | <b>PC</b> .H = Address.H.                                            |
| 4.2        | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1       |                |                                                                      |

|            |                                     |                               |
|------------|-------------------------------------|-------------------------------|
| TCALL (D1) | Cycles: 8                           | Size: 1                       |
| Implied    |                                     |                               |
| Cycle      | R/W                                 | Desc.                         |
| -X.2       | Read <b>PC</b>                      | Store as OpCode.              |
| 1.1        |                                     | Inc. <b>PC</b> .              |
| 1.2        | Read <b>PC</b>                      | Store as Operand.             |
| 2.1        |                                     |                               |
| 2.2        |                                     |                               |
| 3.1        |                                     |                               |
| 3.2        | Write to u8( <b>SP</b> +0)   \$0100 | Push <b>PC</b> .H onto stack. |
| 4.1        |                                     |                               |
| 4.2        | Write to u8( <b>SP</b> -1)   \$0100 | Push <b>PC</b> .L onto stack. |
| 5.1        |                                     | Dec. <b>SP</b> by 2.          |
| 5.2        |                                     |                               |
| 6.1        |                                     |                               |
| 6.2        | Read \$FFC4                         | Store as <b>PC</b> .L.        |
| 7.1        |                                     |                               |
| 7.2        | Read \$FFC5                         | Store as <b>PC</b> .H.        |
| 8.1        |                                     |                               |
| 8.2        | Read <b>PC</b>                      | Store as OpCode.              |
| +X.1       |                                     |                               |

|                                         |                  |                                                                     |
|-----------------------------------------|------------------|---------------------------------------------------------------------|
| CLR1 (D2)                               | Cycles: 4        | Size: 2                                                             |
| Direct Page Bit d.b (Read/Modify/Write) |                  |                                                                     |
| Cycle                                   | R/W              | Desc.                                                               |
| -X.2                                    | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                                     |                  | Inc. <b>PC</b> .                                                    |
| 1.2                                     | Read <b>PC</b>   | Store as Operand.                                                   |
| 2.1                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                                     | Read Address     | Store as Operand.                                                   |
| 3.1                                     |                  | Operand = (Operand & ~\$40)   \$00.                                 |
| 3.2                                     | Write to Address | Write Operand.                                                      |
| 4.1                                     |                  |                                                                     |
| 4.2                                     | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                                    |                  |                                                                     |

|                                 |                |                                                                      |
|---------------------------------|----------------|----------------------------------------------------------------------|
| BBC (D3)                        | Cycles: 5-7    | Size: 3                                                              |
| Direct Page Bit Relative d.b, r |                |                                                                      |
| Cycle                           | R/W            | Desc.                                                                |
| -X.2                            | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1                             |                | Inc. <b>PC</b> .                                                     |
| 1.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 2.1                             |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).  |
| 2.2                             | Read Address   | Store as Operand.                                                    |
| 3.1                             |                | Set Jump to (Operand & (1 << 6)) == (0 << 6).                        |
| 3.2                             |                |                                                                      |
| 4.1                             |                |                                                                      |
| 4.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 5.1                             |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 7.2). |
| 5.2                             |                |                                                                      |
| 6.1                             |                | Address = <b>PC</b> + i8(Operand).<br><b>PC</b> .L = Address.L.      |
| 6.2                             |                |                                                                      |
| 7.1                             |                | <b>PC</b> .H = Address.H.                                            |
| 7.2                             | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1                            |                |                                                                      |

|                                   |                  |                                                                         |
|-----------------------------------|------------------|-------------------------------------------------------------------------|
| MOV (D4)                          | Cycles: 5        | Size: 2                                                                 |
| X-Indexed Direct Page d+X (Write) |                  |                                                                         |
| Cycle                             | R/W              | Desc.                                                                   |
| -X.2                              | Read <b>PC</b>   | Store as OpCode.                                                        |
| 1.1                               |                  | Inc. <b>PC</b> .                                                        |
| 1.2                               | Read <b>PC</b>   | Store as Operand.                                                       |
| 2.1                               |                  | Inc. <b>PC</b> .                                                        |
| 2.2                               |                  |                                                                         |
| 3.1                               |                  | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3). |
| 3.2                               | Read Address     | Discard.                                                                |
| 4.1                               |                  |                                                                         |
| 4.2                               | Write to Address | Write <b>A</b> .                                                        |
| 5.1                               |                  |                                                                         |
| 5.2                               | Read <b>PC</b>   | Store as OpCode.                                                        |
| +X.1                              |                  |                                                                         |

|                                 |                  |                                  |
|---------------------------------|------------------|----------------------------------|
| MOV (D5)                        | Cycles: 6        | Size: 3                          |
| X-Indexed Absolute !a+X (Write) |                  |                                  |
| Cycle                           | R/W              | Desc.                            |
| -X.2                            | Read <b>PC</b>   | Store as OpCode.                 |
| 1.1                             |                  | Inc. <b>PC</b> .                 |
| 1.2                             | Read <b>PC</b>   | Store as Address.L.              |
| 2.1                             |                  | Inc. <b>PC</b> .                 |
| 2.2                             | Read <b>PC</b>   | Store as Address.H.              |
| 3.1                             |                  | Inc. <b>PC</b> .                 |
| 3.2                             |                  |                                  |
| 4.1                             |                  | Pointer = ( <b>X</b> + Address). |
| 4.2                             | Read Pointer     | Discard.                         |
| 5.1                             |                  |                                  |
| 5.2                             | Write to Pointer | Write <b>A</b> .                 |
| 6.1                             |                  |                                  |
| 6.2                             | Read <b>PC</b>   | Store as OpCode.                 |
| +X.1                            |                  |                                  |

|                                 |                  |                                  |
|---------------------------------|------------------|----------------------------------|
| MOV (D6)                        | Cycles: 6        | Size: 3                          |
| Y-Indexed Absolute !a+Y (Write) |                  |                                  |
| Cycle                           | R/W              | Desc.                            |
| -X.2                            | Read <b>PC</b>   | Store as OpCode.                 |
| 1.1                             |                  | Inc. <b>PC</b> .                 |
| 1.2                             | Read <b>PC</b>   | Store as Address.L.              |
| 2.1                             |                  | Inc. <b>PC</b> .                 |
| 2.2                             | Read <b>PC</b>   | Store as Address.H.              |
| 3.1                             |                  | Inc. <b>PC</b> .                 |
| 3.2                             |                  |                                  |
| 4.1                             |                  | Pointer = ( <b>Y</b> + Address). |
| 4.2                             | Read Pointer     | Discard.                         |
| 5.1                             |                  |                                  |
| 5.2                             | Write to Pointer | Write <b>A</b> .                 |
| 6.1                             |                  |                                  |
| 6.2                             | Read <b>PC</b>   | Store as OpCode.                 |
| +X.1                            |                  |                                  |

|                                  |                                                           |                                                                     |
|----------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------|
| MOV (D7)                         | Cycles: 7                                                 | Size: 2                                                             |
| Indirect Y-Indexed [d]+Y (Write) |                                                           |                                                                     |
| Cycle                            | R/W                                                       | Desc.                                                               |
| -X.2                             | Read <b>PC</b>                                            | Store as OpCode.                                                    |
| 1.1                              |                                                           | Inc. <b>PC</b> .                                                    |
| 1.2                              | Read <b>PC</b>                                            | Store as Operand.                                                   |
| 2.1                              |                                                           | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                              | Read Address                                              | Store as Pointer.L.                                                 |
| 3.1                              |                                                           |                                                                     |
| 3.2                              | Read ((Address + 1) & \$FF)   (( <b>PSW</b> & \$20) << 3) | Store as Pointer.H.                                                 |
| 4.1                              |                                                           |                                                                     |
| 4.2                              |                                                           |                                                                     |
| 5.1                              |                                                           | Address = ( <b>Y</b> + Pointer).                                    |
| 5.2                              | Read Address                                              | Discard.                                                            |
| 6.1                              |                                                           |                                                                     |
| 6.2                              | Write to Address                                          | Write <b>A</b> .                                                    |
| 7.1                              |                                                           |                                                                     |
| 7.2                              | Read <b>PC</b>                                            | Store as OpCode.                                                    |
| +X.1                             |                                                           |                                                                     |



|                       |                  |                                                                     |
|-----------------------|------------------|---------------------------------------------------------------------|
| MOV (D8)              | Cycles: 4        | Size: 2                                                             |
| Direct Page d (Write) |                  |                                                                     |
| Cycle                 | R/W              | Desc.                                                               |
| -X.2                  | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                   |                  | Inc. <b>PC</b> .                                                    |
| 1.2                   | Read <b>PC</b>   | Store as Operand.                                                   |
| 2.1                   |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                   | Read Address     | Store as Operand.                                                   |
| 3.1                   |                  |                                                                     |
| 3.2                   | Write to Address | Write <b>X</b> .                                                    |
| 4.1                   |                  |                                                                     |
| 4.2                   | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                  |                  |                                                                     |

|                                   |                  |                                                         |
|-----------------------------------|------------------|---------------------------------------------------------|
| MOV (D9)                          | Cycles: 5        | Size: 2                                                 |
| Y-Indexed Direct Page d+Y (Write) |                  |                                                         |
| Cycle                             | R/W              | Desc.                                                   |
| -X.2                              | Read <b>PC</b>   | Store as OpCode.                                        |
| 1.1                               |                  | Inc. <b>PC</b> .                                        |
| 1.2                               | Read <b>PC</b>   | Store as Operand.                                       |
| 2.1                               |                  | Inc. <b>PC</b> .                                        |
| 2.2                               |                  |                                                         |
| 3.1                               |                  | Address = ((Y + Operand) & \$FF)   ((PSW & \$20) << 3). |
| 3.2                               | Read Address     | Discard.                                                |
| 4.1                               |                  |                                                         |
| 4.2                               | Write to Address | Write <b>X</b> .                                        |
| 5.1                               |                  |                                                         |
| 5.2                               | Read <b>PC</b>   | Store as OpCode.                                        |
| +X.1                              |                  |                                                         |

|                          |                                   |                                                                     |         |
|--------------------------|-----------------------------------|---------------------------------------------------------------------|---------|
| MOVW (DA)                |                                   | Cycles: 5                                                           | Size: 2 |
| Direct Page d (Write 16) |                                   |                                                                     |         |
| Cycle                    | R/W                               | Desc.                                                               |         |
| -X.2                     | Read <b>PC</b>                    | Store as OpCode.                                                    |         |
| 1.1                      |                                   | Inc. <b>PC</b> .                                                    |         |
| 1.2                      | Read <b>PC</b>                    | Store as Operand.                                                   |         |
| 2.1                      |                                   | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |         |
| 2.2                      | Read Address                      | Discard.                                                            |         |
| 3.1                      |                                   |                                                                     |         |
| 3.2                      | Write to Address                  | Write <b>A</b> .                                                    |         |
| 4.1                      |                                   |                                                                     |         |
| 4.2                      | Write to ((Address + 1) & \$00FF) | Write <b>Y</b> .                                                    |         |
| 5.1                      | (( <b>PSW</b> & \$20) << 3)       |                                                                     |         |
| 5.2                      | Read <b>PC</b>                    | Store as OpCode.                                                    |         |
| +X.1                     |                                   |                                                                     |         |

|                                   |                  |                                                                         |
|-----------------------------------|------------------|-------------------------------------------------------------------------|
| MOV (DB)                          | Cycles: 5        | Size: 2                                                                 |
| X-Indexed Direct Page d+X (Write) |                  |                                                                         |
| Cycle                             | R/W              | Desc.                                                                   |
| -X.2                              | Read <b>PC</b>   | Store as OpCode.                                                        |
| 1.1                               |                  | Inc. <b>PC</b> .                                                        |
| 1.2                               | Read <b>PC</b>   | Store as Operand.                                                       |
| 2.1                               |                  | Inc. <b>PC</b> .                                                        |
| 2.2                               |                  |                                                                         |
| 3.1                               |                  | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3). |
| 3.2                               | Read Address     | Discard.                                                                |
| 4.1                               |                  |                                                                         |
| 4.2                               | Write to Address | Write <b>Y</b> .                                                        |
| 5.1                               |                  |                                                                         |
| 5.2                               | Read <b>PC</b>   | Store as OpCode.                                                        |
| +X.1                              |                  |                                                                         |

|          |                |                                                                                 |
|----------|----------------|---------------------------------------------------------------------------------|
| DEC (DC) | Cycles: 2      | Size: 1                                                                         |
| Implied  |                |                                                                                 |
| Cycle    | R/W            | Desc.                                                                           |
| -X.2     | Read <b>PC</b> | Store as OpCode.                                                                |
| 1.1      |                | Inc. <b>PC</b> .                                                                |
| 1.2      |                |                                                                                 |
| 2.1      |                | <b>Y</b> -= 1. <b>N</b> flag = ( <b>Y</b> & \$80), <b>Z</b> flag = ! <b>Y</b> . |
| 2.2      |                | Store as OpCode.                                                                |
| +X.1     |                |                                                                                 |

|          |                |                                                                                        |
|----------|----------------|----------------------------------------------------------------------------------------|
| MOV (DD) | Cycles: 2      | Size: 1                                                                                |
| Implied  |                |                                                                                        |
| Cycle    | R/W            | Desc.                                                                                  |
| -X.2     | Read <b>PC</b> | Store as OpCode.                                                                       |
| 1.1      |                | Inc. <b>PC</b> .                                                                       |
| 1.2      |                |                                                                                        |
| 2.1      |                | <b>A</b> = <b>Y</b> . <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 2.2      |                | Store as OpCode.                                                                       |
| +X.1     |                |                                                                                        |

|                                       |                |                                                                         |
|---------------------------------------|----------------|-------------------------------------------------------------------------|
| CBNE (DE)                             | Cycles: 6-8    | Size: 3                                                                 |
| X-Indexed Direct Page Relative d+X, r |                |                                                                         |
| Cycle                                 | R/W            | Desc.                                                                   |
| -X.2                                  | Read <b>PC</b> | Store as OpCode.                                                        |
| 1.1                                   |                | Inc. <b>PC</b> .                                                        |
| 1.2                                   | Read <b>PC</b> | Store as Operand.                                                       |
| 2.1                                   |                | Inc. <b>PC</b> .                                                        |
| 2.2                                   |                |                                                                         |
| 3.1                                   |                | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3). |
| 3.2                                   | Read Address   | Store as Operand.                                                       |
| 4.1                                   |                | Set Jump to ( <b>A</b> != Operand).                                     |
| 4.2                                   |                |                                                                         |
| 5.1                                   |                |                                                                         |
| 5.2                                   | Read <b>PC</b> | Store as Operand.                                                       |
| 6.1                                   |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 8.2).    |
| 6.2                                   |                |                                                                         |
| 7.1                                   |                | Address = <b>PC</b> + i8(Operand).<br><b>PC</b> .L = Address.L.         |
| 7.2                                   |                |                                                                         |
| 8.1                                   |                | <b>PC</b> .H = Address.H.                                               |
| 8.2                                   | Read <b>PC</b> | Store as OpCode.                                                        |
| +X.1                                  |                |                                                                         |

|          |                |                                                                                                                                            |
|----------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| DAA (DF) | Cycles: 3      | Size: 1                                                                                                                                    |
| Implied  |                |                                                                                                                                            |
| Cycle    | R/W            | Desc.                                                                                                                                      |
| -X.2     | Read <b>PC</b> | Store as OpCode.                                                                                                                           |
| 1.1      |                | Inc. <b>PC</b> .                                                                                                                           |
| 1.2      |                |                                                                                                                                            |
| 2.1      |                | If ( <b>C</b> flag    <b>A</b> > \$99):<br><b>A</b> += \$60, <b>C</b> flag = 1.                                                            |
| 2.2      |                |                                                                                                                                            |
| 3.1      |                | If ( <b>H</b> flag    ( <b>A</b> & \$0F) > \$09):<br><b>A</b> += \$06.<br><b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 3.2      |                | <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> .                                                                           |
|          | Read <b>PC</b> | Store as OpCode.                                                                                                                           |
| +X.1     |                |                                                                                                                                            |



|           |                |                                     |
|-----------|----------------|-------------------------------------|
| CLRV (E0) | Cycles: 2      | Size: 1                             |
| Implied   |                |                                     |
| Cycle     | R/W            | Desc.                               |
| -X.2      | Read <b>PC</b> | Store as OpCode.                    |
| 1.1       |                | Inc. <b>PC</b> .                    |
| 1.2       |                |                                     |
| 2.1       |                | <b>PSW</b> = ( <b>PSW</b> & ~\$40). |
| 2.2       |                | Store as OpCode.                    |
| +X.1      |                |                                     |

|            |                                     |                               |
|------------|-------------------------------------|-------------------------------|
| TCALL (E1) | Cycles: 8                           | Size: 1                       |
| Implied    |                                     |                               |
| Cycle      | R/W                                 | Desc.                         |
| -X.2       | Read <b>PC</b>                      | Store as OpCode.              |
| 1.1        |                                     | Inc. <b>PC</b> .              |
| 1.2        | Read <b>PC</b>                      | Store as Operand.             |
| 2.1        |                                     |                               |
| 2.2        |                                     |                               |
| 3.1        |                                     |                               |
| 3.2        | Write to u8( <b>SP</b> +0)   \$0100 | Push <b>PC</b> .H onto stack. |
| 4.1        |                                     |                               |
| 4.2        | Write to u8( <b>SP</b> -1)   \$0100 | Push <b>PC</b> .L onto stack. |
| 5.1        |                                     | Dec. <b>SP</b> by 2.          |
| 5.2        |                                     |                               |
| 6.1        |                                     |                               |
| 6.2        | Read \$FFC2                         | Store as <b>PC</b> .L.        |
| 7.1        |                                     |                               |
| 7.2        | Read \$FFC3                         | Store as <b>PC</b> .H.        |
| 8.1        |                                     |                               |
| 8.2        | Read <b>PC</b>                      | Store as OpCode.              |
| +X.1       |                                     |                               |

|                                         |                  |                                                                     |
|-----------------------------------------|------------------|---------------------------------------------------------------------|
| SET1 (E2)                               | Cycles: 4        | Size: 2                                                             |
| Direct Page Bit d.b (Read/Modify/Write) |                  |                                                                     |
| Cycle                                   | R/W              | Desc.                                                               |
| -X.2                                    | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                                     |                  | Inc. <b>PC</b> .                                                    |
| 1.2                                     | Read <b>PC</b>   | Store as Operand.                                                   |
| 2.1                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                                     | Read Address     | Store as Operand.                                                   |
| 3.1                                     |                  | Operand = (Operand & ~\$80)   \$80.                                 |
| 3.2                                     | Write to Address | Write Operand.                                                      |
| 4.1                                     |                  |                                                                     |
| 4.2                                     | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                                    |                  |                                                                     |

|                                 |                |                                                                      |
|---------------------------------|----------------|----------------------------------------------------------------------|
| BBS (E3)                        | Cycles: 5-7    | Size: 3                                                              |
| Direct Page Bit Relative d.b, r |                |                                                                      |
| Cycle                           | R/W            | Desc.                                                                |
| -X.2                            | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1                             |                | Inc. <b>PC</b> .                                                     |
| 1.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 2.1                             |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).  |
| 2.2                             | Read Address   | Store as Operand.                                                    |
| 3.1                             |                | Set Jump to (Operand & (1 << 7)) == (1 << 7).                        |
| 3.2                             |                |                                                                      |
| 4.1                             |                |                                                                      |
| 4.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 5.1                             |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 7.2). |
| 5.2                             |                |                                                                      |
| 6.1                             |                | Address = <b>PC</b> + i8(Operand).<br><b>PC</b> .L = Address.L.      |
| 6.2                             |                |                                                                      |
| 7.1                             |                | <b>PC</b> .H = Address.H.                                            |
| 7.2                             | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1                            |                |                                                                      |

|                      |                |                                                                                      |
|----------------------|----------------|--------------------------------------------------------------------------------------|
| MOV (E4)             | Cycles: 3      | Size: 2                                                                              |
| Direct Page d (Read) |                |                                                                                      |
| Cycle                | R/W            | Desc.                                                                                |
| -X.2                 | Read <b>PC</b> | Store as OpCode.                                                                     |
| 1.1                  |                | Inc. <b>PC</b> .                                                                     |
| 1.2                  | Read <b>PC</b> | Store as Operand.                                                                    |
| 2.1                  |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                  |
| 2.2                  | Read Address   | Store as Operand.                                                                    |
| 3.1                  |                | <b>A</b> = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 3.2                  | Read <b>PC</b> | Store as OpCode.                                                                     |
| +X.1                 |                |                                                                                      |

|                    |                |                                                                                      |
|--------------------|----------------|--------------------------------------------------------------------------------------|
| MOV (E5)           | Cycles: 4      | Size: 3                                                                              |
| Absolute !a (Read) |                |                                                                                      |
| Cycle              | R/W            | Desc.                                                                                |
| -X.2               | Read <b>PC</b> | Store as OpCode.                                                                     |
| 1.1                |                | Inc. <b>PC</b> .                                                                     |
| 1.2                | Read <b>PC</b> | Store as Address.L.                                                                  |
| 2.1                |                | Inc. <b>PC</b> .                                                                     |
| 2.2                | Read <b>PC</b> | Store as Address.H.                                                                  |
| 3.1                |                | Inc. <b>PC</b> .                                                                     |
| 3.2                | Read Address   | Store as Operand.                                                                    |
| 4.1                |                | <b>A</b> = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 4.2                | Read <b>PC</b> | Store as OpCode.                                                                     |
| +X.1               |                |                                                                                      |

|                                      |                                                |                                                                                      |
|--------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------|
| MOV (E6)      Cycles: 3      Size: 1 |                                                |                                                                                      |
| Indirect X (X) (Read)                |                                                |                                                                                      |
| Cycle                                | R/W                                            | Desc.                                                                                |
| -X.2                                 | Read <b>PC</b>                                 | Store as OpCode.                                                                     |
| 1.1                                  | Read ( <b>X</b>   (( <b>PSW</b> & \$20) << 3)) | Inc. <b>PC</b> .                                                                     |
| 1.2                                  |                                                |                                                                                      |
| 2.1                                  |                                                |                                                                                      |
| 2.2                                  |                                                | Store as Operand.                                                                    |
| 3.1                                  |                                                | <b>A</b> = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 3.2                                  |                                                | Store as OpCode.                                                                     |
| +X.1                                 |                                                |                                                                                      |

|                                 |                                                              |                                                                                      |
|---------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------|
| MOV (E7)                        | Cycles: 6                                                    | Size: 2                                                                              |
| X-Indexed Indirect [d+X] (Read) |                                                              |                                                                                      |
| Cycle                           | R/W                                                          | Desc.                                                                                |
| -X.2                            | Read <b>PC</b>                                               | Store as OpCode.                                                                     |
| 1.1                             |                                                              | Inc. <b>PC</b> .                                                                     |
| 1.2                             | Read <b>PC</b>                                               | Store as Operand.                                                                    |
| 2.1                             |                                                              | Inc. <b>PC</b> .                                                                     |
| 2.2                             |                                                              |                                                                                      |
| 3.1                             |                                                              | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).              |
| 3.2                             | Read Address                                                 | Store as Pointer.L.                                                                  |
| 4.1                             |                                                              |                                                                                      |
| 4.2                             | Read ((Address + 1) & \$FF)  <br>(( <b>PSW</b> & \$20) << 3) | Store as Pointer.H.                                                                  |
| 5.1                             |                                                              |                                                                                      |
| 5.2                             | Read Pointer                                                 | Store as Operand.                                                                    |
| 6.1                             |                                                              | <b>A</b> = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 6.2                             | Read <b>PC</b>                                               | Store as OpCode.                                                                     |
| +X.1                            |                                                              |                                                                                      |



|           |                |                                                                                                       |
|-----------|----------------|-------------------------------------------------------------------------------------------------------|
| MOV (E8)  | Cycles: 2      | Size: 2                                                                                               |
| Immediate |                |                                                                                                       |
| Cycle     | R/W            | Desc.                                                                                                 |
| -X.2      | Read <b>PC</b> | Store as OpCode.                                                                                      |
| 1.1       |                | Inc. <b>PC</b> .                                                                                      |
| 1.2       | Read <b>PC</b> | Store as Operand.                                                                                     |
| 2.1       |                | Inc. <b>PC</b> . <b>A</b> = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 2.2       | Read <b>PC</b> | Store as OpCode.                                                                                      |
| +X.1      |                |                                                                                                       |

|                    |                |                                                                                      |
|--------------------|----------------|--------------------------------------------------------------------------------------|
| MOV (E9)           | Cycles: 4      | Size: 3                                                                              |
| Absolute !a (Read) |                |                                                                                      |
| Cycle              | R/W            | Desc.                                                                                |
| -X.2               | Read <b>PC</b> | Store as OpCode.                                                                     |
| 1.1                |                | Inc. <b>PC</b> .                                                                     |
| 1.2                | Read <b>PC</b> | Store as Address.L.                                                                  |
| 2.1                |                | Inc. <b>PC</b> .                                                                     |
| 2.2                | Read <b>PC</b> | Store as Address.H.                                                                  |
| 3.1                |                | Inc. <b>PC</b> .                                                                     |
| 3.2                | Read Address   | Store as Operand.                                                                    |
| 4.1                |                | <b>X</b> = Operand. <b>N</b> flag = ( <b>X</b> & \$80), <b>Z</b> flag = ! <b>X</b> . |
| 4.2                | Read <b>PC</b> | Store as OpCode.                                                                     |
| +X.1               |                |                                                                                      |

|                          |                             |                                                |
|--------------------------|-----------------------------|------------------------------------------------|
| NOT1 (EA)                | Cycles: 5                   | Size: 3                                        |
| Absolute Boolean Bit m.b |                             |                                                |
| Cycle                    | R/W                         | Desc.                                          |
| -X.2                     | Read <b>PC</b>              | Store as OpCode.                               |
| 1.1                      |                             | Inc. <b>PC</b> .                               |
| 1.2                      | Read <b>PC</b>              | Store as Address.L.                            |
| 2.1                      |                             | Inc. <b>PC</b> .                               |
| 2.2                      | Read <b>PC</b>              | Store as Address.H.                            |
| 3.1                      |                             | Inc. <b>PC</b> .                               |
| 3.2                      | Read (Address & \$1FFF)     | Store as Operand.                              |
| 4.1                      |                             | Bit = Address >> 13.<br>Operand ^= (1 << Bit). |
| 4.2                      | Write to (Address & \$1FFF) | Write Operand.                                 |
| 5.1                      |                             |                                                |
| 5.2                      | Read <b>PC</b>              | Store as OpCode.                               |
| +X.1                     |                             |                                                |

|                      |                |                                                                                      |
|----------------------|----------------|--------------------------------------------------------------------------------------|
| MOV (EB)             | Cycles: 3      | Size: 2                                                                              |
| Direct Page d (Read) |                |                                                                                      |
| Cycle                | R/W            | Desc.                                                                                |
| -X.2                 | Read <b>PC</b> | Store as OpCode.                                                                     |
| 1.1                  |                | Inc. <b>PC</b> .                                                                     |
| 1.2                  | Read <b>PC</b> | Store as Operand.                                                                    |
| 2.1                  |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                  |
| 2.2                  | Read Address   | Store as Operand.                                                                    |
| 3.1                  |                | <b>Y</b> = Operand. <b>N</b> flag = ( <b>Y</b> & \$80), <b>Z</b> flag = ! <b>Y</b> . |
| 3.2                  | Read <b>PC</b> | Store as OpCode.                                                                     |
| +X.1                 |                |                                                                                      |

|                    |                |                                                                                      |
|--------------------|----------------|--------------------------------------------------------------------------------------|
| MOV (EC)           | Cycles: 4      | Size: 3                                                                              |
| Absolute !a (Read) |                |                                                                                      |
| Cycle              | R/W            | Desc.                                                                                |
| -X.2               | Read <b>PC</b> | Store as OpCode.                                                                     |
| 1.1                |                | Inc. <b>PC</b> .                                                                     |
| 1.2                | Read <b>PC</b> | Store as Address.L.                                                                  |
| 2.1                |                | Inc. <b>PC</b> .                                                                     |
| 2.2                | Read <b>PC</b> | Store as Address.H.                                                                  |
| 3.1                |                | Inc. <b>PC</b> .                                                                     |
| 3.2                | Read Address   | Store as Operand.                                                                    |
| 4.1                |                | <b>Y</b> = Operand. <b>N</b> flag = ( <b>Y</b> & \$80), <b>Z</b> flag = ! <b>Y</b> . |
| 4.2                | Read <b>PC</b> | Store as OpCode.                                                                     |
| +X.1               |                |                                                                                      |

|           |                |                     |
|-----------|----------------|---------------------|
| NOTC (ED) | Cycles: 3      | Size: 1             |
| Implied   |                |                     |
| Cycle     | R/W            | Desc.               |
| -X.2      | Read <b>PC</b> | Store as OpCode.    |
| 1.1       |                | Inc. <b>PC</b> .    |
| 1.2       |                |                     |
| 2.1       |                |                     |
| 2.2       |                |                     |
| 3.1       |                | <b>PSW</b> ^= \$01. |
| 3.2       |                | Store as OpCode.    |
| +X.1      |                |                     |

|          |                |                      |
|----------|----------------|----------------------|
| POP (EE) | Cycles: 4      | Size: 1              |
| Implied  |                |                      |
| Cycle    | R/W            | Desc.                |
| -X.2     | Read <b>PC</b> | Store as OpCode.     |
| 1.1      | Read <b>PC</b> | Inc. <b>PC</b> .     |
| 1.2      |                | Discard.             |
| 2.1      |                |                      |
| 2.2      |                |                      |
| 3.1      |                | Inc. <b>SP</b> by 1. |
| 3.2      |                | Pull to <b>Y</b> .   |
| 4.1      |                |                      |
| 4.2      |                | Store as OpCode.     |
| +X.1     |                |                      |

|            |                |                                             |
|------------|----------------|---------------------------------------------|
| SLEEP (EF) | Cycles: 3      | Size: 1                                     |
| Implied    |                |                                             |
| Cycle      | R/W            | Desc.                                       |
| -X.2       | Read <b>PC</b> | Store as OpCode.                            |
| 1.1        |                | Inc. <b>PC</b> .                            |
| 1.2        |                | Discard.                                    |
| 2.1        |                |                                             |
| 2.2        |                |                                             |
| 3.1        |                | Return to cycle 1 (next half-cycle is 1.2). |



|            |                |                                                                      |
|------------|----------------|----------------------------------------------------------------------|
| BEQ (F0)   | Cycles: 2-4    | Size: 2                                                              |
| Relative r |                |                                                                      |
| Cycle      | R/W            | Desc.                                                                |
| -X.2       | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1        |                | Inc. <b>PC</b> . Set Jump to ( <b>PSW</b> & \$02) == \$02.           |
| 1.2        |                | Store as Operand.                                                    |
| 2.1        |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 4.2). |
| 2.2        |                |                                                                      |
| 3.1        |                | Address = <b>PC</b> + i8(Operand).                                   |
|            |                | <b>PC</b> .L = Address.L.                                            |
| 3.2        |                |                                                                      |
| 4.1        |                | <b>PC</b> .H = Address.H.                                            |
| 4.2        | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1       |                |                                                                      |

|            |                                     |                               |
|------------|-------------------------------------|-------------------------------|
| TCALL (F1) | Cycles: 8                           | Size: 1                       |
| Implied    |                                     |                               |
| Cycle      | R/W                                 | Desc.                         |
| -X.2       | Read <b>PC</b>                      | Store as OpCode.              |
| 1.1        |                                     | Inc. <b>PC</b> .              |
| 1.2        | Read <b>PC</b>                      | Store as Operand.             |
| 2.1        |                                     |                               |
| 2.2        |                                     |                               |
| 3.1        |                                     |                               |
| 3.2        | Write to u8( <b>SP</b> +0)   \$0100 | Push <b>PC</b> .H onto stack. |
| 4.1        |                                     |                               |
| 4.2        | Write to u8( <b>SP</b> -1)   \$0100 | Push <b>PC</b> .L onto stack. |
| 5.1        |                                     | Dec. <b>SP</b> by 2.          |
| 5.2        |                                     |                               |
| 6.1        |                                     |                               |
| 6.2        | Read \$FFC0                         | Store as <b>PC</b> .L.        |
| 7.1        |                                     |                               |
| 7.2        | Read \$FFC1                         | Store as <b>PC</b> .H.        |
| 8.1        |                                     |                               |
| 8.2        | Read <b>PC</b>                      | Store as OpCode.              |
| +X.1       |                                     |                               |

|                                         |                  |                                                                     |
|-----------------------------------------|------------------|---------------------------------------------------------------------|
| CLR1 (F2)                               | Cycles: 4        | Size: 2                                                             |
| Direct Page Bit d.b (Read/Modify/Write) |                  |                                                                     |
| Cycle                                   | R/W              | Desc.                                                               |
| -X.2                                    | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                                     |                  | Inc. <b>PC</b> .                                                    |
| 1.2                                     | Read <b>PC</b>   | Store as Operand.                                                   |
| 2.1                                     |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                                     | Read Address     | Store as Operand.                                                   |
| 3.1                                     |                  | Operand = (Operand & ~\$80)   \$00.                                 |
| 3.2                                     | Write to Address | Write Operand.                                                      |
| 4.1                                     |                  |                                                                     |
| 4.2                                     | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                                    |                  |                                                                     |

|                                 |                |                                                                      |
|---------------------------------|----------------|----------------------------------------------------------------------|
| BBC (F3)                        | Cycles: 5-7    | Size: 3                                                              |
| Direct Page Bit Relative d.b, r |                |                                                                      |
| Cycle                           | R/W            | Desc.                                                                |
| -X.2                            | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1                             |                | Inc. <b>PC</b> .                                                     |
| 1.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 2.1                             |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).  |
| 2.2                             | Read Address   | Store as Operand.                                                    |
| 3.1                             |                | Set Jump to (Operand & (1 << 7)) == (0 << 7).                        |
| 3.2                             |                |                                                                      |
| 4.1                             |                |                                                                      |
| 4.2                             | Read <b>PC</b> | Store as Operand.                                                    |
| 5.1                             |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 7.2). |
| 5.2                             |                |                                                                      |
| 6.1                             |                | Address = <b>PC</b> + i8(Operand).<br><b>PC</b> .L = Address.L.      |
| 6.2                             |                |                                                                      |
| 7.1                             |                | <b>PC</b> .H = Address.H.                                            |
| 7.2                             | Read <b>PC</b> | Store as OpCode.                                                     |
| +X.1                            |                |                                                                      |

|                                  |                |                                                                                      |
|----------------------------------|----------------|--------------------------------------------------------------------------------------|
| MOV (F4)                         | Cycles: 4      | Size: 2                                                                              |
| X-Indexed Direct Page d+X (Read) |                |                                                                                      |
| Cycle                            | R/W            | Desc.                                                                                |
| -X.2                             | Read <b>PC</b> | Store as OpCode.                                                                     |
| 1.1                              |                | Inc. <b>PC</b> .                                                                     |
| 1.2                              | Read <b>PC</b> | Store as Operand.                                                                    |
| 2.1                              |                | Inc. <b>PC</b> .                                                                     |
| 2.2                              |                |                                                                                      |
| 3.1                              |                | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).              |
| 3.2                              | Read Address   | Store as Operand.                                                                    |
| 4.1                              |                | <b>A</b> = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 4.2                              | Read <b>PC</b> | Store as OpCode.                                                                     |
| +X.1                             |                |                                                                                      |

|                                |                |                                                                                      |
|--------------------------------|----------------|--------------------------------------------------------------------------------------|
| MOV (F5)                       | Cycles: 5      | Size: 3                                                                              |
| X-Indexed Absolute !a+X (Read) |                |                                                                                      |
| Cycle                          | R/W            | Desc.                                                                                |
| -X.2                           | Read <b>PC</b> | Store as OpCode.                                                                     |
| 1.1                            |                | Inc. <b>PC</b> .                                                                     |
| 1.2                            | Read <b>PC</b> | Store as Address.L.                                                                  |
| 2.1                            |                | Inc. <b>PC</b> .                                                                     |
| 2.2                            | Read <b>PC</b> | Store as Address.H.                                                                  |
| 3.1                            |                | Inc. <b>PC</b> .                                                                     |
| 3.2                            |                |                                                                                      |
| 4.1                            |                | Pointer = ( <b>X</b> + Address).                                                     |
| 4.2                            | Read Pointer   | Store as Operand.                                                                    |
| 5.1                            |                | <b>A</b> = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 5.2                            | Read <b>PC</b> | Store as OpCode.                                                                     |
| +X.1                           |                |                                                                                      |

|                                |                |                                                                                      |
|--------------------------------|----------------|--------------------------------------------------------------------------------------|
| MOV (F6)                       | Cycles: 5      | Size: 3                                                                              |
| Y-Indexed Absolute !a+Y (Read) |                |                                                                                      |
| Cycle                          | R/W            | Desc.                                                                                |
| -X.2                           | Read <b>PC</b> | Store as OpCode.                                                                     |
| 1.1                            |                | Inc. <b>PC</b> .                                                                     |
| 1.2                            | Read <b>PC</b> | Store as Address.L.                                                                  |
| 2.1                            |                | Inc. <b>PC</b> .                                                                     |
| 2.2                            | Read <b>PC</b> | Store as Address.H.                                                                  |
| 3.1                            |                | Inc. <b>PC</b> .                                                                     |
| 3.2                            |                |                                                                                      |
| 4.1                            |                | Pointer = ( <b>Y</b> + Address).                                                     |
| 4.2                            | Read Pointer   | Store as Operand.                                                                    |
| 5.1                            |                | <b>A</b> = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 5.2                            | Read <b>PC</b> | Store as OpCode.                                                                     |
| +X.1                           |                |                                                                                      |

|                                 |                                                           |                                                                                      |
|---------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------|
| MOV (F7)                        | Cycles: 6                                                 | Size: 2                                                                              |
| Indirect Y-Indexed [d]+Y (Read) |                                                           |                                                                                      |
| Cycle                           | R/W                                                       | Desc.                                                                                |
| -X.2                            | Read <b>PC</b>                                            | Store as OpCode.                                                                     |
| 1.1                             |                                                           | Inc. <b>PC</b> .                                                                     |
| 1.2                             | Read <b>PC</b>                                            | Store as Operand.                                                                    |
| 2.1                             |                                                           | Inc. <b>PC</b> .                                                                     |
| 2.2                             |                                                           |                                                                                      |
| 3.1                             |                                                           | Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                                   |
| 3.2                             | Read Address                                              | Store as Pointer.L.                                                                  |
| 4.1                             |                                                           |                                                                                      |
| 4.2                             | Read ((Address + 1) & \$FF)   (( <b>PSW</b> & \$20) << 3) | Store as Pointer.H.                                                                  |
| 5.1                             |                                                           | Address = ( <b>Y</b> + Pointer).                                                     |
| 5.2                             | Read Address                                              | Store as Operand.                                                                    |
| 6.1                             |                                                           | <b>A</b> = Operand. <b>N</b> flag = ( <b>A</b> & \$80), <b>Z</b> flag = ! <b>A</b> . |
| 6.2                             | Read <b>PC</b>                                            | Store as OpCode.                                                                     |
| +X.1                            |                                                           |                                                                                      |



|                      |                |                                                                                      |
|----------------------|----------------|--------------------------------------------------------------------------------------|
| MOV (F8)             | Cycles: 3      | Size: 2                                                                              |
| Direct Page d (Read) |                |                                                                                      |
| Cycle                | R/W            | Desc.                                                                                |
| -X.2                 | Read <b>PC</b> | Store as OpCode.                                                                     |
| 1.1                  |                | Inc. <b>PC</b> .                                                                     |
| 1.2                  | Read <b>PC</b> | Store as Operand.                                                                    |
| 2.1                  |                | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                  |
| 2.2                  | Read Address   | Store as Operand.                                                                    |
| 3.1                  |                | <b>X</b> = Operand. <b>N</b> flag = ( <b>X</b> & \$80), <b>Z</b> flag = ! <b>X</b> . |
| 3.2                  | Read <b>PC</b> | Store as OpCode.                                                                     |
| +X.1                 |                |                                                                                      |

|                                  |                |                                                                                      |
|----------------------------------|----------------|--------------------------------------------------------------------------------------|
| MOV (F9)                         | Cycles: 4      | Size: 2                                                                              |
| Y-Indexed Direct Page d+Y (Read) |                |                                                                                      |
| Cycle                            | R/W            | Desc.                                                                                |
| -X.2                             | Read <b>PC</b> | Store as OpCode.                                                                     |
| 1.1                              |                | Inc. <b>PC</b> .                                                                     |
| 1.2                              | Read <b>PC</b> | Store as Operand.                                                                    |
| 2.1                              |                | Inc. <b>PC</b> .                                                                     |
| 2.2                              |                |                                                                                      |
| 3.1                              |                | Address = (( <b>Y</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).              |
| 3.2                              | Read Address   | Store as Operand.                                                                    |
| 4.1                              |                | <b>X</b> = Operand. <b>N</b> flag = ( <b>X</b> & \$80), <b>Z</b> flag = ! <b>X</b> . |
| 4.2                              | Read <b>PC</b> | Store as OpCode.                                                                     |
| +X.1                             |                |                                                                                      |

|                                           |                  |                                                                     |
|-------------------------------------------|------------------|---------------------------------------------------------------------|
| MOV (FA)                                  | Cycles: 5        | Size: 3                                                             |
| Direct Page to Direct Page dd, ds (Write) |                  |                                                                     |
| Cycle                                     | R/W              | Desc.                                                               |
| -X.2                                      | Read <b>PC</b>   | Store as OpCode.                                                    |
| 1.1                                       |                  | Inc. <b>PC</b> .                                                    |
| 1.2                                       | Read <b>PC</b>   | Store as Operand.                                                   |
| 2.1                                       |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 2.2                                       | Read Address     | Store as Operand1.                                                  |
| 3.1                                       |                  |                                                                     |
| 3.2                                       | Read <b>PC</b>   | Store as Operand.                                                   |
| 4.1                                       |                  | Inc. <b>PC</b> . Address = (Operand   (( <b>PSW</b> & \$20) << 3)). |
| 4.2                                       | Write to Address | Write Operand1.                                                     |
| 5.1                                       |                  |                                                                     |
| 5.2                                       | Read <b>PC</b>   | Store as OpCode.                                                    |
| +X.1                                      |                  |                                                                     |

|                                  |                |                                                                                      |
|----------------------------------|----------------|--------------------------------------------------------------------------------------|
| MOV (FB)                         | Cycles: 4      | Size: 2                                                                              |
| X-Indexed Direct Page d+X (Read) |                |                                                                                      |
| Cycle                            | R/W            | Desc.                                                                                |
| -X.2                             | Read <b>PC</b> | Store as OpCode.                                                                     |
| 1.1                              |                | Inc. <b>PC</b> .                                                                     |
| 1.2                              | Read <b>PC</b> | Store as Operand.                                                                    |
| 2.1                              |                | Inc. <b>PC</b> .                                                                     |
| 2.2                              |                |                                                                                      |
| 3.1                              |                | Address = (( <b>X</b> + Operand) & \$FF)   (( <b>PSW</b> & \$20) << 3).              |
| 3.2                              | Read Address   | Store as Operand.                                                                    |
| 4.1                              |                | <b>Y</b> = Operand. <b>N</b> flag = ( <b>Y</b> & \$80), <b>Z</b> flag = ! <b>Y</b> . |
| 4.2                              | Read <b>PC</b> | Store as OpCode.                                                                     |
| +X.1                             |                |                                                                                      |

|          |                |                                                                                 |
|----------|----------------|---------------------------------------------------------------------------------|
| INC (FC) | Cycles: 2      | Size: 1                                                                         |
| Implied  |                |                                                                                 |
| Cycle    | R/W            | Desc.                                                                           |
| -X.2     | Read <b>PC</b> | Store as OpCode.                                                                |
| 1.1      |                | Inc. <b>PC</b> .                                                                |
| 1.2      |                |                                                                                 |
| 2.1      |                | <b>Y</b> += 1. <b>N</b> flag = ( <b>Y</b> & \$80), <b>Z</b> flag = ! <b>Y</b> . |
| 2.2      |                | Store as OpCode.                                                                |
| +X.1     |                |                                                                                 |

|          |                |                                                        |
|----------|----------------|--------------------------------------------------------|
| MOV (FD) | Cycles: 2      | Size: 1                                                |
| Implied  |                |                                                        |
| Cycle    | R/W            | Desc.                                                  |
| -X.2     | Read <b>PC</b> | Store as OpCode.                                       |
| 1.1      |                | Inc. <b>PC</b> .                                       |
| 1.2      |                |                                                        |
| 2.1      |                | <b>Y = A. N</b> flag = (Y & \$80), <b>Z</b> flag = !Y. |
| 2.2      |                | Store as OpCode.                                       |
| +X.1     |                |                                                        |

|                |                |                                                                      |
|----------------|----------------|----------------------------------------------------------------------|
| DBNZ (FE)      | Cycles: 4-6    | Size: 2                                                              |
| Relative r (Y) |                |                                                                      |
| Cycle          | R/W            | Desc.                                                                |
| -X.2           | Read <b>PC</b> | Store as OpCode.                                                     |
| 1.1            |                | Inc. <b>PC</b> .                                                     |
| 1.2            |                | Discard.                                                             |
| 2.1            |                | Address = (Operand   (( <b>PSW</b> & \$20) << 3)).                   |
| 2.2            |                |                                                                      |
| 3.1            |                | -- <b>Y</b> . Set Jump to ( <b>Y</b> != 0).                          |
| 3.2            |                | Store as Operand.                                                    |
| 4.1            |                | Inc. <b>PC</b> . If !Jump, end instruction (next half-cycle is 6.2). |
| 4.2            |                |                                                                      |
| 5.1            |                | Address = <b>PC</b> + i8(Operand).<br><b>PC</b> .L = Address.L.      |
| 5.2            |                |                                                                      |
| 6.1            |                | <b>PC</b> .H = Address.H.                                            |
| 6.2            |                | Store as OpCode.                                                     |
| +X.1           |                |                                                                      |

|           |                |                                             |
|-----------|----------------|---------------------------------------------|
| STOP (FF) | Cycles: 3      | Size: 1                                     |
| Implied   |                |                                             |
| Cycle     | R/W            | Desc.                                       |
| -X.2      | Read <b>PC</b> | Store as OpCode.                            |
| 1.1       |                | Inc. <b>PC</b> .                            |
| 1.2       |                | Discard.                                    |
| 2.1       |                |                                             |
| 2.2       |                |                                             |
| 3.1       |                | Return to cycle 1 (next half-cycle is 1.2). |